

FX & FXO

FIX API Rules of Engagement

Version 1.4.3



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Document Control

Date	Version	Author	Description
8/14/2015	1.0.5	M. Mercado	First version of the combined BGC and GFI/GFI Direct FIX specification.
8/25/2015	1.0.6	M. Mercado	Added connection details and re-formatted some message tables.
11/23/2015	1.0.7	N. Chhonker	BGC connection details for internet IP modified for BETA environment.
2/3/2016	1.0.8	M. Mercado	Reformatted Execution Report (8) and News (B) tables; standardize some field names
3/3/2016	1.0.9	M. Mercado	Inserted Introducing Broker Venue Appendix; corrected typos
3/31/2016	1.0.10	M. Mercado	Edited connection details; added EligibleVenues (6003) to (W)
4/29/2016	1.0.11	M. Mercado, S. Agarwal	Added "Regulation Venue Type" sections throughout ROE; edited tags: RegulationVenueType (8505), Structure (9126) and LegDeliveryType (9127); Added New Order Single in section 2.5.1.4
10/03/2016	1.0.12	M. Mercado	Minor correction in Quote Request Reject (AG); (290) is not required; removed "7" from (297)
10/28/2016	1.0.13	M. Mercado	Added value to ManualQuoteIndicator (6289); added MXN-MX cut
12/13/2016	1.0.14	M. Mercado	Updated Appendix H: Tenors
05/04/2017	1.0.15	M. Mercado	Removed "Regulation Venue Type" sections; removed (8505)=4 (IB-SEF) and (8505)=6 (IB-MTF)
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07/5/2017	1.1.1	M. Mercado	Additional MiFID II-related changes outlined in Appendix L
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08/11/2017	1.1.3	M. Mercado	Added SecurityAltID (455), LegSecurityAltID (605), and related tags to New Order – Multileg (AB); added SecurityAltID (455), LegSecurityAltID (605), CFICode (461), LegCFICode (608), and related tags to Execution Report (8)
10/30/2017	1.1.4	N. Nasa S. Agarwal	Added MiFID II related tags - NoTrdTegPublications group to message 35=AB. This is needed for the Fenics Direct business only. Added MiFID II related tags - NoTrdTegPublications group to message 35=8 Removed Benchmark trade type from TradeCaptureReport Connection details in appendix C RUB cut in Appendix E On the Market Data Configuration, IsOwner and Eligible Venues order corrected
12/05/2017	1.1.5	Eric.Jawitz	Removed 454, 455, 456, 461, 24201, 24202, 24203 from (I) message Changed order of 453 group to be 448, 447, 452 Removed tag 54 from 296->295 Added to the AI message 434: 452, 448, 447 2593: 2594, 2595 Moved deferral to leg level for leg and hedge in AE and AB: Separate Deferral and Waiver, Deferral moved to leg level Added 1000 to hedge leg fields in AE and AB Changed Leg to Hedge in field names 10604-10608 Changed the word MTF to Mifid in many places throught the document, changed 8505 3 = MiFID, removed 8505=9 and 97, Changed 6003 to these new numbers Removed repeating group of 453 and 2593 in the G message In NoHedgeLegs 6764 changed 10670 Deferral to 10671 HedgeDeferral
12/07/2017	1.1.5.1	Eric Jawitz S. Agarwal	Changed 10670 Deferral to LegRegPublicationReason Changed 10671 HedgeDeferral to HedgeRegPublicationReason Added valid values for 2595 (Y or N)
29/11/2018	1.1.6	S.Agarwal	Appendix G, added a new value 7 The whole book is published on 35=W without any depth restrictions Appendix L removed. (MiFID highlights)
08/07/2019	1.1.7	S.Agarwal	PremiumCcy (5830) added as an optional field in SecurityList (35=y) Appendix I updated for 9125=3 to be 11:30 MXN rather than 12:30 MXN Enhanced 8588. all-in-price for NDFs and points for FX fws
28/04/2020	1.2	S.Agarwal C.Hidalgo	Added LoTrigger and HiTrigger fields to R, S, AE and 8 messages Added Exotics to 9019, Appendix G and B Removed Appendix E Corrected the name for Fenics Direct platform across the document Removed stage 1 for GFI in Appendix C.

			Added port 20000 in Appendix C
07/08/2020	1.2.1	S.Agarwal	Updated INR cut to be MUM 13:30; KRSE to be SL 15:30 Updated Appendix C with DR connectivity Added optional tag 969 in 35=y
15/09/2023	1.2.2	S.Agarwal	Updated IDR and MXN cuts in Appendix I Updated Appendix C by removing BT Radianz
31/10/2023	1.3	S.Agarwal	Added new cuts for Fixings in Appendix I. Unused cuts removed. Added LegUPICode (2893), HedgeUPICode (10609) – suppressed by default Removed Agency Quote Request and Broker Agent RFQ request. They are not supported
21/12/2023	1.3.1	S.Agarwal	Added LegUPICode (2893) to 35=AE – suppressed by default Added HedgeMIC (7931) to 35=AE and 35=AB – suppressed by default Updated the fixings in Appendix I.
03/01/2024	1.3.2	S.Agarwal	LegUPICode (2893) removed from 35=R
08/08/2024	1.3.3	S.Agarwal	RTN clarification added in 35=AB and 35=AE Appendix I, 9125=8, changed from SG11 to JKT
23/08/2024	1.4	S.Agarwal	Added Package fields (PackageIdentifier (9246), PackagePrice (9247), PackagePriceType (9248), PackagePriceCurrency (9249), PackageType (9250)) to 35=AB and 35=AE Added GFIMidPoint to 35=S, 35=AI, 35=8,35=AE,35=AB These fields are configurable.
19/11/2024	1.4.1	S. Agarwal	New cut mappings added in Appendix I (171 , 172, 173, 174, 175, 178)
10/12/2024	1.4.2	S. Agarwal	Correction to 9248 and 9250 valid values
20/05/2025	1.4.3	S. Agarwal	Fixing References added in Appendix I (156.....182)

1 Introduction

The purpose of this document is to provide details on the various workflow descriptions, connectivity protocols, and message formats exchanged between the application and an electronic trading partner (API client). The FIX 4.4 protocol is the messaging format used by the FIX API. Refer to the FIX Protocol web site (<http://www.fixprotocol.org>) for more information on FIX messaging.

2 Executive Summary

2.1 Front-End Trading Platform

The front-end trading platform facilitates inter-dealer trading for FX and FX options contracts. It supports the hybrid name-give-up brokerage model, combining traditional voice brokerage services with electronic trading technology using both Central Limit Order Book (CLOB) and Request-for-Quote (RFQ) models. Note that the terms RFQ and RFS (Request-for-Stream) are used interchangeably in this document.

2.2 Fenics Direct

Fenics Direct is a fully-electronic dealer-to-client RFQ mechanism which provides its clients the ability to request a direct FXO quote from a specified liquidity bank, followed by execution and STP into Fenics Direct. Matching and routing logic are handled by the front-end.

2.3 Agency Model

The agency model is a fully-electronic market, which allows selected agent broker dealers to RFQ on behalf of their clients to specified liquidity banks. The agency model is anonymous to brokers, utilizing the “Direct RFQ” protocol. Market-making banks and agency banks must go through an on-boarding process before they can have access to the system.

2.4 Available Products

Deliverable and non-deliverable forwards are available on the FIX API, as well as deliverable and non-deliverable options with, up to four call / put/ up or down legs.

2.5 Trading Protocol

The FIX API offers a distinct set of trading protocol for the Central Limit Order Book and Direct RFQ.

2.5.1 Central Limit Order Book (CLOB)

The CLOB is an entry-time-based first-in/first-out protocol. Quotes are anonymous pre-match or pre-execution, but are counterparty name-give-up at match or execution, and afterwards. A CLOB RFQ is a

request which creates a CLOB for a bespoke contract. Once the bespoke contract is created, the price build-up and trade protocol will follow CLOB rules. The platform has specific logic to disseminate quotes in response to a CLOB RFQ, which is an incentive for market makers. RFQs may be created by:

1. Clients using the GUI
2. Brokers acting on behalf of clients
3. Clients with FIX API connections

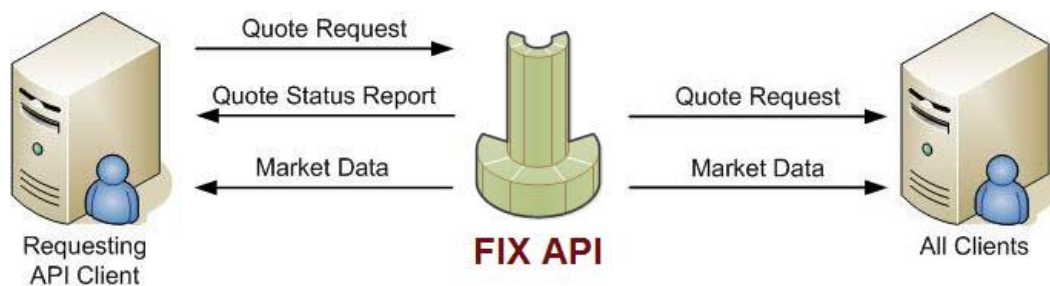
The FIX API uses market data messages to provide both aggregated quotes and information on market development in the CLOB RFQ. Note that this is different from Direct RFQ.

2.5.1.1 CLOB RFQ Rules

1. There is name give-up upon execution.
2. RFQ will create a CLOB for the requested contract for the duration that the contract is live.
3. The originating interest will see all price development.
4. Quotes will be tradeable to the originating interest on a first-look basis for a specified period.
5. Quotes will be ranked by best bid/best offer independently.
6. Quotes and price development are visible only to top-ranked market maker banks.
7. Quotes will be non-tradeable to the ranked market makers until:
 - a. originating interest passes, or
 - b. originating interest trades, or
 - c. originating interest provides a counter, or
 - d. first-look period expires after which the quotes will be tradeable.
8. Quotes will need to meet a minimum percentage of the RFQ size.
9. Quotes can be optionally created with a last-look feature.

2.5.1.2 Example: CLOB RFQ Originating from API Client

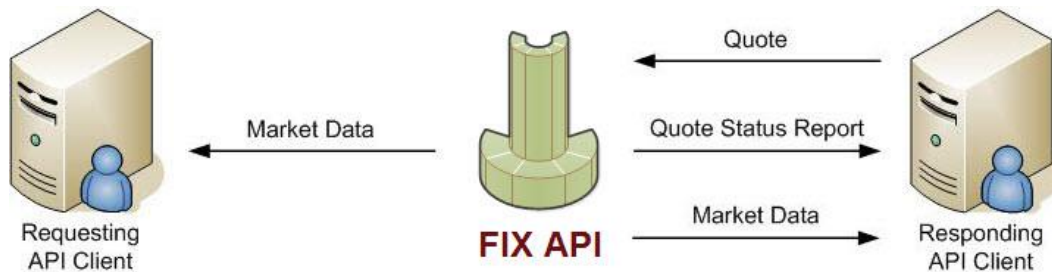
1. API client sends a Quote Request (R) message to create a specific contract.
2. The FIX API publishes to the API client a Quote Status Report (AI) message, acknowledging receipt of the Quote Request (R) message.
3. The FIX API publishes to all API clients a Quote Request (R) message anonymously on behalf of the originating API client.



2.5.1.3 CLOB RFQ Response

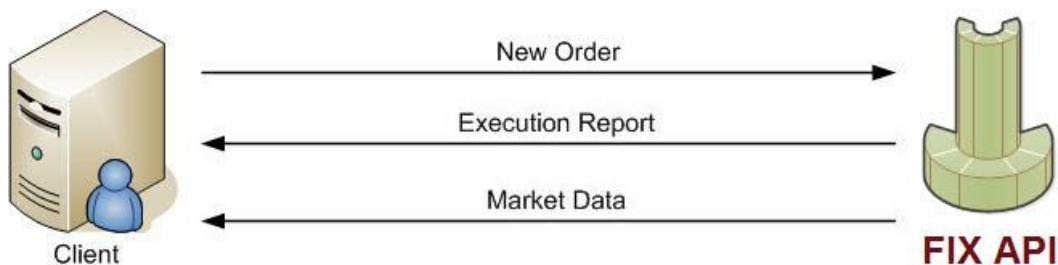
1. Market making banks respond to the RFQ with Quote (S) messages.
2. The FIX API disseminates Market Data (W) messages to ranked market making banks and originating interest for each price update.

3. API clients send Quote (S) messages to the FIX API to contribute prices for the specific interest.
4. API clients receive a Quote Status Report (AI) message for each Quote (S) message they send.
5. Quote Status Report (AI) messages inform API clients that their quotes were either accepted or rejected.
6. The FIX API will publish Market Data (W) messages containing new or updated prices and changes to the tradeable state to top-ranked market makers.



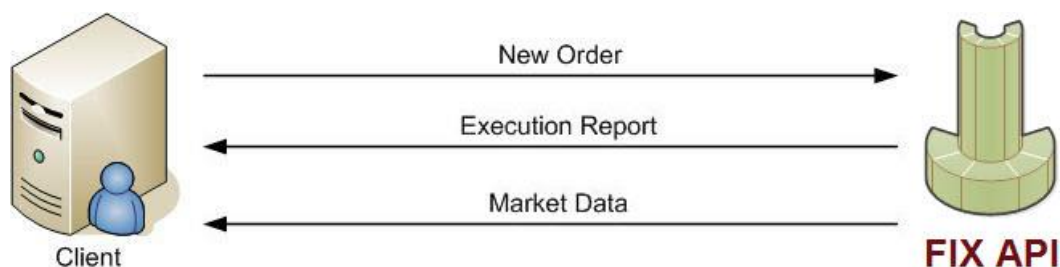
2.5.1.4 New Order or Trade Originating from CLOB

1. When an API client sends a New Order Single (D) message, the FIX API will publish an Execution Report (8) message to the API client.
2. If the order trades, the FIX API will publish an Execution Report (8) message to the API client and counterparties indicating the executed price and size.
3. The FIX API also publishes Market Data (W) messages to top-ranked market makers reflecting the change to the market as the result of the execution.



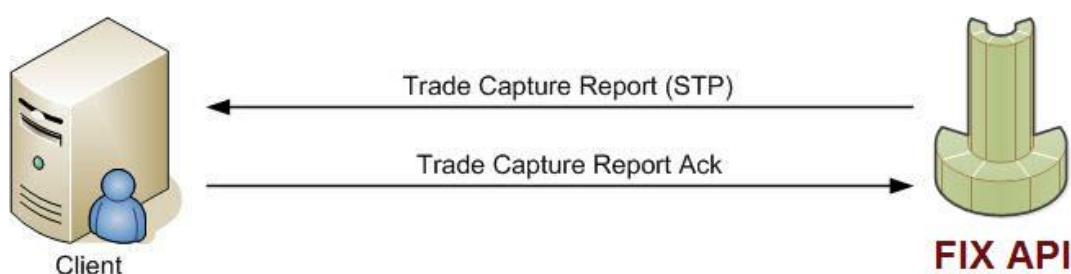
2.5.1.5 New Order or Trade Originating from CLOB RFQ

1. When an API client sends a New Order – Multileg (AB) message, the FIX API will publish an Execution Report (8) message to the API client.
2. If the order trades, the FIX API will publish an Execution Report (8) message to the API client and counterparties indicating the executed price and size.
3. The FIX API also publishes Market Data (W) messages to top-ranked market makers reflecting the change to the market as the result of the execution.



2.5.1.6 FIX API-Initiated STP Details

1. The FIX API publishes STP details in Trade Capture Report (AE) messages.
2. API clients publish Trade Capture Report Ack (AQ) messages to confirm receipt and agreement, or rejection, of the STP details.



2.5.2 Direct RFQ

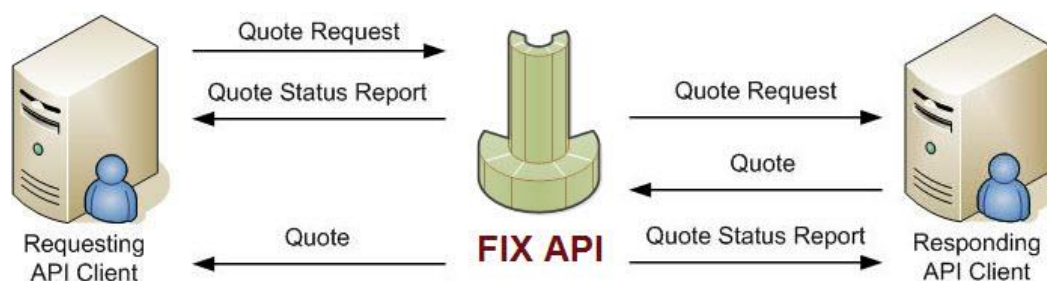
Direct RFQs are made from an originator to a specific market making bank to request a quote (or stream of quotes) on a specific contract. Direct RFQs are visible only to the originator and the market maker, and exclude brokers and other market participants. There is complete transparency of the counterparty names between participants throughout the process.

2.5.2.1 Direct RFQ Rules

1. Requestor and market maker names are known (non-anonymous).
2. Request, quote and execution are known only between requestor and market maker.
3. Requestor and market maker must both be configured properly in the admin to allow the RFQ to take place.
4. Direct RFQ is fully electronic.
5. The responding quote for the RFQ must be for the entire quantity.

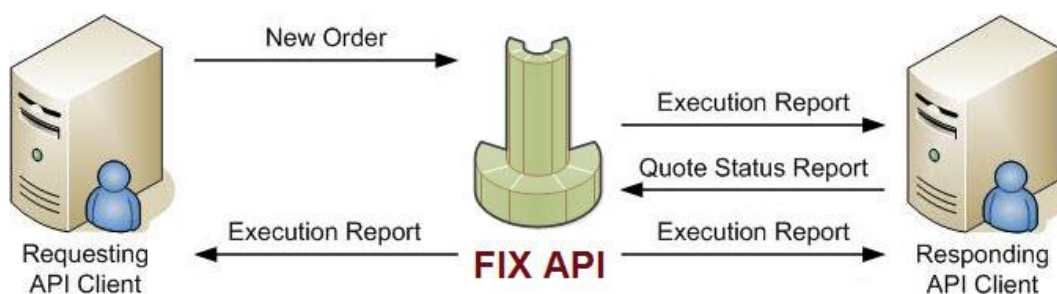
2.5.2.2 Example: Direct Quote Request Initiated by API Client

1. API client sends a Quote Request (R) message to the FIX API for a target market maker bank.
2. The FIX API sends the API client a Quote Status Report (AI) message, acknowledging receipt of the Quote Request (R) message.
3. The FIX API sends a Quote Request (R) message on behalf of originating client to the market maker.
4. Market maker sends a Quote (S) message to the FIX API.
5. The FIX API sends a Quote Status Report (AI) message to the market maker to acknowledge receipt of the Quote (S) message.
6. The FIX API sends the Quote (S) message to the originating API client.



2.5.2.3 New Order Originating from Direct RFQ

1. API client sends a New Order – Multileg (AB).
2. The FIX API sends an Execution Report (8) message to the market maker that requested to trade.
3. Market maker sends a Quote Status Report (AI) message to indicate acceptance or rejection.
4. The FIX API sends an Execution Report (8) message to counterparties indicating the trade status.



2.5.2.4 STP Details for Direct Quote Executions Initiated by FIX API

Where necessary, the FIX API will provide STP details for direct quote executions. It publishes the STP details in a Trade Capture Report (AE) message. The API client, in return, will publish a Trade Capture Report Ack (AR) message to confirm receipt and to agree to, or reject, the STP details.

2.6 Matching Engine

The matching engine contains logic for CLOB, RFQ routing, and order matching. The CLOB price stack and matching priority are timestamp-based. Increases in CLOB size will reset the timestamp, and re-adjust position within a price stack. Decreases in size will not affect price position within the stack. CLOB executions will be filled at best bid/best offer (BBO) based either on price or volatility up to the requested price and size. The matching engine takes into account counterparty credit.

2.7 FX Blotter STP Engine

FX Blotter is a web-based, trade capture tool for transacted deals. It contains the logic for sending to client systems STP details that can be confirmed manually or electronically, via GUI or API, respectively.

2.8 FIX Engine and Messages

The FIX API engine will handle input and output messages between the client, the matching engine, and FX Blotter STP engine. The following are primary messages used in the FIX API.

2.8.1 Quote Request (R)

The Quote Request (R) message is used to create an RFQ for a specific contract.

2.8.2 Quote Request Reject (AG)

The Quote Request Reject (AG) message is used to reject Quote Request (R) messages.

2.8.3 Quote (S)

The Quote (S) message is used to respond to Quote Request (R) messages. The FIX API will send a Quote Status Report (AI) message to acknowledge specific interest quotes.

2.8.4 Quote Cancel (Z)

The Quote Cancel (Z) message is used to remove quotes from the market.

2.8.5 Quote Status Report (AI)

The Quote Status Report (AI) message is published to acknowledge receipt of Quote Request (R), Quote (S), and Quote Cancel (Z) messages.

2.8.6 Market Data (W) – Snapshot/Full Refresh

The FIX API sends unsolicited Market Data (W) messages upon price and size changes. The messages contain aggregate size for every price level and side in the stack for a given contract. The FIX API sends market data messages to API clients to notify them of price and/or size changes. API clients can request a refresh of prices by sending a Market Data Request (V) message. API clients may subscribe to the FIX API by sending a Market Data Request (V) message. The subscription may specify particular currency pairs and products. The subscription will activate the following message types: market data messages, anonymous quote request messages, and news messages.

2.8.7 New Order – Multileg (AB)

API clients can execute on quotes using the New Order – Multileg (AB) message.

2.8.8 Execution Report (8)

API clients will be sent Execution Report (8) messages to accept or reject New Order – Multileg (AB) messages. Market maker banks will be sent Execution Report (8) messages to indicate executions on their quotes. Execution Report (8) messages will be used to allow for “Last-Look” in some RFQ scenarios.

2.8.9 Trade Capture Report (AE)

The Trade Capture Report (AE) message contains trade details for API clients.

2.8.10 News (B)

The News (B) message notifies API clients of a transaction, including those clients who were not directly involved in the trade. It can be requested using the Market Data Request (V) message.

3 Protocol Functionality

3.1 Session Management

The API uses FIX 4.4 protocol to establish and manage these sessions. The session is established between a client and the FIX API through a logon message, and terminated with the corresponding logout message. Additional administrative messages are sent to maintain the health and usability of the connection. The session will start when the client sends a Logon message to the FIX API at the start of the week. The start of the week will commence at 17:00:00 (UTC) on Saturday, and conclude at 15:00:00 (UTC) the following Saturday.

3.1.1 Session Security

The FIX API transmits messages through a Secure Socket Layer (SSL) that provides security and data integrity for messages. All messages are encrypted. The FIX API has a logon process for authenticating client applications based on a username, password, and SenderCompID (49).

3.1.2 Time Synchronization

Clients are required to synchronize the time of their application to the FIX servers. The application time must be equal to the FIX server for log comparison reasons. An hourly synchronization interval, such as the National Institute of Standards and Technology, should be used.

3.1.3 Sequence Reset

The FIX API resets its inbound and outbound sequence numbers to “1” at the start of each week at 17:00:00 (UTC) on Saturday. External clients are required to reset their sequence numbers prior to the weekly session logon. FIX sessions are established at the start of each week by client sending a Logon (A) message. The FIX API will respond with a Logon (A) confirmation with a sequence number of “1”.

3.2 Messages-per-Second Threshold

The FIX API will not support inbound messages that are sent at a frequency greater than the pre-defined messages-per-second threshold. Any set of messages that are sent to the FIX API at a rate greater than the threshold will be rejected. The FIX API will reject the messages by sending a business reject message. The following threshold error message will be found in Text (58) of the Business Message Reject (j) message: *“Number of messages per second exceeds: X (defined threshold)”*.

3.3 Session Sequencing

The FIX API requires separate sequence numbers for incoming and outgoing messages to and from the client for each session. This ensures that all messages to and from the API are in the correct order and can be recovered in the event of an error.

If the client’s local architecture uses a single-server process that maintains multiple FIX sessions with the FIX API and supports multiple applications, the client-side application should allow a session to reset and/or recover sequence numbers. This should not affect other active trading connections on the server or the server process itself.

In order to guarantee message delivery, both the client and the FIX API must maintain the inbound and outbound sequence numbers. The client's responsibilities for maintaining the inbound and outbound sequence numbers include:

1. Resetting the inbound and outbound sequence numbers to "1" prior to the weekly session logon.
2. Incrementing the inbound sequence number by one for each incoming message.
3. Incrementing the outbound sequence number by one for each outgoing message.
4. Issuing a Resend Request when the client detects a sequence gap.
5. Resending any missed or malformed messages when the FIX API detects a sequence gap and issues a Resend Request (2) message.

The API will preserve all inbound and outbound sequence numbers for recovering missed messages during a logon or failover scenario.

Sequence number validation is a vital part of FIX session management. However, a discrepancy in the sequence number series may occur. Recovery of the sequence number is handled differently for different types of FIX messages. For example, if the incoming message type is a Sequence Reset (4) – Reset message, the incoming sequence number is ignored. The NewSeqNo (36) of the Sequence Reset (4) message contains the sequence number of the next message to be transmitted.

3.3.1 Sequence Number Higher Than Expected

If the sequence number on the message submitted is higher than expected and the message is not a Sequence Reset (4):

1. A Resend Request (2) message is sent to request the missing messages.
2. All requested messages should be re-sent.
3. After the missed messages are received and processed, normal message processing continues.

3.3.1.1 Resend Request (2)

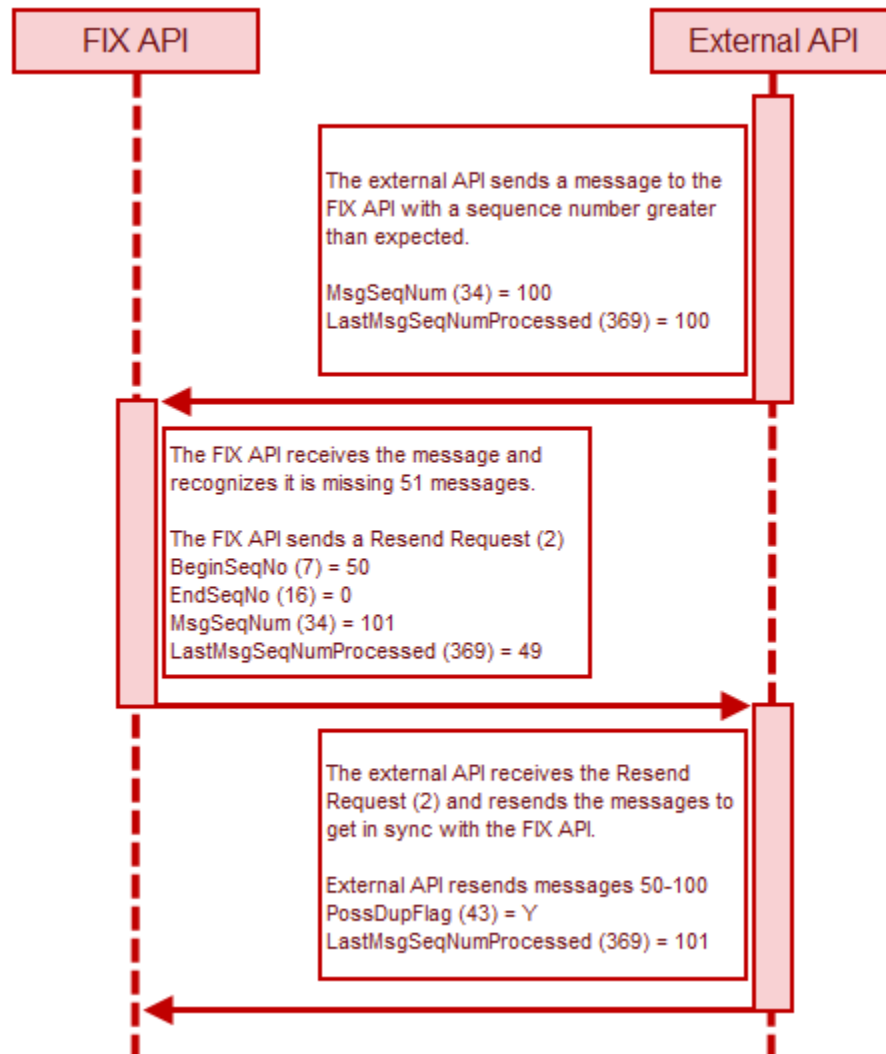
If a Resend Request (2) message with a higher than expected sequence number is received by a party, the party must handle the incoming Resend Request (2) message first. It can then send its own Resend Request (2) message to the counterparty. The FIX API will only allow external clients to request a resend of 1 to 500 messages. If the FIX API receives a Resend Request (2) message for more than 500 messages, the external client will be logged out of the FIX API.

3.3.1.2 Sequence Reset (4) – Reset

If the message is a Sequence Reset (4) – Reset message and the value of NewSeqNo (36) is equal to, or higher than, the expected sequence number, the NewSeqNo (36) is adopted as the new expected sequence number of the party that just processed the Sequence Reset (4) – Reset message. If the value in NewSeqNo (36) is lower than the expected sequence number, a Logout (5) message is sent.

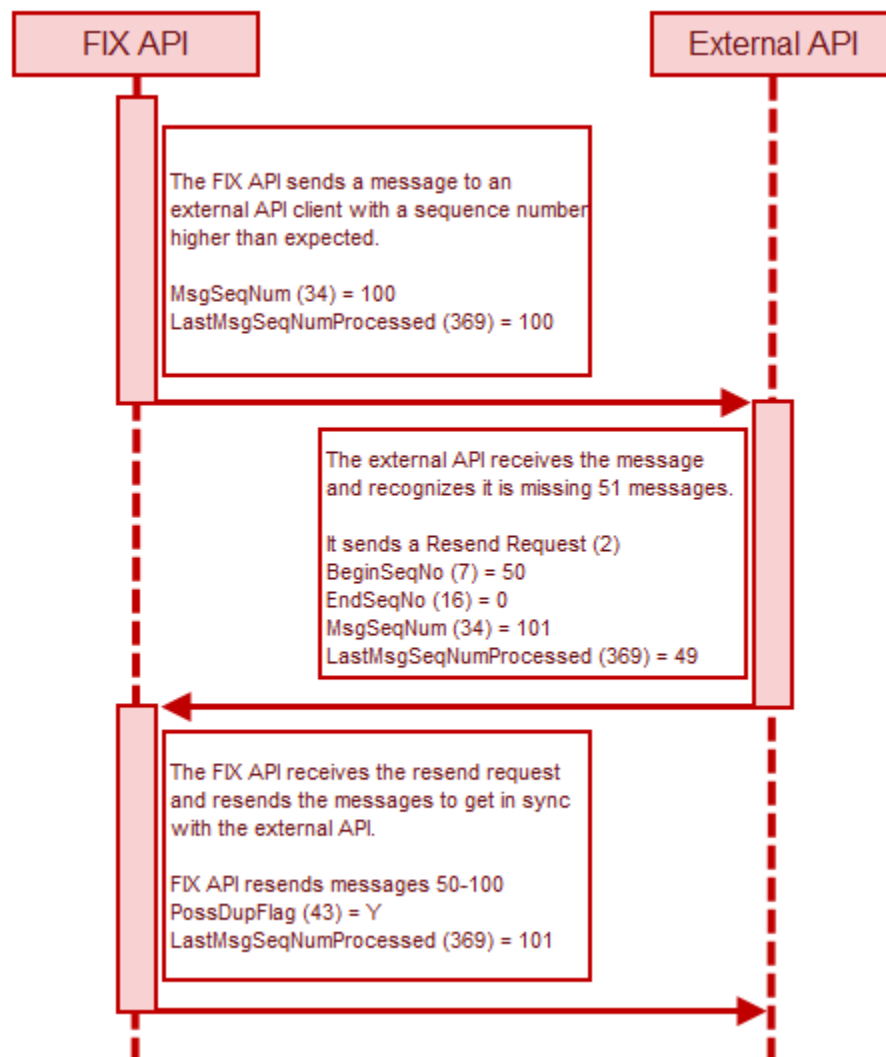
3.3.1.3 FIX API Receives Sequence Number Higher Than Expected

In this example, the FIX API receives a message from an external API that is higher than expected. The FIX API will request the missing messages by sending a Resend Request (2) message to the external API. The external API will receive the Resend Request (2) message and resend the requested message.



3.3.1.4 External Client Receives Sequence Number Higher Than Expected

In this example, an external client receives a message from the FIX API that is higher than expected. The external API will request the missing messages by sending Resend Request (2) message to the FIX API. The FIX API will receive the Resend Request (2) message and resend the requested message.



3.3.2 Sequence Number Lower Than Expected

A serious sequence error is indicated if the incoming message has a sequence number that is lower than expected and the PossDupFlag (43) is not set. In this case, a Logout (5) message is sent and the FIX connection is closed. Certain recovery situations cause the FIX API and client sequence numbers to be out of synchronization. If a client is logged out due to an invalid sequence number, the next expected sequence number is provided in the Logout (5) message. In this case, the client should submit the next expected sequence number in the Logon (A) message.

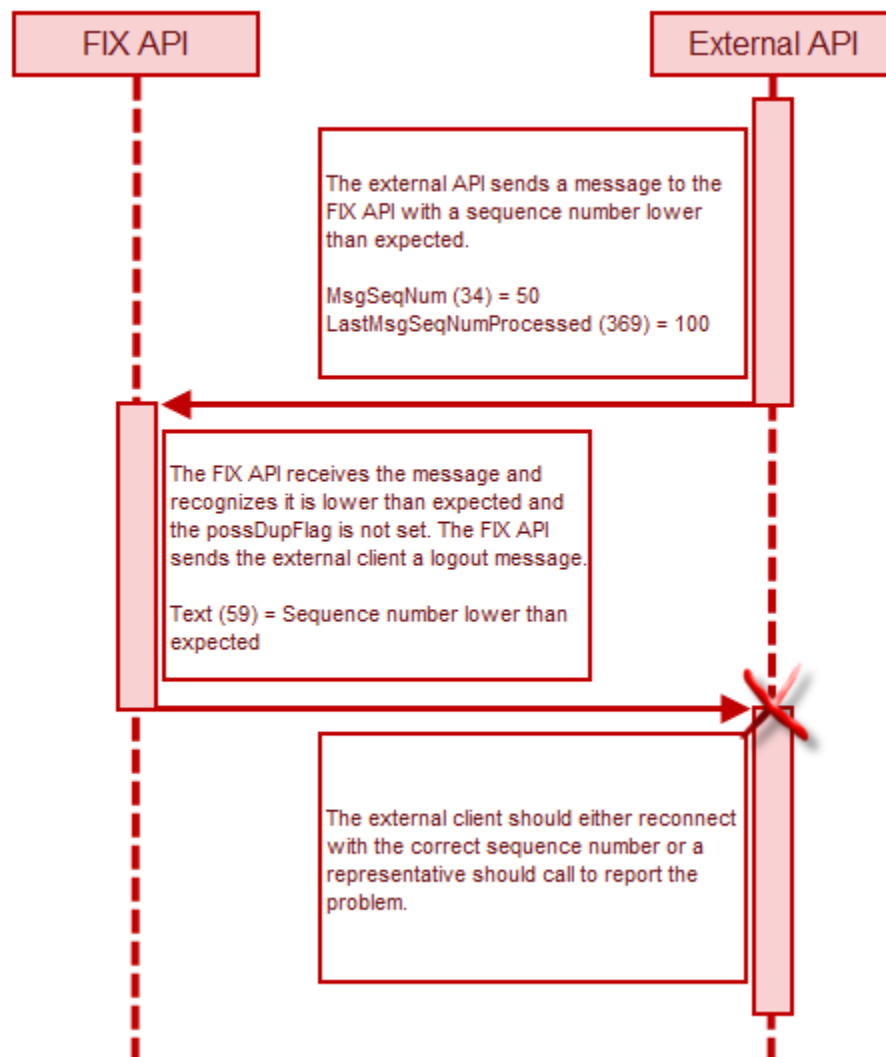
3.3.2.1 Sequence Reset (4) – Reset

If the message is a Sequence Reset (4) – Reset message, and the value in the NewSeqNo (36) is:

1. Equal to or higher than the expected sequence number, the NewSeqNo (36) is adopted as the new expected sequence number of the party that just processed the Sequence Reset (4) – Reset message.
2. Lower than the expected sequence number, a Logout (5) message is sent.

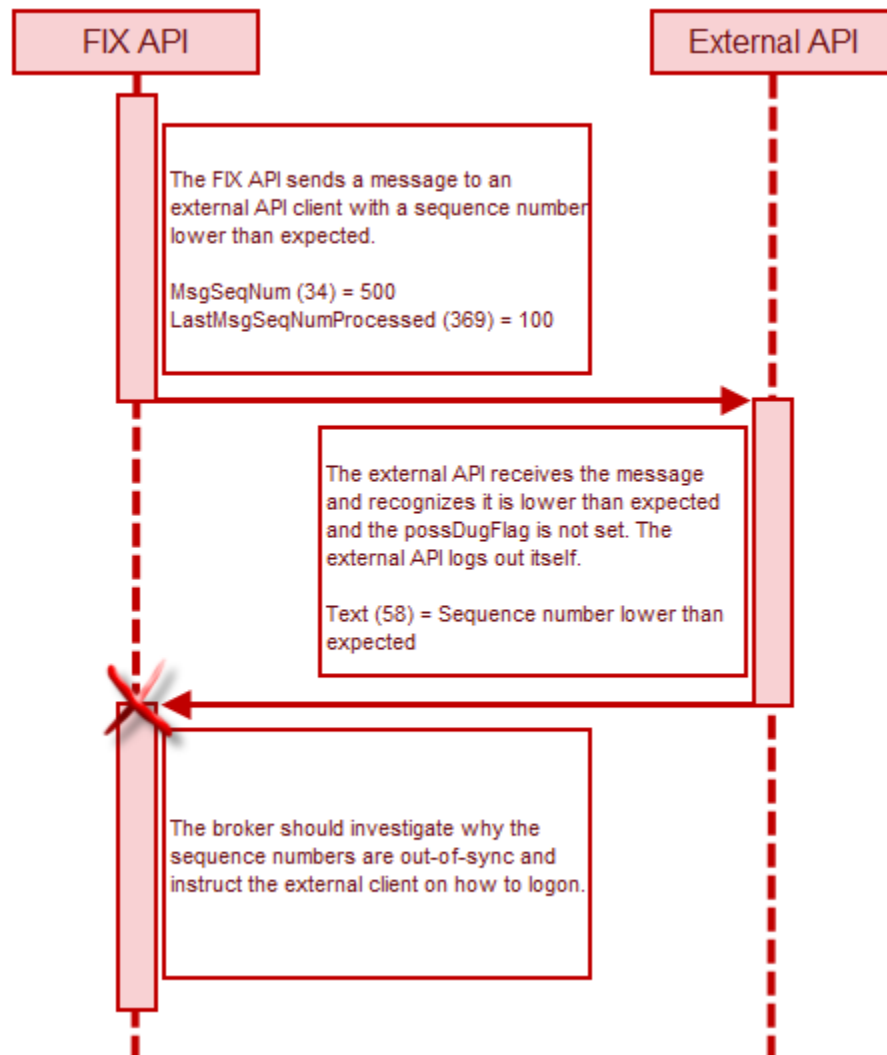
3.3.2.2 FIX API Receives Sequence Number Lower Than Expected

In this example, the FIX API receives a message from an external client that is lower than expected. The FIX API is not able to resend messages that it received from the external client, and for that reason the FIX API should immediately terminate the connection by sending a Logout (5) message. The out-of-sync sequence numbers should be resolved manually by the FIX API's and the external client's technical support teams.



3.3.2.3 External Client Receives Sequence Number Lower Than Expected

In this example, an external client receives a message from the FIX API that is lower than expected. The external API is not able to resend messages that it received from the FIX API, and for that reason the external API should immediately terminate the connection by sending a Logout (5) message. The out-of-sync sequence numbers should be resolved manually by the FIX API's and the external client's technical support teams.



3.4 Message Configuration

A FIX message begins with a standard header, which is followed by the message body consisting of application or administrative messages, and finally ends with a standard trailer.

3.4.1 Standard Header

Preceding each message, session or application, the header is used to identify the message type, length, destination, sequence number, origination point and time. The standard header will provide the routing and institution identifiers necessary on the application messages. The SenderCompID (49) and SenderSubID (50) will be used to identify institution and group or individual sending the message. The TargetCompID (56) and TargetSubID (57) will be used to identify the institution and group or individual that is to receive the message.

3.4.2 Standard Header Message Configuration

Tag	Field Name	Req	Type	Description
8	BeginString	Y	String	Identifies beginning of new message and protocol version; always first field in the message e.g. FIX4.4
9	BodyLength	Y	String	Message length in bytes forwarded to CheckSum (10); always second field in the message
35	MsgType	Y	String	Defines message type; always third field in the message
49	SenderCompID	Y	String	Unique value that identifies the firm sending the message
50	SenderSubID	C	String	
115	OnBehalfOfCompID	C	String	Unique value that identifies the firm for which the message was sent on behalf of
56	TargetCompID	Y	String	Assigned value used to identify receiving firm
57	TargetSubID	C	String	
143	TargetLocationID	C	String	
128	DeliverToCompID	C	String	Unique value that specifies where a message will be routed; the FIX API will use this value when routing client message to specified targets
34	MsgSeqNum	Y	SeqNum	Integer message sequence number
369	LastMsgSeqNumProcessed	N	SeqNum	The last MsgSeqNum (34) value received by the FIX engine and processed by downstream application, such as trading engine or order routing system; can be specified on every message sent; used for detecting a backlog with a counterparty
43	PossDupFlag	N	Boolean	Always required for retransmitted messages, whether prompted by the sending system or as the result of a resend request
97	PossResend	N	Boolean	Required when message may be a duplicate of another message sent under a different sequence number
52	SendingTime	Y*	UTC Timestamp	Time of message transmission in the format of YYYYMMDD-HH:MM:SS e.g. 20060331-16:02:00

Y = Standard FIX required; Y* = FIX API required; C = Conditional; N = Not required

3.4.3 Standard Trailer

When terminating each message, session, or application, a trailer is used to segregate messages. It contains the three-digit character representation of the checksum (10).

3.4.4 Standard Trailer Configuration

Tag	Field Name	Req	Type	Description
10	Checksum	Y	String	Three-byte, simple checksum; always last field in the message

Y = Standard FIX required; Y* = FIX API required; C = Conditional; N = Not required

4 Administrative Messages

The administrative class of messages is intended to address the utility needs of the protocol. Certain administrative messages can be generated from either side of the connection. The following FIX 4.4 administrative messages will be utilized to logon, logout and maintain the FIX session:

Message Type	Description
Logon (A)	Establishes communication session, authenticates connecting client, and initializes the message sequence number
Heartbeat (0)	Monitors status of the communication link and identifies when the last of a string of messages was not received
Test Request (1)	Forces a heartbeat from the opposing application to verify the communication line status
Resend Request (2)	Sent by receiving application to initiate the re-transmission of messages; utilized if a sequence number gap is detected
Sequence Reset (4)	Primarily used to recover from an out-of-sequence condition
Reject (3)	Issued when a message is received but cannot be properly processed due to a session-level rule violation
Logout (5)	Initiates or confirms the normal termination of a session

4.1 Logon (A)

The Logon (A) message is utilized to authenticate a client attempting to establish a connection with the FIX API. The Logon (A) message must be the first message sent by the application requesting to initiate a FIX session. Values for SenderCompID (49) and Password (554) will be provided to the client. To establish a FIX connection, the client will have two types of logons to send depending on the time of day: weekly session logon and the mid-day session logon.

4.1.1 Weekly Session Logon

The weekly session logon will be the first logon message that the client will send to the FIX API every weekend following the weekly server recycle. Clients must reset their inbound and outbound sequence numbers to “1” prior to the weekly session logon.

The client weekly session logon message is expected to have the outbound sequence number reset to “1”, and the ResetSeqNumFlag (141) set to “N”. If the client does not set the outbound sequence number to “1”, the client will be logged out by the FIX API. If the client is logged out after receiving a Logon (A) message, the FIX API will not increment the sequence number. By setting the ResetSeqNumFlag (141) to “N”, the client will receive any messages that were sent by the FIX API while the client was logged out. However, if the ResetSeqNumFlag (141) is set to “Y”, the client will not receive any missed messages. If the client’s inbound sequence number is not reset to “1” prior to the weekly logon message, the FIX API will respond with a lower-than-expected sequence number in the logon confirmation message. As per protocol, the client should submit a Logout (5) message and reset the sequence numbers before re-attempting to logon again.

4.1.2 Weekly Recycle

The FIX API will restart on a weekly basis for maintenance. Maintenance is scheduled for 15:00 – 17:00 (UTC) on Saturdays. At approximately 15:00 (UTC), the FIX API will logoff all connected clients by sending

Logout (5) messages. No client will be able to connect to the FIX API between 15:00 (UTC) and 17:00 (UTC) on this day. After 17:00 (UTC), clients can re-establish connections, but must re-connect with sequence numbers reset to “1”. Clients will not be able to connect unless sequence numbers are reset. Each failed login attempt will be accompanied by an error message. Periodically, the FIX API will be brought down after market hours on Fridays for upgrades.

4.1.3 Mid-Day Session Logon

Mid-day session logons are used for subsequent logons after the weekly session logon. These logons use sequence numbers that continue from the prior session before disconnect. The mid-day session logon cannot have a sequence number set to “1”, and the ResetSeqNumFlag (141) must be set to “N”.

4.1.4 Logon Priority

The FIX API will only allow one logon at a time for every unique username-password combination. External API users that share username-password combinations with GUI users will have API logon priority. The logon rules for external API users and GUI users that share the same username-password are as follows:

1. An external API client will always be able to logon to the FIX API.
2. If a GUI user is logged on and an external API subsequently logs on with the same username-password combination, then the GUI user will be automatically logged off.
3. GUI users will not be able to logon to the application if an external API is logged on with the same username-password confirmation.

4.1.5 Logon Confirmation Message

Upon receipt of Logon (A) message, the FIX API will authenticate the party requesting the connection and issue a Logon (A) confirmation message as acknowledgment that the connection request has been accepted. After sending the Logon (A) confirmation message, the FIX API will issue a Test Request (1) message if the external API client does not send any application or Heartbeat (0) messages. The appropriate response to a Test Request (1) message will be a Heartbeat (0) message. The client must receive the Logon (A) confirmation message prior to sending the Heartbeat (0) and any other subsequent messages.

4.1.6 Logon (A) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType		String	A (Logon)
98	EncryptMethod	Y	Int	Method of encryption; valid value: 0 = None
108	HeartBtInt	Y	Int	Heartbeat interval (seconds); must be set to 10 seconds or less; valid values: <= 10
141	ResetSeqNumFlag	C	Boolean	Indicates whether or not both sides of a session should reset sequence numbers; required for the daily session and mid-day logons
553	Username	Y*	String	User ID or username; required for the daily session and mid-day logons
554	Password	Y*	String	Password set up by an admin and used by clients connecting to the FIX API; required for the daily session and mid-day logons
	Standard Trailer			

Y = Standard FIX required; Y* = FIX API required; C = Conditional; N = Not required

4.2 Heartbeat (0)

During periods of message inactivity, the client will generate Heartbeat (0) messages at regular time intervals. The heartbeat is used to monitor the status of the communication link and to identify incoming sequence number gaps. Heartbeat intervals are declared by the client using the HeartBtInt (108) in the Logon (A) message. HeartBtInt (108) will be set to 10 seconds or less. The heartbeat interval timer will be reset after every message transmission. Client prices will be removed from the GUI if two expected Heartbeat (0) messages are missed consecutively.

4.2.1 Heartbeat (0) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			0 (Heartbeat)
112	TestReqID	N	String	
	Standard Trailer			

Y = Standard FIX required; Y* = FIX API required; C = Conditional; N = Not required

4.3 Test Request (1)

The Test Request (1) message is utilized by the FIX API to force a heartbeat from the client application. The message is useful for verifying the communication line status. It forces a heartbeat from the opposing application to verify the communication line status. The client application will respond to the Test Request (1) message with a Heartbeat (0) message containing the TestReqID (112).

4.3.1 Test Request (1) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			1 (Test Request)
112	TestReqID	Y	String	
	Standard Trailer			

Y = Standard FIX required; Y* = FIX API required; C = Conditional; N = Not required

4.4 Resend Request (2)

The messaging protocol is designed for complete delivery of messages between parties. When an incoming message has a sequence number greater than expected, the messaging protocol specifies steps to be taken for restoring complete message delivery. Upon detection of a gap in the sequence a Resend Request (2) message will be sent to the counterparty. The Resend Request (2) message is sent by the receiving application to initiate the re-transmission of messages. This function will be utilized if a sequence number gap is detected, or if the receiving application lost a message.

A Resend Request (2) message can be used to request a single message, a range of messages, or all messages subsequent to a particular message by using the BeginSeqNo (7) and EndSeqNo (16). To request a single message, BeginSeqNo (7) and EndSeqNo (16) should have the same value. To request a range of messages, BeginSeqNo (7) should be the first message of range, while EndSeqNo (16) the last message of range. To request all messages subsequent to a particular message, BeginSeqNo (7) should be the first message of range, while EndSeqNo (16) should be set to "0". The receiving application must process messages in sequence order. If there is sequence error, the application should ignore the last-received messages, and ask for a resend of all messages at the point of the sequence error.

4.4.1 Resend Request (2) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			2 (Resend Request)
7	BeginSeqNo	Y	SeqNum	
16	EndSeqNo	Y	SeqNum	
	Standard Trailer			

Y = Standard FIX required; Y* = FIX API required; C = Conditional; N = Not required

4.5 Sequence Reset (4)

The Sequence Reset (4) message is used by the sending application to reset the incoming sequence number of the opposing side. The Sequence Reset (4) message can be used in the following situations:

1. During normal resend processing, the sending application may choose not to send a message, the Sequence Reset (4) message can be used to mark the place of that message.
2. During normal resend processing a number of administrative messages are not resent, the Sequence Reset (4) message is used to fill the sequence gap created.
3. In the event of an application failure it may be necessary to force synchronization of sequence numbers on the sending and receiving sides.

The sending application will initiate the sequence reset. The message is used to reset the value of the next sequence number to be transmitted. The Sequence Reset (4) message can only increase the sequence number. If a sequence reset is received attempting to decrease the next expected sequence number the message will be rejected and treated as a serious error (See the *Session Sequencing* segment in the *Messaging Functionality* section for more information). There will be two types of Sequence Reset (4) messages to send: Sequence Reset (4) – Gap Fill and Sequence Reset (4) – Reset.

4.5.1 Sequence Reset (4) – Gap Fill

The Sequence Reset (4) – Gap Fill message is used when the sending application chooses not to resend specific messages or the requested messages are administrative. The Sequence Reset (4) – Gap Fill message must have GapFillFlag (123) set to “Y”. If the Sequence Reset (4) – Gap Fill message is responding to a Resend Request (2) message, PossDupFlag (43) should be set to “Y”.

For “Gap Fill” messages, MsgSeqNum (34) will conform to standard message sequencing rules. The message must contain the next expected sequence number for MsgSeqNum (43) in the message header. NewSeqNo (36) will contain the value of the next sequence number immediately following the messages and/or sequence numbers being skipped. For example, if sequence number “11” represents application message while “5” through “10” represent administrative messages, the series of messages as a result of a Resend Request (2) message may appear as a Sequence Reset (4) – Gap Fill message, with NewSeqNo (26) = 11 followed by message “11”. Administrative messages “5” through “10” are skipped.

4.5.2 Sequence Reset (4) – Reset

The Sequence Reset (4) – Reset message is used in the event of an application failure, to force synchronization of sequence numbers on the sending and receiving sides. The Sequence Reset (4) – Reset message must have GapFillFlag (123) set to “N”.

For “Reset” messages, MsgSeqNum (34) in the header will be ignored on a Sequence Reset (4) – Reset message (i.e. the receipt of a sequence reset message with an out of sequence MsgSeqNum (34) should not generate resend requests). As a result, the following must be recognized for usage of the message:

1. Using the Sequence Reset (4) – Reset message may result in losing messages because MsgSeqNum (34) in the message header is not processed.
2. The Sequence Reset (4) – Reset message should not be used as a normal response to the Resend Request (2) message.
3. If the Sequence Reset (4) – Reset message is sent and is not in response to a Resend Request (2) message, PossDupFlag (43) should be set to “N”. If it is in response to a Resend Request (2) message, PossDupFlag (43) should be set to “Y”.

4.5.3 Sequence Reset (4) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			4 (Sequence Reset)
123	GapFillFlag	N	Boolean	
36	NewSeqNo	Y	SeqNum	
	Standard Trailer			

Y = Standard FIX required; Y* = FIX API required; C = Conditional; N = Not required

4.6 Reject (3)

The Reject (3) message is issued when a message is received which cannot be passed through to the application level, such as a receipt of a message with invalid basic data (e.g. MsgType (35) = &). Rejected messages will be logged and the incoming sequence number incremented. A received message which fails to pass to the application level will be disregarded. Processing of the next valid message will cause detection of a sequence gap and a Resend Request (2) message will be generated. If the sending application chooses to re-transmit the rejected message, it will be assigned a new sequence number. The cause of the failure will be described in Text (58) of the Reject (3) message.

4.6.1 Reject (3) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			3 (Reject)
45	RefSeqNum	Y	SeqNum	MsgSeqNum (34) of rejected message
371	RefTagID	N	Int	The tag number of the field being referenced
372	RefMsgType	N	String	MsgType (35) of the message being referenced
373	SessionCancelRejectReason	N	Int	Code that identifies reason for a session-level rejection; see appendix for valid values
58	Text	N	String	Reject reason; see appendix for descriptions
	Standard Trailer			

Y = Standard FIX required; Y* = FIX API required; C = Conditional; N = Not required

4.7 Logout (5)

Normal termination of the message exchange session is completed using Logout (5) messages. Termination by other means should be considered an abnormal condition and dealt with as an error. Upon receiving a Logout (5) message from an API user, the FIX API will transmit all pending messages to the API user before sending a confirmation Logout (5) message. The API user must wait to receive all pending messages followed by the (5) Logout message before terminating the FIX session. Any live prices in the system owned by the client will be removed when a Logout (5) message is received.

4.7.1 Weekly Sequence Number Reset

The FIX API server will send a Logout (5) message to all clients connected to the FIX API at 15:00:00 (UTC) every Saturday. Upon receipt of the message, API clients will be expected to logout and reconnect to the FIX API with sequences reset to “1”. External clients will be able to reconnect after receiving the message once the FIX API server is available. The FIX API server will be available at 17:00 (UTC) on Saturday. If an external client reconnects with a sequence number that is not “1”, the FIX API will reject the login attempt and will send the API client a reject message that contains details about the failure to connect to the FIX API server.

4.7.2 Logout (5) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			5 (Logout)
58	Text	N	String	An error message for abnormal logout
	Standard Trailer			

Y = Standard FIX required; Y* = FIX API required; C = Conditional; N = Not required

5 Application Messages

Application messages exchange business-related information necessary to support trading. Application messages are generated from both sides of the connection. The following FIX application messages facilitate communication between the FIX API and API clients:

Message Type	Description
Business Message Reject (j)	Rejects an application-level messages
Quote Request (R)	Requests quotes on single or multiple products
Quote Request Reject (AG)	Rejects Quote Request (R) messages
Quote (S)	Sends unsolicited quotes in tradable markets, or solicited quotes in response to Quote Request (R) messages
Quote Status Request (a)	Queries the status of a quote submitted in the market
Quote Status Report (AI)	Responds to Quote (S) and Quote Status Request (a) messages; provides current status of the Quote (S)
Quote Cancel (Z)	Used by originator of quotes to cancel quotes
Market Data – Snapshot/Full Refresh (W)	Send unsolicited data in all tradable markets
Market Data Request (V)	Requests subscription to the FIX API for a specified set of market data; also requests previously sent market data messages
New Order – Multileg (AB)	Submits multileg orders
Order Status Request (H)	Requests the status of an order
Execution Report (8)	Accepts or rejects orders
Trade Capture Report (AE)	Delivers STP details of executed trades
Trade Capture Report Ack (AR)	Acknowledges or rejects Trade Capture Report (AE) messages received from or by the FIX API
Trade Capture Report Request (AD)	Requests one or more trade capture reports
News (B)	Indicates trade details executed in the market, and are sent to all clients

5.1 Business Message Reject (j)

The Business Message Reject (j) message is used to electronically reject an application-level message which fulfills session-level rules and cannot be rejected via any other means. If the message fails a session-level rule, a session-level Reject (3) message will be issued.

If an external liquidity bank's API is not connected, the FIX API will send the Business Message Reject (j) message in response to messages that have the target field set to the disconnected liquidity bank's CompID. The BusinessRejectReason (380) will be sent on the message, along with the error in Text (58), and BusinessRejectRefID (379) of the rejected message.

Value affixed to a not applicable tag will also result in a Business Message Reject (j) message with aforementioned tags populated.

5.1.1 Business Message Reject (j) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			j (Business Message Reject)
45	RefSeqNum	Y*	SeqNum	MsgSeqNum (34) of the rejected message
372	RefMsgType	Y	String	The MsgType (35) of the FIX message being referenced
379	BusinessRejectRefID	C	String	The value of the business-level ID field on the message being referenced
380	BusinessRejectReason	Y	Int	Code to identify reason for a Business Message Reject message. See Appendix for codes

58	Text	Y*	String	Where possible, message to explain reason for rejection. See Appendix for error message text
	Standard Trailer			

Y = Standard FIX required; Y* = FIX API required; C = Conditional; N = Not required

5.2 Quote Request (R)

The Quote Request (R) message is used to request prices for specific bespoke contracts. The requestor will send the message to a target audience; the target audience will respond to the requestor's Quote Request (R) message by sending tradable prices, rejecting the message, or by not responding which will indicate no interest.

The FIX API provides for both CLOB RFQ and Direct RFQ. The table below describes how a request originator could direct their RFQ to the appropriate venue and RFQ type:

Trading Venue	RFQ Type	TargetCompID (56)	OnBehalfOfCompID (115)	SenderCompID (49)	DeliverToCompID (128)	UserID (7936)	BankRequested (8053)
Brokerage	CLOB	Broker ComID	N/A	Requestor's FIXCompID	N/A	N/A	NA
	Direct	Broker CompID	N/A	Requestor's FIXCompID	N/A	N/A	Targeted Bank's FIXCompID
Agency	Direct	Broker CompID	N/A	BA's FIXCompID	NA	Client's TraderID	Targeted Banks' FIXCompIDs

5.2.1 Quote Request via Front-End Application or FIX API

Originating interest may send an anonymous RFQ to the CLOB or a direct request to one or more liquidity bank via the front-end application or the FIX API. When sending via the API, the interest will send the RFQ by setting TargetCompID (56) = broker CompID. Quote request directed to the CLOB will be anonymously routed to banks subscribed to receive requests.

For a dealer-to-dealer direct RFQ, the client is required to use a combination of tags to indicate the targeted banks to which the FIX API will direct the RFQ. The TradingVenue (8600) is required to be set to "2" (Direct). NoBanksRequested (8052) should be set to a numeric value of the number of target banks. BankRequested (8053) will indicate the CompID of the target banks. The FIX API will deliver the request to the designated liquidity banks in the NoBanksRequested (8052) repeating group.

5.2.2 Request for Differential & Vega Neutral Spreads

The FIX API will support Quote Request (R) messages for differential and vega neutral spreads. Neutral legs will have LegQty (687) = 0.

5.2.3 Quote Request ID

The Quote Request (R) identifier, QuoteReqID (131), is a unique identifier as assigned by the client for a quote request they are submitting to the FIX API.

QuoteReqID (131) should conform to the following rules:

1. All Quote Request (R) messages must have a unique value in QuoteReqID (131) for that day, regardless of the trading session the quote request was submitted in.
2. The QuoteReqID (131) value may only contain upper-case letters or numbers.

3. The QuoteReqID (131) value may be up to 128 characters in length.

5.2.4 Delta and Strike Price for FXO Requests

LegDelta (6035) and LegStrikePrice (612) are price fields that determine if the Quote Request (R) message is relative to a specific delta or strike. It is required that if one of the fields is assigned a value in the Quote Request (R) message. ATMF RFQ can be submitted by setting LegStrikePrice (612) = 0.

5.2.5 Position

Position (6351) defines the side of a strategy leg relative to the side of the strategy. If the side of the strategy leg must equal the side of the strategy, then Position (6351) will be set to "1" (same); else if the side of the strategy leg must be opposite the side of the strategy, then Position (6351) will be set to "2" (inverse). For example, a Quote Request (R) message for a straddle will always have two strategy legs; and on each leg the position field will be "1" (same). Position (6351) is always the same on each leg of a straddle because each leg is always in the same direction of the buy or sell interest of the product. Alternatively, the Quote Request (R) message for a risk reversal will always have two strategy legs, and on the first leg Position will be "2" (inverse). The second leg of a risk reversal will always be in the inverse direction to the first leg in other words if you buy the first leg you will sell the second leg.

5.2.6 Strategy ID

LegID (7940) uniquely identifies a strategy leg of a structure for each Quote Request (R) message. The strategy ID is a unique alphanumeric representation of each leg. LegID (7940) will be assigned by the originator of the Quote Request (R) message, and the LegID (7940) will remain the same for all messages that refer to the Quote Request (R) message.

5.2.7 Quote Request (R) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			R (Quote Request)
131	QuoteReqID	Y	String	Unique identifier for quote request
8600	TradingVenue	Y*	Int	Indicates to which trading venue the request is to be forwarded; valid values: 1 = CLOB 2 = Direct 5 = Broker-Agent Direct
9126	Structure	Y*	Int	FX Trading Systems primary Structure information; valid values: see Appendix
75	TradeDate	Y*	Local MktDate	Indicates horizon date
8505	RegulationVenueType	C	Int	Indicates the market type of the quote; conditionally required if trading on SEF or MiFID market; if no value is set, it indicates an IB (8) request in an RFQ for BGC ; valid values: 1 = SEF 3 = MiFID 7 = OTC 8 = IB
8519	OriginatingLEI	C	String	LEI of beneficial owner of swap; required if requesting quote in the SEF market; the FIX API will reject request with LEI that has not been SEF onboarded in the respective venue;

					when using broker agency model, the SenderLEI will be the end-client's LEI	
5159	Clearer		C	String	Specifies the clearer LEI; if tag is not populated, indicates bilateral market	
6661	PBReuterCode		N	String	Assigned Reuter Code used to identify associated Prime Broker or FCM; the FIX API will forward the Prime Broker or FCM's LEI to the market; if LEI does not exist, then trading name is sent	
7936	UserID		C	String	End Client's Account Name; required in broker agent RFQ model; broker agent sends end client's TraderID to the FIX API; on messages to target banks, the FIX API sends end client's LEI (if "Show End Client = Yes"); if "Show End Client = No", then this value will be null	
454	NoSecurityAltId		N	NumIn Group	Number of security ID groups that follow; required for MiFID II-relevant securities	
	455	SecurityAltId	N	String	ISIN of the top-level security; required for MiFID II-relevant securities	
	456	SecurityAltIdSource	N	String	Required for MiFID II-relevant securities ; valid value: 4 = ISIN Identifier	
461	CFIcode		N	String	Indicates the type of top-level security; required for MiFID II-relevant securities	
24201	IlliquidInstrument		N	Boolean	Designated liquidity type based on components and size; required for MiFID II-relevant securities	
24202	SizeThresholdLIS		N	Qty	LIS size threshold; required for MiFID II-relevant securities when IlliquidInstrument (24201) = N	
24203	SizeThresholdSSTI		N	Qty	SSTI size threshold; required for MiFID II-relevant securities when IlliquidInstrument (24201) = N	
8052	NoBanksRequested		C	NumIn Group	Number of liquidity bank targets in the RFQ; required for (8600) = 2; a null value when 8600 = 5 will result in the RFQ to be sent to all allowable liquidity banks	
	8053	BankRequestedCompID	C	String	FIXCompID of target liquidity bank in the RFQ; required if 8052 >= 1	
146	NoRelatedSym		Y	NumIn Group	Number of related symbols (instruments) in request	
	55	Symbol	Y	String	Six-character string that represents the currency pair e.g. "EURUSD"; the first ccy is always the base and the second is always the term	
	6258	Strategy	C	Int	FX trading systems secondary strategy information; FIX API required for FXO; not required when Structure is FX	
	537	QuoteType	Y*	Int	Identifies the type of quote; the FIX API will reject quote request if any other value other than 1 is entered	
	555	NoLegs	Y*	NumIn Group	Number of legs	
		600	LegSymbol	Y*	String	Six-character string that represents the currency pair; the first ccy is always the base and the second is always the term
		6714	LegStrategyInd	C	Int	API required for FXO; valid values: 1 = Call 2 = Put 3 = Up 4 = Down
		9125	Cutoff	Y*	Int	Time of expiration of FX option; valid values: see appendix; for NDFs the cutoff value will indicate the fixing index
		6215	Tenor	Y*	String	FX and FX option maturity; for FX spreads, enter individual leg value instead of spread tenor value; valid values: see appendix
		6203	FixingDate	C	Local MktDate	Required for NDFs
611		LegMaturityDate	C	Local MktDate	Date of option expiration; FIX API required for FXO	
	9127	LegDeliveryType	C	Char	Required when Structure (9126)=65 (FX) to indicate deliverable or non-deliverable; valid values: 1 = Cash (indicates non-deliverable) 2 = Physical (indicates deliverable)	

		743	DeliveryDate	C	Local MktDate	Date of delivery or settlement date; required when Structure (9126)=65 (FX)
		8013	PublishDate	C	Local MktDate	Indicates publish date in YYYYMMDD format
		5020	PremiumDelivery	C	Local MktDate	Premium delivery date; required for premium RFQs. Also, when PremDeliveryType (5475) = F, then 5020 = option delivery date
		6035	LegDelta	C	Float	Delta (%); required for delta-specific interests
		7966	DeltaSpotFwd	N	Char	Define if delta type is spot or forward; valid values: S = Spot (default) F = Forward
		612	LegStrikePrice	C	Price	Strike price; required for strike-specific interests
		5167	LoTrigger	C	Price	Identifies trigger price in absolute terms for the single or low trigger; required on single or double trigger options
		6192	HiTrigger	C	Price	Identifies trigger price in absolute terms for the high trigger; required on double trigger options
		9019	FXOptionStyle	Y*	Int	The style of the FX option; valid values: 1 = American 2 = European Following are supported for Fenics Direct only 21 = Knockin 23 = Reverse Knockin 29 = Double Knockin 30 = Digital 31 = Digital Double Knockout 32 = Digital Knockout 33 = Digital Reverse Knockout 34 = Double One Touch 35 = One Touch 40 = Knockout 42 = Reverse Knockout 48 = Double Knockout 49 = Digital Double Knockin 50 = Digital Knockin 51 = Digital Reverse Knockin 52 = Double No Touch 53 = No Touch 54 = Instant One Touch 59 = European Knockin 60 = European Knockout 61 = Knockin Knockout 62 = Reverse Knockin Knockout
		6351	Position	Y*	Int	The strategy or leg position relative to the other strategies or legs in the structure; valid values: 1 = Same 2 = Inverse
		9904	PriceIndicator	C	Int	Price indicator; valid values: 1 = Pct 2 = Pts
		5235	LegSpotRate	C	Float	F/X spot rate
		9099	ForwardRate	N	Float	Forward rate for option
		5191	LegForwardPoints	C	Float	F/X display swap rate; assume industry standards for swap points per currency pair; for example, a realistic forward points value for a EURUSD call can be 2.7
		9115	DepoRate	C	Float	Depo rate (%); for example, a realistic depo rate value for a EURUSD call can be 5.04
		9073	DepoRateCcy	C	Currency	Depo ccy
		556	LegCurrency	Y*	Currency	Three-character string that represents size currency
		687	LegQty	Y*	Qty	Size of the entry; payout or amount for exotic options; the entry is million based i.e. an entry of 1 equals 1,000,000
		7940	LegID	Y*	String	The strategy or leg identifier for spread price contributions; the max length is 50 bytes
		9034	LegStrategyCcy	C	Currency	Three-character string that represents call/put currency
		604	NoLegSecurityAltId	N	NumIn Group	Number of security ID groups that follow; required for MiFID II-relevant securities

		605	LegSecurityAltId	N	String	ISIN of the leg-level security; required for MiFID II-relevant securities
		606	LegSecurityAltIdSource	N	String	Required for MiFID II-relevant securities ; valid value: 4 = ISIN Identifier
		608	LegCFIcode	N	String	Indicates the type of leg-level security; required for MiFID II-relevant securities
5830	PremiumCcy			C	Currency	Premium currency returned on a premium quote type; required for premium quotes
8589	FwdSettleCcy			C	Currency	Indicates the settlement currency for NDF trade; required when LegDeliveryType (9127) = 1 (Cash)
9016	HedgeTradeType			N	Int	Indicates the type of hedge trade the requestor requires to be included on the quote; API required for FXO; valid values: 0 = None 1 = Spot 2 = Forward
9943	ProductQuoteType			C	Int	Indicates the type of quote to return; valid values: 1 = Volatility 2 = Premium
5475	PremDeliveryType			N	Char	Premium delivery; valid values: S = Spot F = Forward
8522	RequestSource			N	String	The FIX API will indicate if the request is generated via GUI or API; valid values: 1 = GUI 2 = API
60	TransactTime			Y	UTC Timestamp	Timestamp in microseconds of the quote request in YYYYMMDD-HH:MM:SS.ssssss format. Timestamps can be in non microsecond format (based on client preference for MiFID tags)
	Standard Trailer					

Y=FIX required; Y*=API required; C=Conditional; N=Not required

5.3 Quote Request Reject (AG)

The Quote Request Reject (AG) message is used to reject Quote Request (R) messages. This reject message will indicate that there is either a problem with the content of a Quote Request (R) message or that the target audience is not interested in responding to the quote. QuoteRequestRejectReason (658) and Text (58) will describe why the quote is rejected. The Quote Request Reject (AG) message will be transmitted in the following ways:

Sender	Target Audience	Description
Quote Request Receiver	FIX API	The receiver of an RFQ will send a Quote Request Reject message for the RFQ if the receiver can not validate the message or if the receiver is not interested in the message
FIX API	Quote Requestor	The FIX API will validate all Quote Requests and if any message fails validation the API will send the requestor a Quote Request Reject message

5.3.1 Quote Request (R) Message Validation and Rejection

The verification of the content of the Quote Request (R) message will be validated by the business logic server and/or by the target audience's system. If the FIX API or the target audience identifies an error with the content of the message, then the FIX API will send a Quote Request Reject (AG) message to the Quote Request (R) message sender. The quote Request Reject (AG) message will state the rejection reason by using QuoteRequestRejectReason (658) and Text (58).

5.3.2 Delivering Quote Request Reject (R) Messages

All Quote Request Reject (AG) messages will be sent to the FIX API, which will deliver the messages to the quote requester. The client sending the Quote Request Reject (AG) message will set QuoteReqID (131) to what was set in the Quote Request (R) message. The client will also set TargetCompID (56) to the broker CompID and the DeliverToCompID (128) to the target CompID. The target's CompID can be found on the Quote Request (R) message in OnBehalfOfCompID (115).

5.3.3 Quote Request Reject (AG) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			AG (Quote Request Reject)
131	QuoteReqID	Y	String	Unique identifier for quote request
8505	RegulationVenueType	C	Int	Indicates the market type of the quote; valid values: 1 = SEF 3 = MiFID 7 = OTC 8 = IB
5159	Clearer	C	String	Specifies the clearer LEI
8519	OriginatingLEI	C	String	LEI of beneficial owner of swap
658	QuoteRequestRejectReason	Y	Int	Reason Quote Request was rejected; valid values: see appendix
58	Text	Y*	String	Rejection reason when QuoteRequestRejectReason = Rejected
146	NoRelatedSym	Y	NumIn Group	Number of related symbols (instruments) in Request
55	Symbol	Y	String	Six-character string that represents the currency pair e.g. "EURUSD"; the first ccy is always the base and the second is always the term
	Standard Trailer			

Y=FIX required; Y*=API required; C=Conditional; N=Not required

5.4 Quote (S)

The Quote (S) message is used by banks to respond to RFQ's in terms of volatility and premium on FXO, and an all-in rate for FX spot and forward products. The target of a Quote (S) message will be able to trade with the sender of the message at the prices and details specified.

5.4.1 Response to Quote Request (R) Message

A Quote (S) message will only contain the details for one side (bid or offer) of the Quote Request (R) message. If a client responds to a Quote Request (R) message with a pair of Quote (S) messages having opposite sides, then both quotes will be live. Quotes must reference the Quote Request (R) message using QuoteReqIDs (131) and LegIDs (7940). If MQSizes (5359) on the Quote (S) message is not provided, then the value will default to the LegQty (687) from the Quote Request (R) message.

5.4.2 Quote Update

A new Quote (S) message will replace an existing quote message of the same product, side, tenor and sender. A quote update must have a new QuoteID (117). If the FIX API receives a quote that references a previous Quote ID, the Quote (S) message will be rejected. The QuoteReqID (131) must be referenced when updating a quote that was in response to an RFQ.

5.4.3 Quote ID

QuoteID (117) is a unique identifier assigned by the client for a quote they are sending to the FIX API. It should conform to the following rules:

1. All Quote (S) messages must have a unique value in QuoteID (117) for that day, regardless of the trading session the quote was submitted, or if the submission is for a modification.
2. The QuoteID (117) value may only contain upper-case letters or numbers.
3. The QuoteID (117) value may be up to 24 characters in length.

5.4.4 Valid Until Time

The ValidUntilTime (62) will be used to specify the exact date and time expiration of the quote. The ValidUntilTime (62) will only apply to quotes that are in response to Quote Request (R) messages. The field will comply to the following format: YYYYMMDD-HH:MM:SS. For example, October 6th, 2008 at 1 PM will be expressed as "20081006-13:00:00". If ValidUntilTime (62) is not provided by the quote sender, the FIX API will apply a default time to the message. For a quote in response to Direct RFQ, the target will be able to receive subsequent quote messages to update the previous quote on a per-side basis. ValidUntilTime (62) for both sides of the tradable quote will equal the ValidUntilTime (62) on the last quote message received.

5.4.5 Quote (S) Message Configuration

Tag	Field Name		Req	Type	Description
	Standard Header				
35	MsgType				S (Quote)
117	QuoteID		Y	String	Unique identifier for the quote; established by the client
131	QuoteReqID		Y	String	Unique identifier for quote request
8519	OriginatingLEI		C	String	LEI of beneficial owner of swap; required for SEF; the FIX API will reject LEI that has not been SEF onboarded for the respective venue
5159	Clearer		C	String	Specifies the clearer LEI; if tag does not match value in the request for quote message, the quote will be rejected
6661	PBReuterCode		C	String	Assigned Reuter Code used to identify the associated Prime Broker/FCM; required if quote provider uses a Prime Broker/FCM
9126	Structure		Y*	Int	FX trading systems primary structure information; see appendix for valid values
55	Symbol		Y	String	Six-character string that represents the currency pair
453	NoPartyID		N	NumIn Group	Number of party distinctions as required for MiFID II-relevant securities
	448	PartyID	N	String	Value for party role; required for MiFID II-relevant securities
	447	PartyIDSource	N	Char	Required for MiFID II-relevant securities; Valid value: D = Proprietary ID
	452	PartyRole	N	Int	Required for MiFID II-relevant securities; valid values: 11 = Order Origination Trader, execution within firm 16 = Executing System, algorithmic system identifier 122 = Investment Decision Maker
54	Side		Y*	Char	Side of the entry; indicates the direction the quote should be posted; for FX spreads, the side indicates the direction of the far leg; valid values: 1 = Bid 2 = Offer
2593	NoOrderAttributes		N	NumIn Group	Number of regulatory order attributes; required for MiFID II-relevant securities.
	2594	OrderAttributeType	N	Int	Optionally required for MiFID II-relevant securities if NoOrderAttributes (2593) > 0; valid values: 2 = Liquidity provision activity order 4 = Algorithmic order
	2595	OrderattributeValue	N	String	Attribute value; required for MiFID II-relevant securities ; valid Values: Y: Yes N: No

60	TransactTime		Y*	UTC Timestamp	Timestamp in microseconds of quote creation by client in YYYYMMDD-HH:MM:SS.ssssss format. Timestamps can be in non microsecond format (based on client preference for MiFID tags)
6436	Premium		C	Price	Structure premium; required for FXO premium quotes; represents the perspective of the receiver of the quote; if the receiver of the quote will pay premium, then this value will be negative; likewise, if the receiver of the quote will receive premium, then this value will be positive
8503	PremiumMid		N	Price	Dealer submitted overall premium mid of requested ccy
6289	ManualQuoteIndicator		C	Char	Indicates if this quote requires manual intervention; valid value: A = Automatically Live (default if value is not provided) M = Manual
62	ValidUntilTime		C	UTC Timestamp	The time when the quote will expire
6120	NoMQEntries		Y	Int	Count of market quote entries to follow
	7940	LegID	Y	String	The strategy or leg identifier; “NA” for non-RFQ quotes; the max length is 50 bytes
	5678	Volatility	C	Float	Price volatility; there should not be more than 3 significant digits after the decimal point; the system will not match if there are more than 15 significant digits after the decimal
	8588	FwdPrice	C	Price	All-in rate for NDFs Points for FX products; this field is required for deliverable and non-deliverable FX
	8515	VolMid	N	Price	Dealer provided mid-price
	8518	GFIMid	N	Price	GFI calculated Mid-Price (only applicable for Fenics Direct)
	5359	MQSize	N	Float	Size of entry
	5235	LegSpotRate	C	Float	FX spot rate
	9099	ForwardRate	N	Float	Forward rate
	5191	LegForwardPoints	C	Float	FX display swap rate; assume industry standards for swap points per currency pair; for example, a realistic forward points value for a EURUSD call can be 2.7
	9115	DepoRate	N	Float	Depo rate (%); for example, a realistic depo rate value for a EURUSD call can be 5.04
	9073	DepoRateCcy	N	Currency	Depo rate currency
	5844	LegPremPrice	C	Price	Leg price; required for premium quotes; this field will be sent as a formatted value: For Pct LegPrice = Raw % Price * 100 For Pts LegPrice = Raw Pts Price * 10^(spot decimal places)
	6035	LegDelta	C	Float	Delta (%)
	6354	MQStrikePrice	C	Price	Strike price
	5167	LoTrigger	C	Price	Identifies trigger price in absolute terms for the single or low trigger; required on single or double trigger options
	6192	HiTrigger	C	Price	Identifies trigger price in absolute terms for the high trigger; required on double trigger options
	7092	LegPremium	C	Price	Leg premium; required for premium quotes; used to capture leg level trade details; the value will not be validated against Premium (6436)
7464	NoHedgeLegs		N	NumIn Group	The number of hedges
	9074	HedgeCurrency	Y*	Currency	Currency of the hedge amount; only required when NoHedgeLegs (7464) > 0
	9016	HedgeTradeType	C	Int	Indicates the type of hedge trade executed for offsetting option risk; only required when the NoHedgeLegs > 0; valid values: 1 = Spot 2 = Forward
	6666	HedgeDirection Indicator	C	Int	Hedge direction; valid values: 1 = Buying 2 = Selling
	6036	HedgeAmount	C	Amt	Hedge amount; entry value in millions i.e. an entry of 1 equals 1,000,000
	9657	HedgeRate	C	Float	Hedge exchange rate; required when the NohedgeLegs (7464) > 0
	9112	HedgeDelivery	C	LocalMkt Date	Delivery date for the delta hedge; required if there is a hedge leg
	6426	HedgeSymbol	C	String	Currency pair for hedge leg
9536	Trader		N	String	The name of the trader who submitted the price
	Standard Trailer				

Y=FIX required; Y*=API required; R=RFQ response required; C=Conditional; N=Not required

5.5 Quote Status Report (AI)

The Quote Status Report (AI) message will be published to API clients to acknowledge receipt of all Quote Request (R) messages, Quote Request Reject (AG) messages, Quote (S) messages (new and modified), Quote Status Request (a) messages, and Quote Cancel (Z) messages. The message will also be used by the FIX API to alert clients of updates to their published Quote (S) messages. The Quote Status Report (AI) messages will include QuoteStatus (297) to indicate status. If any message sent to the FIX API is not immediately processed, the FIX API will set QuoteStatus (297) = 10 (pending) on the Quote Status Report (AI) message.

5.5.1 Response to Quote (S) Messages for Specific Interests

The Quote Status Report (AI) message will be published to the liquidity bank in response to a Quote (S) message. QuoteStatus (297) will indicate if the quote has been accepted (= "0") or rejected (= "5"). If the quote is accepted, the price will be entered into the market. If the quote is rejected, the Quote Status Report (AI) message will include the business reason for the rejection in CancelRejectReason (9466).

5.5.2 Response to Quote Cancel (Z) Messages

The Quote Status Report (AI) message will be published to the client in response to a Quote Cancel (Z) message. The Quote Cancel (Z) message is used to either cancel Quote (S) message or Quote Request (R) message. The Quote Status Report (AI) message will reference QuoteID (117) with QuoteStatus (297) = 1 (canceled by client) indicating the price or request for quote was canceled at the request of the client.

5.5.3 Response to Quote Status Request (a) Messages

The Quote Status Report (AI) message will reference the requested QuoteID (117) or APIQuoteID (5001) with the appropriate QuoteStatus (297) value. If both tags are provided on the Quote Status Report (AI) message, then the message will be rejected and the following text will be in the reject message: "QuoteID (117) and APIQuoteID (5001) are mutually exclusive." The Quote Status Report (AI) message will also reference the QuoteStatusReqID (649) that caused the message to be generated. If the requested QuoteID (117) or APIQuoteID (5001) does not exist, QuoteStatus (297) = 9 (quote not found) will be returned to the client.

5.5.4 Response to Execution

The Quote Status Report (AI) message will be published in response to an execution. The Quote Status Report (AI) message will be sent to the owner of the quote that was executed. The QuoteStatus (297) value will be dependent on the execution type (full or partial). If the execution exhausts the entire size of the quote (full execution), QuoteID (117) will be referenced and QuoteStatus (297) will be "6" (removed from market). A partially executed price, without the OCO protection parameter set, will produce two Quote Status Report (AI) messages. Both Quote Status Report (AI) messages will reference the same QuoteID (117). One Quote Status Report (AI) message will have QuoteStatus (297) = 6 (removed from market), with the partially executed size. The other Quote Status Report (AI) message will have the QuoteStatus (297) = 0 (accepted) for the remaining live size in the market. The quote for the remaining size will keep the same QuoteID (117).

5.5.5 Quote Status Sent at End-of-Day

The FIX API will remove client quotes from the market automatically at the end-of-day. The Quote Status Report (AI) message will reference QuotelD (117) with QuoteStatus (297) = 3 (canceled by FIX) to indicate the quote was removed due to the end-of-day process.

5.5.6 Response to Quote Request (R) Message

The Quote Status Report (AI) message will be published to the clients in response to Quote Request (R) message. QuoteStatus (297) will indicate "0" (accepted) or "5" (rejected). If the Quote Request (R) message is accepted, the request will be routed to the specified target audience. The specified target audience will be found in DeliverToCompID (128). If the quote is rejected, the Quote Status Report (AI) message will include the business reason for the rejection in CancelRejectReason (9466).

The Quote Status Report (AI) message in response to a quote request will neither contain the product nor pricing details of the quote request. Along with the required fields, the Quote Request (R) message will only contain fields that refer to the status of the quote request: QuoteReqID (131), QuoteStatus (297), CancelRejectCode (9466) and Text (58).

5.5.7 Client Response to Canceled Message

If a client responds to a canceled message, they will receive a Quote Status Report (AI) message indicating that the message was canceled by the client. The Quote Status Report (AI) message will indicate QuoteStatus (297) = 1 (canceled by client). The Quote Status Report (AI) message will also indicate that CancelRejectCode (9466) = 1 (canceled by client).

5.5.8 Rejections/Cancellations by Agency Broker-Dealer

NoRFQCancelReject (8054) tracks the number of rejections and cancellations on the RFQ message. This value is incremented each time the FIX API receives an additional Quote Request Reject (AG) or Quote Cancel (Z) message. An update will be sent to the broker agent each time a cancellation or rejection is received by FIX API. If an RQF to multiple target banks fails because of an incorrect mapping in the admin console, the FIX API will send out QuoteStatus (297) = 21 (target mapping not found). If there is only one target bank and it is not mapped, the Quote Request Reject (AG) message will be sent to the broker agent. The broker agent will be notified which target bank has the erroneous mapping in BankRequested (8053). If a target bank rejects the RFQ, the FIX API sends out a Quote Status Report (AI) message indicating QuoteStatus (297) = 22 (canceled by target bank).

5.5.9 Quote Status Report (AI) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			AI (Quote Status Report)
117	QuotelD	Y	String	Unique identifier for the Quote, established by the client
131	QuoteReqID	N	String	Unique identifier for Quote Request (R) message
5001	APIQuotelD	C	String	Unique identifier for the received accepted Quote (S) message; established by the FIX API; included in the Market Data (W) message
649	QuoteStatusReqID	C	String	Unique identifier for the Quote Status Request ID; only required if the message is in response to a request from the client

297	QuoteStatus		Y*	Int	Identifies the status of the quote acknowledgement; valid values: 0 = Accepted 1 = Canceled by Client 3 = Canceled by FIX 5 = Rejected 6 = Removed from Market 9 = Quote Not Found 10 = Pending 21 = Target Mapping Not Found 22 = Canceled by Target Bank
9466	CancelRejectCode		C	Int	The rejection or cancel code for QuoteStatus (297) = 5 (Rejected) or 1 / 3 / 22 (Canceled)
8505	RegulationVenueType		C	Int	Indicates the market type of the quote; valid values: 1 = SEF 3 = MiFID 7 = OTC 8 = IB
8519	OriginatingLEI		C	String	LEI of beneficial owner of swap
5159	Clearer		C	String	Specifies the clearer LEI
6661	PBReuterCode		C	String	Assigned LEI used to identify the associated prime broker / FCM; if LEI is not available, then trading name is used; used if quote provider uses a prime broker/ FCM
453	NoPartyID		N	NumIn Group	Number of security ID groups that follow; required for MiFID II-relevant securities
	448	PartyID	N	String	Value for party role; required for MiFID II-relevant securities
	447	PartyIDSource	N	Char	Required for MiFID II-relevant securities; Valid value: D = Proprietary ID
	452	PartyRole	N	Int	Required for MiFID II-relevant securities; valid values: 11 = Order Origination Trader, execution within firm 16 = Executing System, algorithmic system identifier 122 = Investment Decision Maker
2593	NoOrderAttributes		N	NumIn Group	Number of regulatory order attributes; required for MiFID II-relevant securities.
	2594	OrderAttributeType	N	Int	Optionally required for MiFID II-relevant securities if NoOrderAttributes (2593) > 0; valid values: 2 = Liquidity provision activity order 4 = Algorithmic order
	2595	OrderattributeValue	N	String	Attribute value; required for MiFID II-relevant securities securities ; valid values: Y = Yes N = No
8052	NoBanksRequested		C	NumIn Group	Number of liquidity bank targets in the RFQ
	8053	BankRequested	C	String	FIXCOMPID of target liquidity bank in the RFQ
8054	NoRFQCancelReject		N	Int	This tracks the number of rejects and cancellations in the BA RFQ
9126	Structure		Y*	Int	FX Trading Systems primary Structure information; see appendix for valid values
55	Symbol		Y	String	Six-character string that represents the currency pair
54	Side		Y	Char	Side of the entry; indicates the direction the quote was posted; required for non-RFQ request; valid values: 1 = Bid 2 = Offer
6436	Premium		C	Price	Structure premium
8503	PremiumMid		N	Price	
5830	PremiumCcy		C	Currency	Currency of the premium amount
6289	ManualQuoteIndicator		C	Char	Indicates if this quote requires manual intervention; required in response to a Quote Request (R); valid value: A = Automatically Live
62	ValidUntilTime		C	UTC Timestamp	The time when the quote will expire
6120	NoMQEntries		C	Int	Count of market quote entries to follow
Repeating	7940	LegID	Y*	String	The strategy or leg identifier; max length = 50 bytes
	5678	Volatility	C	Float	Price volatility
	8588	FwdPrice	N	Price	All-in Rate for NDFs Points for FX products

	8515	VolMid	N	Price	Dealer provided mid-price
	8518	GFIMid	N	Price	GFI calculated Mid-Price (only applicable for Fenics Direct)
	5359	MQSize	Y*	Float	Size of entry
	5235	LegSpotRate	C	Float	F/X spot rate
	9099	ForwardRate	N	Float	Forward rate
	5191	LegForwardPoints	C	Float	F/X display swap rate; assume industry standards for swap points per currency pair; for example, a realistic forward points value for a EURUSD call can be 2.7
	9115	DepoRate	C	Float	Depo rate (%); for example, a realistic depo rate value for a EURUSD call can be 5.04
	9073	DepoRateCcy	C	Currency	Depo rate currency
	5844	LegPremPrice	C	Price	Leg price; will be sent as a formatted value: • For Pct LegPrice = Raw % Price * 100 • For Pts LegPrice = Raw Pts Price * 10^(spot decimal places)
	6035	LegDelta	C	Float	Delta (%)
	6354	MQStrikePrice	C	Price	Strike price
	5167	LoTrigger	C	Price	Identifies trigger price in absolute terms for the single or low trigger; required on single or double trigger options
	6192	HiTrigger	C	Price	Identifies trigger price in absolute terms for the high trigger; required on double trigger options
	7092	LegPremium	C	Price	Leg premium; required for FXO premium quotes; used to capture leg level trade details; will not be validated against Premium (6436)
58	Text		C	String	When possible, will contain the cancel / reject reason; see appendix for descriptions
60	TransactTime		Y*	UTC Timestamp	Timestamp in microseconds of execution/order creation in YYYYMMDD-HH:MM:SS.ssssss format. Timestamps can be in non microsecond format (based on client preference for MiFID tags)
7464	NoHedgeLegs		N	NumIn Group	The number of hedges
Repeating Group	9074	HedgeCurrency	Y*	Currency	Currency of the hedge amount; only required when NoHedgeLegs (7464) > 0
	9016	HedgeTradeType	N	Int	Indicates the type of hedge trade executed for offsetting option risk; valid values: 1 = Spot 2 = Forward
	6666	HedgeDirection Indicator	N	Int	Hedge direction; valid values: 1 = Buying 2 = Selling
	6036	HedgeAmount	N	Amt	Hedge amount; entry value in millions e.g. an entry of 1 equals 1,000,000
	9657	HedgeRate	N	Float	Hedge exchange rate
	9112	HedgeDelivery	N	LocalMkt Date	Delivery date for the delta hedge
	6426	HedgeSymbol	N	String	Currency pair for hedge leg
9536	Trader		N	String	The trader identifier who submitted the price
	Standard Trailer				

Y=FIX required; Y*=API required; C=Conditional; N=Not required

5.6 Quote Status Request (a)

The Quote Status Request (a) message is used to allow a client to query the status of published quotes. A client will query the status of a request by sending the Quote Status Request (a) message with at least one of the following fields: QuoteID (117), APIQuoteID (5001) or Symbol (55). The FIX API will reply to this message with a Quote Status Report (AI) message.

5.6.1 For All Active Quotes

The Quote Status Request (a) message will be used by a client to request all active quotes that are currently in the trading system from their institution. To do so, the client will set Symbol (55) to

“ALL_ALL”. If the client is requesting all active quotes, then the following fields will be excluded from the Quote Status Request (a) message: QuoteID (117) and APIQuoteID (5001). Including either QuoteID (117) or APIQuoteID (5001) will force the FIX API to look for a specific quote.

5.6.2 For Specific Quotes

The Quote Status Request (a) message will also be used by the client to request the last state of a Quote (S) message. To do so, the client will provide an ID for one of the following fields: QuoteID (117) or APIQuoteID (5001). If the client is requesting the status of a specific quote, the QuoteID (117) or APIQuoteID (5001) must contain the appropriate value. If the QuoteID (117) or APIQuoteID (5001) is invalid, the Quote Status Report (AI) message will return QuoteStatus (297) = 9 (quote not found). To request a specific quote request, only one ID, QuoteID (117) or APIQuoteID (5001), will be provided. All other values will be ignored.

5.6.3 Quote Status Request (a) Identifier

QuoteStatusReqID (649) is a unique identifier assigned by client. The QuoteStatusReqID (649) will conform to the following rules:

1. All Quote Status Request (a) messages must have a unique value in QuoteStatusReqID (649) for that day, regardless of the trading session the request was submitted in.
2. The QuoteStatusReqID (649) value may only contain upper-case letters or numbers.
3. The QuoteStatusReqID (649) value may be up to 24 characters in length.

5.6.4 Quote Status Request (a) Message Configuration

Tag	Field Name		Req	Type	Description
	Standard Header				
35	MsgType				a (Quote Status Request)
649	QuoteStatusReqID		Y*	String	Unique identifier generated by the client
117	QuoteID		C	String	Unique identifier for the Quote (S) established by the client
5001	APIQuoteID		C	String	Unique identifier for the Quote Status Report (AI) established by the API
55	Symbol		Y	String	Six-character string that represents the currency pair; to request all quotes across all currency pairs, use ALL_ALL
555	NoLegs		N	NumIn Group	Number of legs
	600	LegSymbol	Y	String	Six-character string that represents the currency pair e.g. “EURUSD”; the first ccy is always the base and the second is always the term; “[N/A]” is an acceptable value
	624	LegSide	N	Char	Used to request a specific side: 1 = Buy 2 = Sell
	Standard Trailer				

Y=FIX required; Y*=API required; C=Conditional; N=Not required

5.7 Quote Cancel (Z)

The Quote Cancel (Z) message is used to cancel quotes or quote requests. The objective is for the sender of the Quote Cancel (Z) message to successfully remove their own message from the trading system. If any client responds to a canceled message, that response will be rejected by the FIX API. The Quote Cancel (Z) message will be transmitted in the following ways:

Sender	Target Audience	Cancel Types	Description
API Client	FIX API	Cancel Quote Cancel Quote Request	The sender will use the message to cancel a quote or request for a quote that the sender’s system sent to the FIX API

FXI API	API Clients	Cancel Quote Cancel Quote Request	The FIX API will act as a hub by routing Quote Cancel messages to the appropriate target audiences
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5.7.1 Quote Cancellation for API Client

API clients will be able to cancel their own quote messages by sending Quote Cancel (Z) messages. To cancel a quote, the client will specify the quote's QuoteID (117) in CancelID (9262) and set QuoteCancelType (298) to "1" (cancel quote).

The FIX API will receive the Quote Cancel (Z) message from the client and return a Quote Status Report (AI) message for acknowledgement. If the Quote Status Report (AI) message is returned with QuoteStatus (297) = 1 (canceled by client), the FIX API will route the cancel message to the appropriate targets. The appropriate targets will be all of the clients that received the original Quote (S) message sent by the client. The FIX API client will send the Quote Cancel (Z) message to the appropriate targets by setting SenderCompID (49) to broker CompID, TargetCompID (56) to target CompID, and the OnBehalfOfCompID (115) to sender CompID.

If a trader uses the front-end to cancel a quote that was sent by the trader's liquidity bank API client, then the FIX API will notify the trader's API that the quote was canceled by sending a Quote Status Report (AI) message. The FIX API will also forward the Quote Cancel (Z) message to the appropriate target.

5.7.2 Quote Cancel (Z) Message to Cancel Quote Request

Users will be able to cancel their own Quote Request (R) messages. To cancel a quote request, the client should send a Quote Cancel (Z) message to the FIX API and will specify the quote's ID in CancelID (9262). The client should also set QuoteCancelType (298) = 2 (Cancel Quote Request).

The FIX API will receive the Quote Cancel (Z) message from the client and will send a Quote Status Report (AI) message to the client to indicate the validity of the message. If the Quote Status Report (AI) message is returned with QuoteStatus (297) = 1 (canceled by client), the FIX API will route the cancel message to the appropriate targets. The appropriate targets will be the clients that received the original Quote Request (R) message sent by the client. The FIX API client will send the Quote Cancel (Z) message to the appropriate targets by setting SenderCompID (49) to broker CompID, TargetCompID (56) to target CompID, and OnBehalfOfCompID (115) to sender CompID.

5.7.3 Cancel Quote Request with Live Quotes

API clients will be able to cancel their own Quote Request (R) messages that have live quotes associated with the request. The API client will send a Quote Cancel (Z) message for the Quote Request (R) to the FIX API, and the FIX API will send a Quote Status Report (AI) message to the owner of the quote in response to the Quote Request (R) to give cancel notification. The FIX API will also send the target of the quote request a Quote Cancel (Z) message to cancel the quote request.

5.7.3.1 Authorized User Rejects Quote Request OBO API Client

Authorized front-end traders will be able to reject Quote Request (R) messages on behalf of their own bank's API client. If a front-end trader rejects a quote request, the front-end will send a Quote Request Reject (AG) message to the FIX API. The FIX API will forward the Quote Request Reject (AG) message to the sender of the Quote Request (R) and will send the bank's external API client a Quote Cancel (Z)

message to cancel the quote request. The QuoteCancelType (298) will be set to 2 (cancel quote request), the TargetCompID (56) will be set to the bank's CompID, and there will not be an OnBehalfOfCompID (115) value.

5.7.3.2 Quote Canceled Upon Quote Modification

In response to a quote request, a Quote (S) message will be canceled when a subsequent quote, having the same side, is sent in response to the same quote request. The FIX API client will send a Quote Status Report (AI) message to the quote sender to indicate that their quote was canceled. The FIX API will also send a Quote Cancel (Z) message and the new quote to the specified target. The target will acknowledge both the cancel and the new quote by sending quote status messages.

5.7.3.3 Automatic End of Day Quote Cancells: Quotes and Quote Requests

At the designated end-of-day, the FIX API server will send Quote Cancel (Z) messages for all live quotes and quote requests to all external API clients connected to the FIX API. The Quote Cancel (Z) messages will have "ALL_ALL" in the Symbol (55) in order to cancel all live messages for all currency pairs.

5.7.3.4 Automatic Logout Quote Cancells: Quotes and Quote Requests

Upon every external API logout, the FIX API will cancel all of the external client's live quotes and quote requests. The FIX API will cancel the live messages by sending quote cancels to the clients that were the targets of the messages. The Quote Cancel (Z) messages will have "ALL_ALL" in Symbol (55) in order to cancel all live messages for all currency pairs.

5.7.4 Quote Cancel (Z) Message Configuration

Tag	Field Name		Req	Type	Description
	Standard Header				
35	MsgType				Z (Quote Cancel)
131	QuoteReqID		C	String	Unique identifier for Quote Request (R) message; required in response to an RFQ
117	QuoteID		Y	String	Unique identifier established by the client
9262	CancelID		N	String	Unique identifier of the quote being canceled
8505	RegulationVenueType		N	Int	Indicates the market type of the quote; if venue type is specified, the FIX API will cancel all quotes for the venue when Symbol (55) = ALL_ALL; valid values: 1 = SEF 3 = MiFID 7 = OTC 8 = IB
298	QuoteCancelType		Y	Int	Identifies the type of request; valid values: 1 = Cancel Live Quote 2 = Cancel Quote Request 4 = Pass on Interest (used by the quote requestor to indicate a pass signal on an anonymous RFQ)
295	NoQuoteEntries		N	NumIn Group	The number of securities (instruments) whose quotes are to be canceled; valid value = 1
Repeating Group	55	Symbol	Y	String	Six-character string that represents the currency pair; to cancel prices across all currency pairs, use "ALL_ALL"
	9126	Structure	N	Int	FX trading systems primary structure information; see appendix for valid values
	9127	LegDeliveryType	C	Char	Required if using Structure (9126) to cancel prices in deliverable forward and NDF pods; valid values: 1 = Cash 2 = Physical
	811	PriceDelta	N	Float	Indicates if the cancel is for the 25-delta or 10-delta structures; required for risk reversal and butterfly structures; see appendix for valid values
	555	NoLegs	N	NumIn Group	Number of legs

		600	LegSymbol	Y	String	Six-character string that represents the currency pair e.g. "EURUSD"; the first ccy is always the base and the second is always the term; "[N/A]" is acceptable value
		624	LegSide	N	Char	Used to request a specific side: 1 = Buy 2 = Sell
		Standard Trailer				

Y=FIX required; Y*=API required; C=Conditional; N=Not required

5.8 Market Data (W) – Snapshot/Full Refresh

The Market Data (W) – Snapshot/Full Refresh message is used to electronically stream prices, in terms of volatility for FXO; or spot and forward points, or all-in rate for FX products from the front-end via the FIX API. Published price messages will not require a confirmation of receipt from the client. It will be the responsibility of the heartbeat to ensure a connection is present. Market Data (W) messages will be published for indirect (anonymous) quotes on specific interests and quotes on standard products in the front-end.

5.8.1 MD Publications for Anon Specific Interests and STP Products

The Market Data (W) message will be an unsolicited message, which will contain the bid and offer prices for a specific interest or a standard product for which a tradable market exists in the front-end.

The message sent for an anonymous specific interest or standard product will provide the product details in the structure level of the message. The price contributions for each strategy or leg of Structure (9126) will be provided in the NoMDEntries (268) repeating group.

5.8.2 Market Data for Standard Products

The Market Data (W) message will be published in response to a change in the market for any of the standard products. The change in the market does not have to be on the best bid or best offer level and can reflect a change of a price in the depth. The message will be able to store multiple prices for multiple tenors that belong to a single standard product. The whole book is published on this message without any depth restrictions

5.8.3 Market Data for Specific Interests

The Market Data (W) message that is publishing data for a FXO spread specific interests will contain a volatility contribution for each strategy or leg, depending on Structure (9126). The Market Data (W) message will link the volatilities contributed to the strategy or leg of a spread specific interest using the QuoteReqID (131) and LegID (7940) fields. However, differential prices will not have volatility information.

The Market Data (W) message will be structured to provide the best prices, as well as any additional prices. The combination of LegID (7940), MDEntryType (269), Tenor (6215) and MDEntryPositionNo (290) will be used to determine the position of the market data record.

- **LegID (7940):** Uniquely identifies a strategy leg and will be used with QuoteReqID (131) to reference the leg on the original RFQ.
- **MDEntryPositionNo (290):** Indicates the position of a price within a stack of prices. The field accepts positive integers ranging from "1" to "N", such that "1" indicates the best price and the greatest value, while "N" indicates the worst price.

- **IsOwner (5331):** Indicates “Y” if the price belongs to the receiving client’s trading group. This can be a price entered in using the API or GUI.

5.8.4 Market Data Updates

The Market Data (W) message will be used to broadcast price updates that are already in the system. If a price change occurs in the front-end, the Market Data (W) message will contain the entire stack for the updated tenor and side (bid or offer) of the contract. MDSOURCE (5177) will be set to “1” (update) to broadcast updated prices.

If the client sends a subscription message per currency pair and there is no data in that Currency Pair at all, the “No Market Data” is sent. If, however, there are updates available for that Currency Pair, then the system will send what is live for the products that have live prices. All other products for that currency pair can then be considered as No Market Data.

5.8.5 Market Data Cancellations

The Market Data (W) message will be used to broadcast canceled prices already in the system. If there is no longer a tradable price for which one previously existed, a (W) message will be sent without Price (270) and Size (271) set to “0”. For differentials, the differential price and size will not be sent. MDSOURCE (5177) will be set to “1” (withdrawal) to broadcast canceled prices. If there is no market data for a tenor, then “No Market Data” will be in Text (58).

5.8.6 Market Data Execution

The (W) message will be used to broadcast the state of a specific market, after an execution on prices that were previously broadcasted in a (W) message. The (W) message will contain all live prices for the specific market; the executed prices will not be present in the message. MDSOURCE (5177) will be set to “0” (execution) to indicate that prices were executed and removed from the market. Full execution details will be captured in the News (B) message, which will be sent at the same time as the market data execution message. The News (B) message will contain trade details including, but not limited to, side (aggressor’s perspective), structure, tenor, delta, price, and quantity.

5.8.7 Specific Interests, Differential and Vega Neutral Updates

In the Market Data (W) message, specific interests, differential, and vega neutral spreads will not contain granular level leg details. The leg details for the prices will implicitly reference those leg details from the related Quote Request (R) message. For differential and vega neutral spreads, the bid and offer will implicitly reference the position (same or inverse) from the specified RFQ. For instance, a bid will indicate that the quote is buying legs that have Position (6351) = 1 (same) (on the referenced RFQ), and selling legs that have Position (6351) = 2 (inverse). Likewise, an offer will indicate that the quote is selling legs that have Position (6351) = 1 (same), and buying legs that have Position (6351) = 2 (inverse). Both Quote (S) and Quote Request (R) messages may contain granular leg level details for differential and vega neutral spreads. If the differential value is not provided in the Quote (S) message, then the granular leg price details will be converted to differential details for the Market Data (W) message.

5.8.8 Market Data (W) Message Configuration

Tag	Field Name	Req	Type	Description
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	Standard Header				
35	MsgType				W (Market Data)
262	MDReqID		C	String	Market Data unique request ID; populated if the messages is in response to a Market Data Request (V) message
131	QuoteReqID		C	String	Unique identifier for the request; required only for anonymous RFQs
8505	RegulationVenueType		C	Int	Indicates the market type of the quote; valid values: 1 = SEF 3 = MiFID 7 = OTC 8 = IB
5159	Clearer		C	String	Specifies the clearer LEI
55	Symbol		Y	String	Six-character string that represents the currency pair e.g. "EURUSD"; the first ccy is always the base and the second is always the term
9126	Structure		Y	Int	FX trading systems primary structure information; required only for FXO and FX standard pods; see appendix for valid values
9127	LegDeliveryType		C	Char	Required when Structure (9126)=65 (FX) to indicate deliverable or non-deliverable; valid values: 1 = Cash (indicates non-deliverable) 2 = Physical (indicates deliverable)
9072	CallOrPutOver		C	Int	Call or put indication; required only for standard pod risk reversals; valid values: 1 = Calls Over Puts 2 = Puts Over Calls
5232	CallOrPutOverCcy		C	Currency	Three-character string that represents call/put currency; required only for standard pod risk reversals
6035	LegDelta		C	Float	Delta (%); required only for standard pods
9125	Cutoff		C	Int	Time of expiration of FX option and fixing index for NDF; required only for FXO standard products and all FX products; see appendix for valid values
5177	MDSource		Y*	Int	Market data source; valid values: 0 = Execution 1 = Withdrawal/Entry/Update 2 = Request 3 = Not in Top Ranking 4 = Market Halt
268	NoMDEntries		Y	NumIn Group	Number of entries in the message
	269	MDEntryType	Y	Char	Type of market data entry; must be the first field in this repeating group; valid values: 0 = Bid 1 = Offer
	7940	LegID	C	String	The strategy or leg identifier for spread price contributions; required for RFQ-related quotes; not required for FXO standard pods and FX products; max length = 50 bytes
	537	QuoteType	Y	Int	Valid values: 0=Indicative 1=Tradable
	6215	Tenor	C	String	FX and FX option maturity; only required for standard pods; see appendix for valid values
	8528	FXAPISecurityID	C	String	Security identifier; only utilized for standard pods
	5475	PremDeliveryType	N	Char	Premium delivery; valid values: S = Spot F = Forward
	270	MDEntryPx	C	Price	Price of the entry in terms of volatility (%) for FXO products, and all-in rate for FX
	271	MDEntrySize	C	Qty	Quantity for the entry in terms of millions e.g. an entry of 1 equals 1,000,000
	9102	FXMarketType	C	Int	NDF indicator; valid values: 1 = Onshore 2 = Offshore
	432	ExpireDate	C	LocalMkt Date	Date of expiration; only required for FXO standard pods
	6203	FixingDate	C	LocalMkt Date	Date of the comparison between the nondeliverable FX rate and prevailing spot rate
	9121	FarFixingDate	C	LocalMkt	Fixing date for the second leg of NDF

				Date	
	743	DeliveryDate	C	LocalMkt Date	Delivery / settlement date
	8014	FarDeliveryDate	C	LocalMkt Date	Far delivery / settlement date; required for deliverable forward and NDF spreads
	290	MDEntryPositionNo	N	Int	The position of a bid or offer, numbered from most competitive to least competitive per market side beginning with "1"
	58	Text	C	String	Free text field; used to specify no market data
	9773	OCOProtection	N	Int	OCO price protection parameter; standard products only; valid values: 0 = Off 1 = On
	6003	EligibleVenues	C	String	Denotes all eligible venues of the market maker; valid values: 1 = SEF Only 3 = MiFID Only 7 = OTC Only 13 = SEF or MiFID 17 = SEF or OTC 37 = MiFID or OTC 137 = SEF, MiFID or OTC
	5331	IsOwner	C	Boolean	Will indicate "Y" if the price belongs to the receiving client's trading group (can be a price which is entered in using the API or entered in using the GUI); valid value: Y = Your Trading Groups Price
60	TransactTime		Y*	UTC Timestamp	Timestamp in microseconds of market data in YYYYMMDD-HH:MM:SS.ssssss format. Timestamps can be in non microsecond format (based on client preference for MiFID tags)
	Standard Trailer				

Y=FIX required; Y*=API required; C=Conditional; N=Not required

5.9 Market Data Request (V)

The Market Data Request (V) message will be used by external liquidity bank API clients to subscribe to receive Market Data (W), anonymous Quote Request (R), and News (B) messages. The Market Data Request (V) message will be sent after an external API client successfully logs into the FIX API. To make a market data subscription, the external API will specify a set of currency pairs in the NoRelatedSym (146) repeating group. The external API will only receive market data messages (market data, anonymous RFQs, and news messages) for the currency pairs that were listed in the Market Data Request (V) message.

The Market Data Request (V) message will also be used to give clients the ability to sync their market data should the market data become distorted on the client side. A sync request will indicate SubscriptionRequestType (263) = 1 (market snapshot). All Market Data Request (V) messages should have a valid Symbol (55) value.

5.9.1 Full Market Sync and Subscription to All Currency Pairs

A full market sync request or a subscription to all authorized currency pairs can be made by specifying the currency pair in Symbol (55) as "ALL_ALL" in the Market Data Request (R) message.

5.9.2 Specific Markets Sync

If a select few markets need to be synced, the client can specify the desired currency pair in Symbol (55) to request market data for all structures for that currency pair. Subscribing to market data by Structure (tag 9126) is not supported.

5.9.3 Market Data Request (V) Message Identifier

The Market Data Request (V) message identifier, MDReqID (262), is a unique identifier assigned by the client for a market data request they are submitting to the trading system.

The MDReqID (262) should conform to the following rules:

1. All Market Data Request (V) messages must have a unique value in MDReqID (262) for that day, regardless of the trading session the request was submitted.
2. MDReqID (262) value may only contain letters (upper case) or numbers.
3. MDReqID (262) value may be up to 24 characters in length.

5.9.4 Market Data Request (V) Message Configuration

Tag	Field Name		Req	Type	Description
	Standard Header				
35	MsgType				V (Market Data Request)
262	MDReqID		Y	String	Unique identifier the request
263	SubscriptionRequestType		Y	Char	Subscription request type; valid value: 1 = Subscribe to receive snapshots and updates (Market Data (W), Anonymous Quote Request (R), and News (B))
264	MarketDepth		Y	Int	Depth of market for book; valid value: 0 = Full Book
267	NoMDEntries		Y	NumIn Group	Number of entry types the request message
	269	MDEntryType	Y	Char	List of all the types of market data entries that the client requesting the market data is interested in receiving; valid values: 0 = Bid 1 = Offer
146	NoRelatedSym		Y	NumIn Group	Specifies the number of repeating currency pair groups
	55	Symbol	Y	String	Six-character string that represents the currency pair; to request all currency pairs, use "ALL_ALL"
	9126	Structure	N	Int	FX trading systems primary structure information; see appendix for valid values
	811	PriceDelta	N	Float	Indicates if risk reversal or butterfly request is a 25-delta or 10-delta request; see appendix for valid values
	9125	Cutoff	N	Int	Time of expiration of FX option. Valid values; see appendix for valid values
	Standard Trailer				

Y=FIX required; Y*=API required; C=Conditional; N=Not required

5.10 Market Data Request Reject (Y)

The Market Data Request Reject (Y) message is used to reject Market Data Request (V) messages. The rejection reasons could be due to requesting market data on markets not supported, or making market data request using tags not supported in the Market Data Request (V) message.

5.10.1 Market Data Request Reject (Y) Message Configuration

Tag	Field Name		Req	Type	Description
	Standard Header				
35	MsgType				Y (Market Data Request Reject)
58	Text		Y	String	The reason for Market Data Request rejection. See the Appendix for descriptions.
262	MDReqID		Y	String	Specifies the Market Data Request ID
	Standard Trailer				

Y=FIX required; Y*=API required; C=Conditional; N=Not required

5.11 New Order – Single (D)

API users can send unsolicited orders to the central limit order book (CLOB) with the New Order – Single (D) message. Subsequent updates or cancellation can be submitted via Order Cancel Replace Request (G) or Order Cancel Request (F) messages, respectively. The FIX API will acknowledge each message with either an Execution Report (8) or Order Cancel Reject (9) message.

5.11.1 Submit Standing Order

When submitting an unsolicited CLOB order, the New Order – Single (D) message must provide the FXAPISecurityID (8528) that identifies the contract. A list of forexAPISecurityIDs will be provided to clients upon receiving the Security List Request (x). The New Order – Single (D) message will have TimeInForce (59) = 1 (Good ‘til Cancel) or 6 (Good ‘til Date). When selecting (59) = 6 (Good ‘til Date), the expiration date must be provided in ExpireTime (126).

5.11.2 Execute a Standing Order

The New Order – Single (D) message can also be used to execute on a standing CLOB order. The FIX API client can specify the ClOrdID (11), FXAPISecurityID (8528), FwdPrice (8588) or Volatility (5678), Side (54), MQSize (5359), TimeInForce (59) and PremDeliveryType (5475). OriginatingLEI (8519) is required when executing on SEF orders. The New Order – Single (D) message can only be used to execute standing CLOB orders. Tradable quotes submitted via the Quote (S) can be executed using the New Order - Multileg (AB) message.

5.11.3 New Order – Single (D) Message Configuration

Tag	Field Name		Req	Type	Description
	Standard Header				
35	MsgType				D (New Order – Single)
11	ClOrdID		Y	String	Unique identifier for the order
55	Symbol		Y	String	Six-character string that represents the currency pair
8528	FXAPISecurityID		Y	String	Contract identifier for CLOB orders
8519	OriginatingLEI		C	String	LEI of beneficial owner of the swap; required when forexAPISecurityID (8528) belongs to SEF venue
6661	PBReuterCode		C	String	LEI of prime broker or FCM; required for prime-brokered or cleared orders
453	NoPartyID		N	NumIn Group	Number of party distinctions as required for MiFID II-relevant securities
	448	PartyID	N	String	Value for party role; required for MiFID II-relevant securities
	447	PartyIDSource	N	Char	Required for MiFID II-relevant securities; Valid value: D = Proprietary ID
	452	PartyRole	N	Int	Required for MiFID II-relevant securities; valid values: 11 = Order Origination Trader, execution within firm 16 = Executing System, algorithmic system identifier 122 = Investment Decision Maker
54	Side		Y	Char	Requestor side of the order; valid values: 1 = Buy 2 = Sell
2593	NoOrderAttributes		N	NumIn Group	Number of regulatory order attributes; required for MiFID II-relevant securities.
	2594	OrderAttributeType	N	Int	Optionally required for MiFID II-relevant securities if NoOrderAttributes (2593) > 0; valid values: 2 = Liquidity provision activity order 4 = Algorithmic order
	2595	OrderattributeValue	N	String	Attribute value; required for MiFID II-relevant securities securities ; valid values: Y = Yes N = No

60	TransactTime	Y	UTC Timestamp	Timestamp in microseconds of the message in YYYYMMDD-HH:MM:SS.ssssss format. Timestamps can be in non microsecond format (based on client preference for MiFID tags)
8588	FwdPrice	C	Price	All-in rate for NDFs Points for FX products
5678	Volatility	C	Float	Price volatility for FXO
40	OrdType	Y	Char	Valid value: 2 = Limit (indicates CLOB order)
5359	MQSize	Y	Float	Order size
59	TimeInForce	Y	Char	Valid values: 1 = Good 'til Cancel (GTC; used to indicate a CLOB order) 3 = Immediate or Cancel (IOC; used when trying to execute on a standing order) 6 = Good 'til Date (GTD)
126	ExpireTime	C	UTC Timestamp	The time when the order will expire; required if TimeInForce (59) = 1 (GTD)
5475	PremDeliveryType	C	Char	Premium Delivery; required when executing on a standing order; valid values: S = Spot F = Forward B = Both
9102	FXMarketType	N	Int	Used only when executing on a standing order; valid values: 1 = Onshore 2 = Offshore 3 = Both
9536	Trader	N	String	Identifies the trader submitting the order; broker- assigned UserID
	Standard Trailer			

5.12 Order Cancel/Replace Request (G) Message

The Order Cancel/Replace Request (G) message is used to modify existing CLOB orders with remaining unfilled quantity. When submitting the (G) message, the ClOrdID (11) of the referenced order is the value for OrigClOrdID (41). The following fields are the only fields that may be updated as part of order modification: Volatility (5678), FwdPrice (8588), MQSize (5359) and ExpireTime (126). A request to replace MQSize (5359) to a level less than CumQty (14) will not be allowed, and thus will be rejected with an Order Cancel Reject (9) message. All other needed changes (including OriginatingLEI) must be done via Order Cancel Request (F), followed by New Order – Single (D) with the correct parameters. FIX API clients streaming orders will submit a (D) message for the initial order, but use the (G) messages for subsequent order updates.

5.12.1 Order Cancel/Replace Request (G) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			G (Order Cancel/Replace Request)
41	OrigClOrdID	Y	String	ClOrdID (11) of the referenced order that is being canceled or replaced
11	ClOrdID	Y	String	Unique identifier of replacement order assigned by client; will be used in ClOrdID (11) of Order Cancel Reject (9) if the replacement request is rejected
55	Symbol	Y	String	Required by FIX but not used by the API; valid value: N/A
54	Side	Y	Char	Side of order to the requestor; should match the original order's side; valid values: 1 = Buy 2 = Sell
60	TransactTime	Y	UTC Timestamp	Timestamp in microseconds of the message in YYYYMMDD-HH:MM:SS.ssssss format. Timestamps can be in non microsecond format (based on client preference for MiFID tags)
40	OrdType	Y	Char	Valid value: 2 = Limit

5359	MQSize	Y	Float	Order size
5678	Volatility	C	Float	Price volatility; required for FXO
8588	FwdPrice	C	Price	All-in-rate for NDFs
59	TimeInForce	N	Char	Valid value: 1 = Good 'til Cancel (Default) 6 = Good 'til Date
126	ExpireTime	N	UTC Timestamp	The time when the order will expire
	Standard Trailer			

5.13 Order Cancel Request (F)

The Order Cancel Request (F) message requests the cancellation of all the remaining quantity of a CLOB order. Any order parameter changes besides size and price will require cancellation of the original order and submission of a new order using the New Order – Single (D) message.

5.13.1 Order Cancel Request (F) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			F (Order Cancel Request)
41	OrigClOrdID	Y	String	ClOrdID (11) of the referenced order that is being canceled or replaced
11	ClOrdID	Y	String	Unique identifier of Order Cancel Request as assigned by client; will be used in ClOrdID (11) field of the Order Cancel Reject (9) message if the replacement request is rejected
	Instrument		Component Block	
55	Symbol	Y	String	Required by FIX but not used by the FIX API; valid value: NA
54	Side	Y	Char	Side of the initial order; FIX API will reject if the side specified is not the same as in the New Order – Single (D) message; valid values: 1 = Buy 2 = Sell
60	TransactTime	Y	UTC Timestamp	Timestamp in microseconds of the message in YYYYMMDD-HH:MM:SS.ssssss format. Timestamps can be in non microsecond format (based on client preference for MiFID tags)
	Standard Trailer			

5.14 Order Cancel Reject (9)

The Order Cancel Reject (9) message is the response to failed Order Cancel Request (F) or Order Cancel/Replace Request (G) messages. The message will indicate the current status of the order. FIX API will return only four values in OrdStatus (39): “New” to indicate that the live order has not been partially or fully filled, “Partially Filled” to indicate that the order has remaining balance, “Rejected” to indicate when the order does not exist, or “Filled” to indicate that the order has already been fully filled (resulting in a race condition).

5.14.1 Order Cancel Reject (9) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			9 (Order Cancel Reject)
37	OrderID	Y	String	Unique identifier of order assigned by FIX API; when CxlRejReason (102) = 1 (Unknown Order), value “NONE” will be specified
41	OrigClOrdID	Y	String	ClOrdID (11) of the previously accepted order which could not be replaced or canceled
11	ClOrdID	Y	String	Unique identifier of replacement order assigned by client

39	OrdStatus	Y	Char	Identifies the current status of the order; valid value: 0 = New 1 = Partially Filled 2 = Filled 8 = Rejected
434	CxlRejResponseTo	Y	Char	Identifies the type of request that the Order Cancel Reject is in response to; valid values: 1 = Order Cancel Request (F) 2 = Order Cancel/Replace Request (G)
102	CxlRejReason	Y	Int	Identify reason for rejection; valid values: 1 = Unknown Order [OrdStatus (39) will be 8 (Rejected)] 3 = Order Already in Pending Cancel or Pending Replace Status 99 = Other
58	Text	N	String	Free format text string which specifies the rejection reason
60	TransactTime	N	UTC Timestamp	Timestamp in microseconds specified by trader in the original Order Cancel Request or Order Cancel/Replace Request in YYYYMMDD-HH:MM:SS.ssssss .format. Timestamps can be in non microsecond format (based on client preference for MiFID tags)
	Standard Trailer			

5.15 New Order – Multileg (AB)

The New Order – Multileg (AB) message is used to execute on orders on instruments that have one or more strategy legs in Quote (S) messages. The message will have TimeInForce (59) = 3 (Immediate or Cancel) which means that the order will not appear in the CLOB. QuoteID (117) should refer to an ID of a Quote (S) message. QuoteReqID (131) is only required on messages that are in response to Quote (S) messages that were, in turn, generated for Quote Request (R) messages. The (AB) messages must reference individual LegID (7940) entries from (S) messages. All (AB) messages will generate Execution Report (8) messages which will indicate whether the orders were filled, canceled, or rejected. NoLegs (555) will contain price details per LegID (7940). For direct RFQs, a USI or UTI namespace and value for each trade leg or hedge (where hedge is requested). Legs will use LegNamespace (8500) and LegUsiValue (8501) or LegUtiValue (8524). Hedges will use HedgeNamespace (8508) and HedgeUsiValue (8509) or HedgeUtiValue (8526). The minimum quantity, MinQty (110), will be used to calculate the smallest notional size that will be executed on each order leg. If MQSize (5359) of any leg is less than the minimum quantity, the (AB) message will be rejected.

5.15.1 Client Order Identifier

A unique client order identifier, ClOrdID (11), will be generated by the FIX API when an (AB) message is sent. The ClOrdID (11) will conform to the following rules:

1. All new messages will have a unique value in this field for that day, regardless of the trading session the report was submitted.
2. The value will only contain upper-case letters and/or numbers up to 24 characters in length.

5.15.2 New Order – Multileg (AB) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			AB (New Order – Multileg)
11	ClOrdID	Y	String	Unique identifier for the order
131	QuoteReqID	C	String	Unique identifier for Quote Request (R); required for direct RFQ
8504	RegulationVenue	C	String	LEI of execution venue; required for SEF trades

8505	RegulationVenueType		C	Int	Indicates market type; trade will be prevented if the sender is not entitled for the market type; valid values: 1 = SEF 3 = MiFID 7 = OTC 8 = IB
7705	MIC		C	String	Market Identifier Code required for MiFID II-relevant securities for Fenics Direct when RegulationVenueType (8505) = 3 (MiFID)
454	NoSecurityAltId		N	NumIn Group	Number of security ID groups that follow; required for MiFID II-relevant securities
	455	SecurityAltId	N	String	ISIN of the security; required for MiFID II-relevant securities
	456	SecurityAltIdSource	N	String	Required for MiFID II-relevant securities ; valid value: 4 = ISIN Identifier
461	CFIcode		N	String	Indicates the type of top-level security; required for MiFID II-relevant securities for Fenics Direct
8519	OriginatingLEI		C	String	LEI of beneficial owner of swap; required for SEF venue
5159	Clearer		C	String	Specifies the clearer LEI
8019	SwapDataRepo		C	String	LEI of SDR
6661	PBReuterCode		C	String	LEI of prime broker or FCM; required for prime-brokered or cleared orders
7936	EndClientUserName		C	String	Account name of the end-client; in the agency model, the broker agent sends trader ID of the end-client; on messages that target banks with Last-Look, FIX API sends LEI or trading name of the end-client when Show End Client" = "Yes"; otherwise, LEI or trading name of the broker agent is sent
9126	Structure		Y*	Int	Primary trade structure; refer to appendix for valid values
9102	FXMarketType		C	Int	Used only when executing on a standing order; omission of field will indicate the client's willingness to trade either; valid values: 1 = Onshore 2 = Offshore
110	MinQty		N	Qty	Minimum quantity of the order; indicates that a partial execution is desired; not required in RFQ responses; the minimum value will only apply to the first leg size of the multi-legged instrument
453	NoPartyID		N	NumIn Group	Number of party distinctions as required for MiFID II-relevant securities
	448	PartyID	N	String	Value for party role; required for MiFID II-relevant securities
	447	PartyIDSource	N	Char	Required for MiFID II-relevant securities; Valid value: D = Proprietary ID
	452	PartyRole	N	Int	Required for MiFID II-relevant securities; valid values: 11 = Order Origination Trader, execution within firm 16 = Executing System, algorithmic system identifier 122 = Investment Decision Maker
54	Side		Y	Char	Requestor side of the order; indicates the direction of the first leg; not required in RFQ responses; valid values: 1 = Buy 2 = Sell
2593	NoOrderAttributes		N	NumIn Group	Number of regulatory order attributes; required for MiFID II-relevant securities.
	2594	OrderAttributeType	N	Int	Optionally required for MiFID II-relevant securities if NoOrderAttributes (2593) > 0; valid values: 2 = Liquidity provision activity order 4 = Algorithmic order
	2595	OrderattributeValue	N	String	Attribute value; required for MiFID II-relevant securities securities ; valid values: Y = Yes N = No
8506	TradeGroupID		Y*	String	Swap Link ID Used for RTN (EMIR) RTN=CONCAT(8504,8506)

6139	TotalNumOfTrades		Y	Int	Number of trades associated with the TradeGroupID (8506)
55	Symbol		Y	String	Six-character string that represents the currency pair e.g. "EURUSD"; the first currency is always the base and the second currency is always the term; "NA" is an acceptable value when multiple currencies are in the request
555	NoLegs		Y*	NumIn Group	Number of legs
	600	LegSymbol	Y*	String	Six-character string that represents the currency pair
	7940	LegID	C	String	Unique strategy or leg identifier with maximum length of 50; required for RFQ
	5678	Volatility	C	Float	Price volatility for FXO
	8588	FwdPrice	C	Price	All-in rate for NDFs Points for FX products; required for deliverable forwards and NDFs if not in response to RFQ
	8515	VolMid	N	Price	Dealer-provided mid-price
	8518	GFIMid	N	Price	GFI calculated Mid-Price (only applicable for Fenics Direct)
	8500	LegNamespace	C	String	Issuer namespace
	8501	LegUsiValue	C	String	Unique Swap Identifier (USI); provided on SEF trades for Dodd Frank reporting
	8524	LegUtiValue	C	String	Unique Trade Identifier (UTI); provided on all trades for EMIR reporting
	8516	LegReportingParty	C	String	Identifies if client is the reporting counterparty on the trade
	5359	MQSize	C	Float	Size of entry in multiples of 1 million e.g. 1 = 1,000,000
	5844	LegPremPrice	C	Price	Leg price; will be sent as a formatted value: - For Pct: LegPremPrice (5844) = Raw % Price * 100 - For Pts: LegPremPrice (5844) = Raw Pts Price * 10^(spot decimal places)
	6215	Tenor	C	String	FX and FXO maturity; not required in response to FXO RFQ; refer to appendix for valid values
	9127	LegDeliveryType	C	Char	Required when Structure (9126)=65 (FX) to indicate deliverable or non-deliverable; valid values: 1 = Cash (indicates non-deliverable) 2 = Physical (indicates deliverable)
	6714	LegStrategyInd	C	Int	Not required in response to direct RFQ; valid values: 1 = Call 2 = Put 3 = Up 4 = Down
	556	LegCurrency	C	Currency	Three-character string representing the currency of the notional size; not required in response to RFQ
	9034	LegStrategyCcy	C	Currency	Three-character string representing call or put currency
	604	NoLegSecurityAltId	N	NumIn Group	Number of security ID groups that follow; required for MiFID II-relevant securities
	605	LegSecurityAltId	N	String	ISIN of the leg-level security; required for MiFID II-relevant securities
	606	LegSecurityAltIdSource	N	String	Required for MiFID II-relevant securities ; valid value: 4 = ISIN Identifier
	608	LegCFIcode	N	String	Indicates the type of leg-level security; required for MiFID II-relevant securities for Fenics Direct
	2893	LegUPICode	N	String	Uniquely identifies the product of a security using ISO 4914 standard, Unique Product Identifier (UPI). Service provider – ANNA; required for MiFID II-relevant securities
	10670	LegRegPublicationReason* (only for Fenics Direct RFQ)	N	Int	Valid values: 6 = Deferral due to "Large in Scale" 7 = Deferral due to "Illiquid Instrument" 8 = Deferral due to "Size Specific"
7464	NoHedgeLegs		N	NumIn Group	Number of hedges
	9074	HedgeCurrency	Y*	Currency	Currency of the hedge amount; required when NoHedgeLegs (7464) >= 1

	9016	HedgeTradeType	N	Int	Indicates the type of hedge trade executed for offsetting option risk; valid values: 1 = Spot 2 = Forward
	6666	HedgeDirectionIndicator	N	Int	Hedge direction; valid values: 1 = Bought 2 = Sold
	7931	HedgeMIC	C	String	Market Identifier Code required for MiFID II-relevant hedge leg when hedge leg is MiFID regulated
	6036	HedgeAmount	N	Amt	Hedge amount
	9657	HedgeRate	Y*	Float	Hedge exchange rate; required when NoHedgeLegs (7464) >= 1
	9112	HedgeDelivery	N	LocalMkt Date	Delivery date for the delta hedge
	6426	HedgeSymbol	N	String	Currency pair for hedge leg
	8508	HedgeNamespace	C	String	Issuer namespace; required when NoHedgeLegs (7464) >= 1 on SEF trade
	8509	HedgeUsiValue	C	String	Unique identifier for each hedge leg; required when NoHedgeLegs (7464) >= 1 on SEF trades
	8526	HedgeUtiValue	C	String	Unique identifier for each hedge leg; required when NoHedgeLegs (7464) >= 1 on EMIR trades
	10604	NoHedgeSecurityAltId	N	NumIn Group	Number of security ID groups that follow; required for MiFID II-relevant securities
	10605	HedgeSecurityAltId	N	String	ISIN of the hedge-level security; required for MiFID II-relevant securities
	10606	HedgeSecurityAltIdSource	N	String	Required for MiFID II-relevant securities ; valid value: 4 = ISIN Identifier
	10608	HedgeCFIcode	N	String	Indicates the type of hedge-level security; required for MiFID II-relevant securities for Fenics Direct
	10609	HedgeUPIcode	N	String	Uniquely identifies the product of a security using ISO 4914 standard, Unique Product Identifier (UPI). Service provider – ANNA; required for SEF, MTF and OTF relevant securities
	10671	HedgeRegPublicationReason * (only for Fenics Direct RFQ)	N	Int	Valid values: 6 = Deferral due to “Large in Scale” 7 = Deferral due to “Illiquid Instrument” 8 = Deferral due to “Size Specific”
2668	NoTrdRegPublications		N	NumIn Group	Trade publication reasons that are required by regulatory agencies * Only supported for Fenics Direct business
	2669	TrdRegPublicationType	N	Int	Valid values: 0 = Pre-trade transparency waiver
	2670	TrdRegPublicationReason	N	Int	Valid values: 4 = No public price quoted as instrument is illiquid (not supported in the current implementation) 9 = No public price and/or size quoted as transaction is "large in scale" 101 = Transaction size above “Size Specific to Instrument” (SSTI) threshold (not supported in the current implementation)
60	TransactTime		Y	UTC Timestamp	Timestamp in microseconds of the message in YYYYMMDD-HH:MM:SS.ssssss format. Timestamps can be in non microsecond format (based on client preference for MiFID tags)
40	OrdType		Y	Char	Valid value: 1 = Market
8503	PremiumMid		N	Price	Premium mid-point
15	Currency		C	Currency	Three-character string that represents call or put currency; indicates the currency for which the risk reversal is bid over; required for risk reversal structures; not required in response to RFQ
117	QuoteID		C	String	Unique identifier for Quote (S); required for direct RFQ
9246	PackageIdentifier		N	Char	Describes if the trade is a package deal or not
9247	PackagePrice		N	Price	Price of the overall Package

9248	PackagePriceType	N	String	Valid values: Monetary (an amount of money, ex: Options quoted in price) Decimal (a rate, ex: Options quoted in vol)
9249	PackagePriceCurrency	N	Currency	Three-character string representing the package price currency
9250	PackageType	N	String	Valid values: Spread (if the price is a spread) Price (if the price is not a spread)
59	TimeInForce	Y*	Char	Specifies how long the order will remain in the market; valid value: 3 = Immediate or Cancel
6436	Premium	C	Price	Structure premium
5830	PremiumCcy	C	Currency	Three-character string that represents the premium currency
9125	Cutoff	C	Int	Time of expiration of FXO, and fixing index for NDF; required for both FX and FXO except in response to a direct RFQ; refer to appendix for valid values
9072	CallOrPutOver	C	Int	Call or put; required for risk reversal structures; indicates if a risk reversal is quoted as calls over puts or puts over calls; not required in response to RFQ; valid values: 1 = Calls Over 2 = Puts Over
811	PriceDelta	C	Float	Delta; required for risk reversal and butterfly structures; not required in response to RFQ; refer to appendix for valid values
5475	PremDeliveryType	N	Char	Premium delivery type; validation will occur for spot or forward; If not blank, then it matches the value defined in the RFQ or defined for Quote (S); valid values: S = Spot F = Forward B = Both
	Standard Trailer			

Y = Standard FIX Required; Y* = FIX API Required; C = Conditional; N = Not Required

5.16 Order Status Request (H)

The Order Status Request (H) message is used by the client to generate an order Execution Report (8) that reflects the current status of the order. If there were multiple trade executions on the order, then the (8) message will reflect the summary of those executions. The Execution report will have ExecType (150) = I (Order Status). The Order Status Request (H) message must reference a valid OrderID (37) or ClOrdID (11) for a specific order status return.

When requesting order status for order entered via New Order Single message, the client must specify the ClOrdID (11). Where ClOrdID is not available, the client will specify the ClOrdID= 'NA' and indicate the OrderID (37). If there are no execution report messages that met the specified criteria, then the returned execution report will be rejected stating "Order <id #> Not Found".

5.16.1 Order Status Request Identifier

A unique Order Status Request identifier, OrdStatusReqID (790), will be generated by the client when (H) messages are sent. The OrdStatusReqID (790) will conform to the following rules:

1. All new messages will have a unique value in this field for that day, regardless of the trading session the report was submitted.
2. The value will only contain upper-case letters and/or numbers up to 24 characters in length.

5.16.2 Order Status Request (H) Message Configuration

Tag	Field Name	Req	Type	Description
FX & FXO FIX API Rules of Engagement				

	Standard Header			
35	MsgType			H (Order Status Request)
37	OrderID	N	String	Unique identifier for the executed order assigned by FIX API
11	ClOrdID	Y	String	Unique identifier of replacement order assigned by client
55	Symbol	Y	String	Required by FIX but not used by the API
790	OrdStatusReqID	Y*	String	Used to uniquely identify a specific Order Status Request message.
54	Side	Y*	Char	Side of order; FIX required by is not verified by the FIX API; valid values: 1 = Bought 2 = Sold
	Standard Trailer			

Y=FIX required; Y*=API required; C=Conditional; N=Not required

5.17 Execution Report (8)

The Execution Report (8) messages are sent to a client in response to transaction event messages such as quotes, new order single, new order multileg, and executions. The (8) message will communicate the status of the order in the OrdStatus (39) field. There may be multiple execution report messages in response to Quote (S), New Order – Single (D), or New Order – Multileg (AB) messages. Each (8) message will have unique identifiers, ExecID (17) and ClOrdID (t11).

As order an acknowledgement message, the (8) message will provide the order status of the New Order – Single (D) message received. As a trade execution message, the (8) message will provide the details of the execution, similar to the details provided in Quote (S) or New Order – Multileg (AB) message. Specific trade or calculated details, such as counterparty codes, premium, etc., will not be provided in this message but in the Trade Capture Report (AE) which should be used as the source for complete transaction affirmation and STP post-execution.

5.17.1 Order Status

An Execution Report message will be sent to the sender of a standing order or quote every time an execution occurs on a Quote (S). If the size on the Quote (S) message is filled entirely, the (8) message will be sent to the client with OrdStatus (39) = 2 (Filled). If the size on the Quote (S) message is partially filled, (8) message will be sent to the client with OrdStatus (39) = 1 (Partially Filled). The remaining size of Quote (S) will be the LeavesQty (151) value on the (8) message.

If the order is filled at different prices, multiple (8) messages will be sent. Each message will have OrdStatus (39) = 1 (Partially Filled). The last (8) message with LeavesQty (151) = 0 (zero) will have OrdStatus (39) = 2 (Filled). The total MQSize (5359) of all (8) messages for the same ClOrdID (11), should equal MQSize (5359) on the New Order – Single (D) message.

5.17.1.1 In Response to New Order – Single (D) Message

An Execution Report (8) message will be sent as an order acknowledgement or trade execution notification to the sender of a New Order – Single (D) message. In the message, ClOrdID (11) and OrderID (37) will be the only reference to the order that was executed.

5.17.1.2 In Response to Quote (S) Message

An Execution Report (8) message will be sent to the sender of a Quote (S) message when there is an execution on the quote. The NoMQEntries (6120) repeating group will contain the price details at the strategy leg level. LegID (7940) will refer to specific strategy legs found in the Quote (S) message.

5.17.1.3 In Response to New Order – Multileg (AB) Message

An Execution Report (8) message will be sent to the sender of a New Order – Multileg (AB) message when there are executions on the quote. The NoMQEntries (6120) repeating group will be populated with the trade details. LegID (7940) will refer to specific strategy legs found in the New Order – Multileg (AB) message.

5.17.2 Order Status – Reject

An Execution Report (8) message will be sent by the FIX API in response to a New Order – Single (D) or New Order – Multileg (AB) message to reject the order. Although expected rarely, an order rejection can occur for multiple scenarios, including an Execution Report (8) message not being found in response to an Order Status Request (H) message, the price no longer being available or the message having a required field outside a specified parameter. In such cases, the message will be sent to the client with OrdStatus (39) = 8 (Rejected). The message will provide the details of the rejected order, and the reason for the rejection in OrdRejReason (103).

5.17.3 Last-Look on Executions for Liquidity Banks

The FIX API will allow liquidity bank APIs to confirm or reject executions on quotes they originally submitted. A bank that is configured for last-look on executions will receive a new order Execution Report (8) message in response to the quote that was originally submitted by that bank. The bank will have a pre-defined time to respond to the new order Execution Report (8) message to initiate the execution. It will respond to the new order Execution Report (8) message by sending a condensed message (see below). The condensed Execution Report (8) message will be used to accept or reject the new order Execution Report (8) message. The FIX API will only validate the following fields for the condensed Execution Report (8) message: ClOrdID (11) and OrderStatus (39). The FIX API will form a complete Execution Report (8) message upon receiving the condensed Execution Report (8) message. The complete message will be sent to both the liquidity bank API and the sender of the new order Execution Report (8) message.

5.17.4 Condensed Execution Report (8) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			8 (Execution Report)
11	ClOrdID	Y	String	Unique identifier of the order assigned by FIX API
117	QuoteID	Y	String	Unique identifier of Quote (S) assigned by the client
17	ExecID	Y	String	Unique identifier of the execution message
54	Side	Y*	Char	Valid values: 1 = Bought 2 = Sold
	Standard Trailer			

Y=FIX required; Y*=API required; C=Conditional; N=Not required

The following fields are required for the condensed Execution Report message: ExecType (150), OrderStatus (39), Symbol (55), Structure (9126), CutOff (9125), OrderQty (38), LeavesQty (151), CumQty (14), TransactionTime (60), and AvgPx (6).

When a Quote Request (R) message is directed at the CLOB market where requestor is not disclosed (e.g. on a clearable trade), FIX API will not forward the requestor's FIXCompID in the OnBehalfOfCompID (115) on the New Order – Multileg (AB) message.

An Execution Report (8) message may be sent in response to a New Order – Multileg (AB) message to confirm cancellation of an “unfilled” size. The remaining size of a new order, the difference resulting from a partially executed order, will be canceled automatically by the API. An Execution Report (8) message will be sent to the client, with OrdStatus (39) = 4 (Canceled). The canceled size will be the Size (53) with LeavesQty (151) = 0. The total of Size (53) of all Execution Report (8) messages for the same ClOrdID (t11), should equal the Size (53) on the New Order –Multileg (AB) message.

5.17.5 Special Field Descriptions

5.17.5.1 Execution Report Identifier

A unique Execution Report identifier, ExecID (17), will be generated by the API when Execution Report messages are sent. The ExecID will conform to the following rules:

1. All new Execution Report message will have a unique value in the ExecID field for that day, regardless of the trading session the report was sent.
2. The ExecID value will contain alphanumeric characters separated by a period.

5.17.5.2 Order Identifier

A unique Order identifier, OrderID (37), will be generated by the API when Execution Report messages are sent in response to New Order Single messages. The OrderID will conform to the following rules:

1. All accepted New Order Single messages will have a unique value in the OrderID field for that day, regardless of the trading session the order was submitted.
2. The OrderID value will only contain letters (upper case) or numbers.

5.17.6 Execution Report (8) Message Configuration

Tag	Field Name		Req	Type	Description
	Standard Header				
35	MsgType				8 (Execution Report)
17	ExecID		Y	String	Unique identifier of execution message
790	OrdStatusReqID		C	String	Used to uniquely identify an execution report message sent in response to a specific Order Status Request (H) message
37	OrderID		Y	Int	Unique identifier for the order established by the API
11	ClOrdID		Y	Int	Unique identifier for the order established by the client
41	OrigClOrdID		C	String	ClOrdID (11) of the previously accepted order; required in response to ExecType (150) = 5 (Replace) or 4 (Canceled)
117	QuoteID		C	String	Unique identifier for Quote (S), established by the client; required for Fenics Direct RFQ
5001	APIQuoteID		C	String	Unique identifier for the Quote Status Report (AI), established by the FIX API
8505	RegulationVenueType		C	Int	Indicates the market type of the quote; valid values: 1 = SEF 3 = MiFID 7 = OTC 8 = IB <i>Note that trade notification will use 1, 3, 5 and 7 only</i>
454	NoSecurityAltId		N	NumIn Group	Number of security ID groups that follow; required for MiFID II-relevant securities
	455	SecurityAltId	N	String	ISIN of the security; required for MiFID II-relevant securities
	456	SecurityAltIdSource	N	String	Required for MiFID II-relevant securities ; valid value: 4 = ISIN Identifier
461	CFIcode		N	String	Indicates the type of top-level security; required for MiFID II-relevant securities
8520	IntroducingBroker		C	String	LEI of broker offering access to the SEF, OTF or OTC venues; will be populated if IB was involved in the trade

8527	SubmittingMember		C	String	LEI of SEF or MIFID member that is crossing trade into SEF or MIFID; will be populated if trade was arranged into SEF or MIFID venue
8519	OriginatingLEI		C	String	LEI of beneficial owner of swap; this is the firm's own LEI sent in Quote Request (R) or Quote (S)
5159	Clearer		C	String	Specifies the clearer LEI
8019	SwapDataRepo		C	String	Indicates the SDR's LEI
6661	PBReuterCode		C	String	Prime Broker/FCM's LEI where available; else the trading name; if counterparty has a PB/FCM, then it is populated, else this is empty
7936	UserID		C	String	This identifies the LEI of the trading counterparty; his applies to all trades executed via pods and RFQ; required in broker agent RFQ model; if "Show End Client"= Yes, target bank sees end client's LEI (or trading name if LEI does not exist); if "Show End Client"=No, target bank sees broker agent's LEI/trading name; the broker agent sees target bank's LEI or trading name
150	ExecType		Y	Char	Describes the specific execution report (i.e. pending cancel) while OrdStatus (39) will always identify the current status (i.e. partially filled); valid values: 0 = New 1 = Partial Fill 2 = Fill 4 = Canceled 5 = Replace 8 = Rejected C = Expired
39	OrdStatus		Y	Char	Identifies current status of order; valid values: 0 = New 1 = Partially Filled 2 = Filled 4 = Canceled 8 = Rejected
103	OrdCanRejCode		C	Int	Code to identify reason for order cancellation or order rejection; will only be required if OrdStatus (39) = 4 (Canceled) or 8 (Rejected); see appendix for valid values
55	Symbol		Y	String	Six-character string that represents the currency pair e.g. "EURUSD"; the first ccy is always the base and the second is always the term
9126	Structure		Y*	Int	FX trading systems primary structure information; see appendix for valid values
9125	Cutoff		Y*	Int	Time of expiration of FX option and fixing index for NDF; see appendix for valid values
15	Currency		C	Currency	Call/put currency; three-character string required for risk reversal structures; indicates the currency for which the risk reversal is bid over
9072	CallOrPutOver		C	Int	Required for risk reversal structures; indicates if a risk reversal is quoted as "Calls Over Puts" or "Puts Over Calls"; valid values: 1 = Calls Over 2 = Puts Over
811	PriceDelta		C	Float	Delta; required for risk reversal and butterfly structures; indicates if a risk reversal or butterfly is quoted as 25 delta or 10 delta; see appendix for valid values
453	NoPartyID		N	NumIn Group	Number of party distinctions as required for MiFID II-relevant securities
	448	PartyID	N	String	Value for party role; required for MiFID II-relevant securities
	447	PartyIDSource	N	Char	Required for MiFID II-relevant securities; Valid value: D = Proprietary ID
	452	PartyRole	N	Int	Required for MiFID II-relevant securities; valid values: 11 = Order Origination Trader, execution within firm 16 = Executing System, algorithmic system identifier 122 = Investment Decision Maker

54	Side		Y*	Char	Valid values: 1 = Bought 2 = Sold
2593	NoOrderAttributes		N	NumIn Group	Number of regulatory order attributes; required for MiFID II-relevant securities.
	2594	OrderAttributeType	N	Int	Optionally required for MiFID II-relevant securities if NoOrderAttributes (2593) > 0; valid values: 2 = Liquidity provision activity order 4 = Algorithmic order
	2595	OrderattributeValue	N	String	Attribute value; required for MiFID II-relevant securities securities ; valid values: Y = Yes N = No
38	OrderQty		Y	Qty	Order quantity
151	LeavesQty		Y	Qty	Size remaining on a partial execution
14	CumQty		Y	Qty	Total quantity executed
6	AvgPx		Y	Price	Execution average price; required by FIX but not used by the FIX API
58	Text		C	String	The reason for order cancellation or rejection; see appendix for descriptions
60	TransactTime		Y*	UTC Timestamp	Timestamp in microseconds of the message in YYYYMMDD-HH:MM:SS.ssssss format. Timestamps can be in non microsecond format (based on client preference for MiFID tags)
6436	Premium		C	Price	Structure premium
8503	PremiumMid		N	Price	Dealer provided premium mid-point
5830	PremiumCcy		C	Currency	Currency of the premium amount
8589	FwdSettleCcy		C	Currency	Indicates the settlement ccy for NDF trade; required when LegDeliveryType (9127)=1 (Cash)
6120	NoMQEntries		N	Int	Count of market quote entries to follow
	7940	LegID	Y*	String	The strategy or leg identifier; “NA” will be a valid value for executions on pods; the max length is 50 bytes
	5678	Volatility	C	Float	Price volatility
	8588	FwdPrice	C	Price	All-in rate for NDFs Points For FX product, the value is populated if execution was done on price
	8515	VolMid	N	Price	Dealer provided mid-price
	8518	GFIMid	N	Price	GFI calculated Mid-Price (only applicable for Fenics Direct)
	8500	LegNamespace	C	String	Issuer namespace; required for direct RFQ SEF trades
	8501	LegUsiValue	C	String	Unique Swap Identifier; trade identifier used in DFA reporting; required for direct RFQ SEF trades
	8524	LegUtiValue	C	String	Unique Trade Identifier; trade identifier used in EMIR reporting
	8516	LegReportingParty	Y	Char	Will be provided for all trades
	5359	MQSize	C	Float	Size of entry
	5235	LegSpotRate	C	Float	FX spot rate; this value is populated if it was entered in the Quote (S)
	9099	ForwardRate	N	Float	Forward rate for option
	5191	LegForwardPoints	C	Float	FX display swap rate; assume industry standards for swap points per currency pair; for example, a realistic forward points value for a EURUSD call can be 2.7; for FX products, this value is populated if it was entered in the Quote (S)
	9115	DepoRate	C	Float	Depo rate (%); for example, a realistic depo rate value for a EURUSD call can be 5.04
	9073	DepoRateCcy	C	Currency	Depo rate currency
	9904	PriceIndicator	C	Int	Price indicator; valid values: 1 = Percent 2 = Points
	5844	LegPremPrice	C	Price	Leg price; this field will be sent as a formatted value: - For Pct LegPrice = Raw % Price * 100 - For Pts LegPrice = Raw Pts Price * 10^(spot decimal places)
	6035	LegDelta	C	Float	Delta (%)
	6354	MQStrikePrice	C	Price	Strike price

	5167	LoTrigger	C	Price	Identifies trigger price in absolute terms for the single or low trigger; required on single or double trigger options
	6192	HiTrigger	C	Price	Identifies trigger price in absolute terms for the high trigger; required on double trigger options
	7092	LegPremium	C	Price	Leg premium; required for premium quotes; used to capture leg level trade details; the value will not be validated against Premium (6436)
	6215	Tenor	C	String	Option maturity; see appendix for valid values
	6714	LegStrategyInd	C	Int	Not required in standard pod execution; valid values: 1 = Call 2 = Put 3 = Up 4 = Down
	9017	LegSizeCurrency	C	String	Size currency; three-character string
	9034	LegStrategyCcy	C	Currency	Call/put currency; three-character string; not required in standard pod execution
	604	NoLegSecurityAltId	N	Num InGroup	Number of security ID groups that follow; required for MiFID II-relevant securities
	605	LegSecurityAltId	N	String	ISIN of the leg-level security; required for MiFID II-relevant securities
	606	LegSecurityAltIdSource	N	String	Required for MiFID II-relevant securities ; valid value: 4 = ISIN Identifier
	608	LegCFIcode	N	String	Indicates the type of leg-level security; required for MiFID II-relevant securities
	2893	LegUPICode	N	String	Uniquely identifies the product of a security using ISO 4914 standard, Unique Product Identifier (UPI). Service provider – ANNA; required for SEF, MTF and OTF relevant securities
7464	NoHedgeLegs		N	Num InGroup	The number of hedges
	9074	HedgeCurrency	Y*	Currency	Currency of the hedge amount; only required when the NoHedgeLegs (7464) >=1
	9016	HedgeTradeType	N	Int	Indicates the type of hedge trade executed for offsetting option risk; valid values: 1 = Spot 2 = Forward
	6666	HedgeDirection Indicator	N	Int	Hedge direction; valid values: 1 = Bought 2 = Sold
	6036	HedgeAmount	N	Amt	Hedge amount; entry value in millions i.e. an entry of 1.5 equals 1,500,000
	9657	HedgeRate	C	Float	Hedge exchange rate; required if NoHedgeLegs (7464) >=1
	9112	HedgeDelivery	N	Local MktDate	Delivery date for the delta hedge (YYYYMMDD)
	6426	HedgeSymbol	N	String	Currency pair for hedge leg
	8517	HedgeReportingParty	C	Char	Identifies if client is the reporting party for the hedge leg; required for Direct RFQ
	8508	HedgeNamespace	C	String	Issuer namespace; required for Direct RFQ SEF execution where NoHedgeLegs (7464) >=1
	8509	HedgeUSIValue	C	String	Unique Swap Identifier for hedge leg; required for Direct RFQ SEF execution where NoHedgeLegs (7464) >=1
	8526	HedgeUTIValue	C	String	Unique Trade Identifier value for hedge leg; required for allDirect RFQ where NoHedgeLegs (7464) >=1
	10604	NoLHedgeSecurityAltId	N	NumIn Group	Number of security ID groups that follow; required for MiFID II-relevant securities
	10605	HedgeSecurityAltId	N	String	ISIN of the hedge-level security; required for MiFID II-relevant securities
	10606	HedgSecurityAltIdSource	N	String	Required for MiFID II-relevant securities ; valid value: 4 = ISIN Identifier
	10608	HedgeCFIcode	N	String	Indicates the type of hedge-level security; required for MiFID II-relevant securities
	10609	HedgeUPICode	N	String	Uniquely identifies the product of a security using ISO 4914 standard, Unique Product Identifier (UPI). Service provider – ANNA; required for SEF, MTF and OTF relevant securities

2668	NoTrdRegPublications		N	NumIn Group	Trade publication reasons that are required by regulatory agencies
	2669	TrdRegPublicationType	N	Int	Valid values: 0 = Pre-trade transparency waiver
	2670	TrdRegPublicationReason	N	Int	Valid values: 4 = No public price quoted as instrument is illiquid (not supported in the current implementation) 9 = No public price and/or size quoted as transaction is "large in scale" 101 = Transaction size above "Size Specific to Instrument" (SSTI) threshold (not supported in the current implementation)
	Standard Trailer				

Y=FIX required; Y*=API required; C=Conditional; N=Not required

5.18 Trade Capture Report (AE)

Trade Capture Report messages are sent electronically by the FIX API to report clients' trade details. The message identifies trade details by leg of each executed trade, including spot, premium and hedge. In some instances, such as corrections and cancels, multiple (AE) messages will be sent in response to a single execution. In addition, (AE) messages will contain the necessary details of STP messages.

5.18.1 Trade Capture Report Message Publication

The standard method of transmitting the (AE) message will be after an execution occurs and the brokerage desk has applied the relevant market rates. There is no manual intervention on the client side. A second method will be available to the client upon request. This method requires manual intervention from the client. The message will not be sent to the client until they have verified the details on the FX Blotter. Once the client (or client and counterparty, depending on the client's preference) has confirmed the details of the trade on FX Blotter, the message will be sent. The API will receive a Trade Capture Report Ack (AR) message from the client in response. New (AE) messages will have TradeReportTransType (487) = 3 (Release). Messages that are amendments to previous messages will have TradeReportTransType (487) = 2 (Replace), and will reference the ID of the amended message in TradeReportRefID (572). Canceled messages will have TradeReportTransType (487) = 1 (Cancel). Messages are first canceled before being amended (replaced).

5.18.2 Trade Capture Report (AE) Message Confirmation

The FIX API will receive an (AR) message from the client in response to an (AE) message by either accepting or rejecting the STP details. An acknowledgement message with a TrdRptStatus (939) = 0 (Accepted) confirms that the STP details are correct. An acknowledgement message with a TrdRptStatus (939) = 1 (Rejected) indicates one or more of the values in the STP message is incorrect. Rejection details must be specified in the Text (58) field of the Trade Capture Report Ack (AR) message. Although this field is free-format in syntax, the description should be explicit enough for the clerks who will make the necessary corrections to the trade.

5.18.3 Trade Capture Report Identifier

A unique Trade Capture Report identifier, TradeReportID (571), will be generated by the FIX API when an (AE) message is sent. The TradeReportID (571) will conform to the following rules:

1. All new messages will have a unique value in this field for that day, regardless of the trading session the report was submitted.

- The value will only contain upper-case letters and/or numbers up to 24 characters in length.

5.18.4 Trade Capture Report (AE) Message Configuration

Tag	Field Name	Req	Type	Description
	Header			
35	MsgType			AE (Trade Capture Report)
571	TradeReportID	Y	String	Unique identifier for the Trade Capture Report
117	QuoteID	C	String	Unique identifier for Quote (S); established by the client; required field for electronic trades
131	QuoteReqID	C	String	Unique identifier for Quote Request (R); required field for electronic trades
9126	Structure	Y*	Int	Primary trade structure; refer to appendix for valid values
487	TradeReportTransType	Y*	Int	Identifies message transaction type; valid values: 1 = Cancel 2 = Replace 3 = Release
568	TradeRequestID	N	String	Unique identifier for Trade Capture Report Request (AD)
855	SecondaryTrdType	N	Int	Required for MiFID II-relevant securities; valid values:65 = Package trade
150	ExecType	N	Char	Type of execution being reported; valid values: 0 = New 4 = Canceled 5 = Replace D = Restated
8504	RegulationVenue	N	String	LEI of execution venue
8520	IntroducingBroker	C	String	LEI of introducing broker offering access to the SEF or MIFID venue; required if trade was arranged by the IB
8527	SubmittingMember	C	String	LEI of SEF or MIFID member that is crossing trade onto SEF or MIFID; required if trade was arranged by the IB
572	TradeReportRefID	C	String	The TradeReportID (571) that is being referenced for correction or cancellation; will be provided when the Trade Capture Report is being referenced for correction or cancellation
17	ExecID	N	String	Unique identifier of the execution message
570	PreviouslyReported	Y	Boolean	Indicates if the Trade Capture Report was previously reported; all (AE) messages sent in response to (AR) messages will have (570) = Y; valid values: Y = Yes N = No
8506	TradeGroupID	Y*	String	Swap Link ID Used for RTN (EMIR) RTN=CONCAT(8504,8506)
6139	TotalNumOfTrades	Y	Int	Number of trades associated with the TradeGroupID (8506)
55	Symbol	Y	String	Six-character string that represents the currency pair e.g. "EURUSD"; the first currency is always the base and the second currency is always the term
32	LastQty	Y	Qty	Trade quantity; expected value = 0; required by FIX but not used by the API
31	LastPx	Y	Price	Trade price; expected value = 0; required by FIX but not used by the API
75	TradeDate	Y	LocalMkt Date	Indicates date of trade in YYYYMMDD format
8505	RegulationVenueType	Y	Int	Valid values: 1 = SEF 3 = MiFID 7 = OTC
5159	Clearer	C	String	Specifies the clearer LEI
8507	ClearingException	Y	Char	Indicates if a regulatory mandated clearable contract will not be sent to clearing; valid values: Y = Yes N = No
8019	SwapDataRepo	C	String	LEI of SDR

7705	MIC		C	String	Market Identifier Code required for MiFID II-relevant securities when RegulationVenueType (8505) = 3 (MiFID)	
6642	TradeBookingType		N	Char	Describes if the trade occurred via STP or booked manually; only included in messages in response to Trade Capture Report Request (AR); valid values: S = STP M = Booked manually	
9246	PackageIdentifier		N	Char	Describes if the trade is a package deal or not	
9247	PackagePrice		N	Price	Price of the overall Package	
9248	PackagePriceType		N	String	Valid values: Monetary (an amount of money, ex: Options quoted in price) Decimal (a rate, ex: Options quoted in vol)	
9249	PackagePriceCurrency		N	Currency	Three-character string representing the package price currency	
9250	PackageType		N	String	Valid values: Spread (if the price is a spread) Price (if the price is not a spread)	
2668	NoTrdRegPublications		N	NumIn Group	Trade publication reasons that are required by regulatory agencies	
	2669	TrdRegPublicationType	N	Int	Valid values: 0 = Pre-trade transparency waiver	
	2670	TrdRegPublicationReason	N	Int	Valid values: 4 = No public price quoted as instrument is illiquid 9 = No public price and/or size quoted as transaction is "large in scale" 101 = Transaction size above “Size Specific to Instrument” (SSTI) threshold	
555	NoLegs		Y*	NumIn Group	Number of legs	
	600	LegSymbol	Y	String	Six-character string that represents the currency pair	
	8500	LegNamespace	Y	String	Broker namespace	
	8501	LegUsiValue	C	String	Unique Swap Identifier (USI); provided on SEF trades for Dodd Frank reporting	
	8524	LegUtiValue	Y	String	Unique Trade Identifier (UTI); provided on all trades for EMIR reporting	
	8516	LegReportingParty	Y	Char	Identifies if client is the reporting counterparty on the trade leg; provided for all trades; valid values: Y = Yes N = No	
	604	NoLegSecurityAltId	N	NumIn Group	Number of security ID groups that follow; required for MiFID II-relevant securities	
		605	LegSecurityAltId	N	String	ISIN of the leg-level security; required for MiFID II-relevant securities
		606	LegSecurityAltIdSource	N	String	Required for MiFID II-relevant securities ; valid value: 4 = ISIN Identifier
	608	LegCFIcode	N	String	Indicates the type of leg-level security; required for MiFID II-relevant securities	
	2893	LegUPIcode	N	String	Uniquely identifies the product of a security using ISO 4914 standard, Unique Product Identifier (UPI). Service provider – ANNA; required for MiFID II-relevant securities	
	8521	BlockSize	C	Char	Indicates if the notional constitutes a block size; conditionally required for SEF trades; valid values: Y = Yes N = No	
	654	LegRefID	Y*	String	Unique identifier for a specific leg order	
	624	LegSide	Y*	Char	Side of the leg; valid values: 1 = Bought 2 = Sold	
	1057	AggressorIndicator	N	Boolean	Valid values: Y = Order initiator is aggressor N = Order initiator is passive	
	687	LegQty	Y*	Qty	Size of the leg in multiples of 1 million e.g. 1 = 1,000,000	
	556	LegCurrency	Y	Currency	Three-character string representing the currency of the notional size	
	6714	LegStrategyInd	Y*	Int	Call or put; valid values: 1 = Call	

					2 = Put 3 = Up 4 = Down
9034	LegStrategyCcy	Y*	Currency	String	Three-character string representing call or put currency
6215	Tenor	C	String	String	FX and FXO maturity; refer to appendix for valid values
611	LegMaturityDate	Y*	LocalMkt Date	Date	Date of expiration in YYYYMMDD format
6203	FixingDate	C	LocalMkt Date	Date	Indicates fixing date in YYYYMMDD format; required for NDF For NDOs, expiry date is sent
743	DeliveryDate	Y*	LocalMkt Date	Date	Date of delivery in YYYYMMDD format; settlement date for spot and forward trades
8013	PublishDate	C	LocalMkt Date	Date	Indicates publish date in YYYYMMDD format; required for NDF
9125	Cutoff	Y	Int	Int	Time of expiration of FXO, and fixing index for NDF; refer to appendix for valid values
9019	FXOptionStyle	N	Int	Int	Option style; valid values: 1 = American 2 = European Following are supported for Fenics Direct only 21 = Knockin 23 = Reverse Knockin 29 = Double Knockin 30 = Digital 31 = Digital Double Knockout 32 = Digital Knockout 33 = Digital Reverse Knockout 34 = Double One Touch 35 = One Touch 40 = Knockout 42 = Reverse Knockout 48 = Double Knockout 49 = Digital Double Knockin 50 = Digital Knockin 51 = Digital Reverse Knockin 52 = Double No Touch 53 = No Touch 54 = Instant One Touch 59 = European Knockin 60 = European Knockout 61 = Knockin Knockout 62 = Reverse Knockin Knockout
612	LegStrikePrice	Y*	Price	Price	Strike price
6035	LegDelta	Y*	Float	Float	Delta; refer to appendix for valid values
5235	LegSpotRate	C	Float	Float	FX spot rate; required for FXO
5191	LegForwardPoints	C	Float	Float	FX display swap rate; required for FXO; assume industry standards for swap points per currency pair e.g. a realistic forward point value for a EURUSD call is 2.7
9115	DepoRate	Y*	Float	Float	Depo rate in percent; assume industry standards for depo rate per currency pair e.g. a realistic depo rate value for a EURUSD call is 5.04
9073	DepoRateCcy	Y*	Currency	Currency	Three-character string that represent the depo rate currency
5678	Volatility	Y*	Float	Float	Volatility rate in percent
8588	FwdPrice	C	Price	Price	All-in rate for NDFs Points for FX products; required for deliverable forwards and NDFs
8515	VolMid	N	Price	Price	Dealer provided mid-price
8518	GFIMid	N	Price	Price	GFI calculated Mid-Price (only applicable for Fenics Direct)
8589	FwdSettleCcy	C	Currency	Currency	Three-character string that represents the settlement currency; required when LegDeliveryType (9127) = 1 (Cash)
9904	PriceIndicator	Y*	Int	Int	Price indicator; valid values: 1 = Pct 2 = Pts
566	LegPrice	Y*	Price	Price	Price
5830	PremiumCcy	C	Currency	Currency	Three-character string that represents the premium currency
5020	PremiumDeliveryDate	C	LocalMkt Date	Date	Premium delivery date in YYYYMMDD format

	6034	PremiumAmount	C	Amt	Premium amount
	9127	LegDeliveryType	Y	Char	Required when Structure (9126)=65 (FX) to indicate deliverable or non-deliverable; valid values: 1 = Cash (indicates non-deliverable) 2 = Physical (indicates deliverable)
	5167	LoTrigger	C	Price	Identifies trigger price in absolute terms for the single or lower trigger; required on single and double trigger options
	6192	HiTrigger	C	Price	Identifies trigger price in absolute terms for the high trigger; required on double trigger options
	8511	FxRateSource	N	String	Identifies the primary fixing rate source for NDF
	8512	RateSourcePage	N	String	Identifies the page of published rate
	10670	LegRegPublicationReason	N	Int	Valid values: 6 = Deferral due to "Large in Scale" 7 = Deferral due to "Illiquid Instrument" 8 = Deferral due to "Size Specific"
7464	NoHedgeLegs		N	NumIn Group	The number of hedges
	9074	HedgeCurrency	Y*	Currency	Currency of the hedge amount; required when NoHedgeLegs (7464) >= 1
	8508	HedgeNamespace	C	String	Issuer namespace; required when NoHedgeLegs (7464) >= 1 on SEF trade
	8509	HedgeUsiValue	C	String	Unique identifier for each hedge leg; required when NoHedgeLegs (7464) >= 1 on SEF trades
	8526	HedgeUtiValue	Y	String	Unique identifier for each hedge leg; required when NoHedgeLegs (7464) >= 1 on EMIR trades
	8517	HedgeReportingParty	Y	Char	Identifies if customer is the reporting counterparty on the trade leg; valid values: Y = Yes N = No Required when NoHedgeLegs (7464) >= 1 on SEF trade
	10604	NoHedgeSecurityAltId	N	NumIn Group	Number of security ID groups that follow; required for MiFID II-relevant securities
	10605	HedgeSecurityAltId	N	String	ISIN of the hedge-level security; required for MiFID II-relevant securities
	10606	HedgeSecurityAltIdSource	N	String	Required for MiFID II-relevant securities ; valid value: 4 = ISIN Identifier
	10608	HedgeCFICode	N	String	Indicates the type of hedge-level security; required for MiFID II-relevant securities
	10609	HedgeUPICode	N	String	Uniquely identifies the product of a security using ISO 4914 standard, Unique Product Identifier (UPI). Service provider – ANNA; required for SEF, MTF and OTF relevant securities
	6666	HedgeDirectionIndicator	N	Int	Hedge direction; valid values: 1 = Bought 2 = Sold
	7931	HedgeMIC	C	String	Market Identifier Code required for MiFID II-relevant hedge leg when hedge leg is MiFID regulated
	6036	HedgeAmount	N	Amt	Hedge amount in multiples of 1 million e.g. 1.5 = 1,500,000
	9657	HedgeRate	C	Float	Hedge exchange rate; required when NoHedgeLegs (7464) >= 1
	9112	HedgeDelivery	N	LocalMkt Date	Delivery date for the delta hedge in YYYYMMDD format
	6426	HedgeSymbol	N	String	Six-character string that represents the currency pair for hedge leg
	5790	HedgeFixingDate	N	LocalMkt Date	Hedge fixing date in YYYYMMDD format; available for NDF currency pairs
	8513	HedgeRateSource	N	String	Identifies the primary fixing rate source for NDF
	8514	HedgeRateSourcePage	N	String	Identifies the page of the published rate
	10671	HedgeRegPublicationReason	N	Int	Valid values: 6 = Deferral due to "Large in Scale" 7 = Deferral due to "Illiquid Instrument" 8 = Deferral due to "Size Specific"
60	TransactTime		Y	UTC Timestamp	Timestamp in microseconds of execution or order creation in YYYYMMDD-HH:MM:SS.sssss format. Timestamps can be in non microsecond format (based on client preference for MiFID tags)
768	NoTrdRegTimestamps		N	NumIn	Regulatory timestamp block required for MiFID II-relevant securities

				Group	
	769	TrdRegTimestamp	N	UTC Timestamp	Timestamp in microseconds of execution in YYYYMMDD-HH:MM:SS.ssssss format
	770	TrdRegTimestampType	N	String	Regulatory timestamp type; valid value: 1 = Execution time
454	NoSecurityAltId		N	NumIn Group	Number of security ID groups that follow; required for MiFID II-relevant securities
	455	SecurityAltId	N	String	ISIN of the security; required for MiFID II-relevant securities
	456	SecurityAltIdSource	N	String	Required for MiFID II-relevant securities ; valid value: 4 = ISIN Identifier
461	CFICode		N	String	Indicates the type of top-level security; required for MiFID II-relevant securities
9536	Trader		N	String	Identifies the client on the trade
6030	ConfirmedByUser		N	String	The name of the user who confirmed the trade
552	NoSides		Y	NumIn Group	Number of sides
	54	Side	Y	Char	Valid values: B = As defined (for multi-leg instruments) Refer to LegSide (624) in the NoLegs (555) repeating group; for trades in response to an RFQ, reference the leg position as defined in an originating RFQ
	37	OrderID	Y	String	Unique identifier for each chain of orders
	11	ClOrdID	C	String	Unique identifier for the executed order
	1	Account	Y*	String	Counterparty account up to 79 characters long; valid format: (Counterparty Trading Name or LEI) + " " + (Broker Code) + " " + "WORLD" + " " + (Counterparty Code) Note that if client specifies a counterparty code mapping, that code will be appended after "WORLD"
	8510	AccountIDSource	Y	Int	Identifies source of the Account (1); valid values: 1 = LEI 3 = Trading Name
	6810	CustomerAccount	Y*	String	Customer legal trading name up to 79 characters long; valid format: (Customer Legal Trading Name or LEI) + " " + (Broker Code) + " " + "WORLD" + " " + (Counterparty Code) Note that if client specifies a counterparty code mapping, that value will be appended after "WORLD"
	8502	CustAcctIDSource	Y	Int	Identifies source of CustomerAccount (6810); valid values: 1 = LEI 3 = Trading Name
	8020	CustomerPrimeBrokerTN	C	String	Customer prime broker trading name (or LEI)
	8021	CounterPartyPrimeBrokerTN	C	String	Counterparty prime broker trading name (or LEI)
	136	NoMiscFees	N	NumIn Group	Required if any miscellaneous fees are reported indicates number of repeating entries; the fees will be those defined by the customer receiving the message
	137	MiscFeeAmt	C	Amt	Brokerage amount; required if NoMiscFees (136) > 0
	138	MiscFeeCurr	C	Currency	Three-character string that represents the brokerage amount currency; required if NoMiscFees (136) > 0
	139	MiscFeeType	C	Int	Required if NoMiscFees (136) > 0; valid value: 7 = Other
	453	NoPartyID	N	NumIn Group	Number of party distinctions as required for MiFID II-relevant securities
	448	PartyID	N	String	Value for party role; required for MiFID II-relevant securities
	447	PartyIDSource	N	Char	Required for MiFID II-relevant securities; Valid value: D = Proprietary ID
	452	PartyRole	N	Int	Required for MiFID II-relevant securities; valid values: 11 = Order Origination Trader, execution within firm 16 = Executing System, algorithmic system identifier 122 = Investment Decision Maker
	Standard Trailer				

Y = Standard FIX Required; Y* = FIX API Required; C = Conditional; N = Not Required

5.19 Trade Capture Report Ack (AR)

All Trade Capture Report (AE) messages should have corresponding Trade Capture Report Ack (AR) messages. Trade Capture Report Ack (AR) messages will be required from the client prior to the end of day for all (AE) messages. Any (AE) message that is not acknowledged, including those in the buffer queue, is assumed to be confirmed via another channel, and thus automatically confirmed by the brokerage at end-of-day.

5.19.1 Trade Capture Report Ack Message Sent by Client

The (AR) messages sent by the client acknowledge the receipt of (AE) message sent by the FIX API. The message will confirm or deny the details of the (AE) message via TrdRptStatus (939) = 0 (Accept) or 1 (Reject), respectively. The Text (58) field will be populated when TrdRptStatus = 1 (Reject). The text should briefly describe the reason for the rejection e.g. “incorrect spot rate” or “incorrect delivery date”. The (AR) message can be used to electronically confirm or reject trades in FX Blotter for authorized users. Electronically rejected trades will be blocked in FX Blotter and will require manual intervention to unblock the trades.

5.19.2 Trade Capture Report Ack Message Sent by the FIX API

The (AR) messages sent by the FIX API acknowledge the receipt of counterparty trade confirmations or rejections. The message will confirm or deny the details of the (AE) message via TrdRptStatus (939) = 0 (Accept) or 1 (Reject), respectively. The Text (58) field will be populated when TrdRptStatus (939) = 1 (Reject). The text should briefly describe the reason for the rejection e.g. “incorrect spot rate” or “incorrect delivery date”. Shortly after the (AR) rejection message is sent, an amended (AE) message will be sent by the FIX API with the new details of the trade.

5.19.3 Trade Capture Report Ack (AR) Message Configuration

Tag	Field Name	Req	Type	Description
	Header			
35	MsgType			AR (Trade Capture Report Ack)
571	TradeReportID	Y	String	Unique identifier for the Trade Capture Report Ack
9536	Trader	N	String	Identifies the client who made this trade
150	ExecType	Y	Char	Type of execution being reported; valid values: 0 = New 4 = Canceled 5 = Replace
939	TrdRptStatus	Y*	Int	Trade report status; valid values: 0 = Accepted 1 = Rejected
751	TradeReportRejectReason	Y	Int	Reason for rejection of (AE) message; valid values: 0 = Successful (Default) 1 = Invalid party information 2 = Unknown instrument 3 = Unauthorized to report trades 4 = Invalid trade type 99 = Other
55	Symbol	Y	String	Six-character string that represents the currency pair
58	Text	C	String	Rejection reason when TrdRptStatus (939) = 1 (Rejected); free format text
1	Account	N	String	Counterparty trading name from the FIX API up to 79 characters long
	Standard Trailer			

Y = Standard FIX Required; Y* = FIX API Required; C = Conditional; N = Not Required

5.20 Trade Capture Report Request (AD)

The Trade Capture Report Request (AD) messages are used to request Trade Capture Reports. The (AD) message can be submitted using the following unique identifiers: QuoteID (117), OrderID (37), ClOrdID (11) or TradeReportID (571). If more than one parameter is submitted in a request, the trade that satisfies the combination of IDs will be returned. If the combinations of IDs do not match any trade, no report will be returned.

5.20.1 Trade Capture Report Request Identifier

A unique Trade Capture Report Request identifier, TradeRequestID (568), will be generated by the FIX API when an (AD) message is sent. The TradeRequestID (568) will conform to the following rules:

1. All new messages will have a unique value in this field for that day, regardless of the trading session the report was submitted.
2. The value will only contain upper-case letters and/or numbers up to 24 characters in length.

5.20.2 Trade Capture Report Request (AD) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			AD (Trade Capture Report Request)
568	TradeRequestID	Y	String	Unique identifier for the Trade Capture Report Request
569	TradeRequestType	Y	Int	Type of request; valid values: 0 = All trades 1 = Matched trades matching criteria provided on request
117	QuoteID	C	String	Unique identifier for Quote (S)
571	TradeReportID	C	String	Unique identifier for the Trade Capture Report (AE)
37	OrderID	C	String	Unique identifier for the order that was previously generated by the FIX API
11	ClOrdID	C	String	Unique identifier for the order that was previously generated by the client
580	NoDates	C	Int	Number of date ranges provided; if this field is not provided, then the assumed trade date is the current date; valid values: 1 = Date specified in TradeDate (75) is trade date 2 = Dates specified in TradeDate (75) are the start and end dates of the range
75	TradeDate	C	LocalMkt Date	Used when reporting other than current day trades; in YYYYMMDD format; conditionally required if NoDates (580) > 0
	Standard Trailer			

Y = Standard FIX Required; Y* = FIX API Required; C = Conditional; N = Not Required

5.21 Trade Capture Report Request Ack (AQ)

All Trade Capture Report Request (AD) messages must have corresponding Trade Capture Report Request Ack (AQ) messages. The (AQ) message is used to acknowledge the receipt of the request. TradeRequestResult (749) = 0 (Successful) confirms that the request was accepted, while TradeRequestResult (749) = 1 (Invalid or unknown instrument) confirms that the request was rejected. TradeRequestStatus (750) indicates the current state of the trade request. The (AQ) message will also indicate the number of (AE) messages to expect in response to the request.

5.21.1 TCR Request Ack (AQ) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			AQ (Trade Capture Report Request Ack)
568	TradeRequestID	Y	String	Unique identifier for the request ack
569	TradeRequestType	Y	Int	Type of request; valid values:

				0 = All trades 1 = Match trades on matching criteria provided on request
748	TotalNumTradeReports	N	Int	Number of trade Ccapture reports that will be returned
749	TradeRequestResult	Y	Int	Result of request; valid values: 0 = Successful 1 = Invalid or unknown instrument
750	TradeRequestStatus	Y	Int	Status of request; valid values: 0 = Accepted 1 = Completed 2 = Rejected
55	Symbol	Y		Six-character string that represents the currency pair e.g. "EURUSD"; the first currency is always the base and the second currency is always the term; "NA" is an acceptable value when multiple currencies are in the request
	Standard Trailer			

Y=FIX required; Y*=API required; C=Conditional; N=Not required

5.22 News (B)

The FIX News message is a general message which will indicate high level trade details executed in the market. The News message will not provide any client specific information, only the currency pair, structure, cutoff, direction, price and size of the execution.

The News Message will either contain summary level execution details or granular leg level execution details. Summary level execution details will be provided for non-specific and all differential and vega neutral products. For summary level executions on both differential and non-differential spreads, the NoLegs (555) repeating group will contain the following fields: Volatility (5678), LegQty (687), LegSymbol (600), Cutoff (9125), Tenor (6215) or LegMaturityDate (611), LegDelta (6035) if applicable.

5.22.1 News (B) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			See standard header section
35	MsgType			B (News)
42	OrigTime	N	UTC Timestamp	Time of trade execution.
131	QuoteReqID	C	String	Unique identifier for quote request; conditional on the execution being on a specific interest
8528	FXAPISecurityID	N	String	Broker security identifier; available only on CLOB trades
8505	RegulationVenueType	C	Int	Indicates the market type of the quote; valid values: 1 = SEF 3 = MiFID 7 = OTC 8 = IB
5159	Clearer	N	String	Specifies the clearer LEI
8019	SwapDataRepo	N	String	SDR LEI
148	Headline	Y	String	Specifies the headline text; will read: "Execution Alert"
9126	Structure	Y*	Int	FX Trading Systems primary structure information; see appendix for valid values
9072	CallOrPutOver	C	Int	Required for risk reversal structures; indicates if a risk reversal is quoted as "Calls Over Puts" or "Puts Over Calls"; valid values: 1 = Calls Over 2 = Puts Over
5232	CallOrPutOverCcy	C	Currency	Call/put currency; three-character string; conditional on structure being a risk reversal
54	Side	Y	Char	Indicates if the contract was paid or given on the screen from the aggressor's perspective; in the case of spreads it refers to position of leg 1 indicated by Position (6351)=1 (Same); note, however, that this may be arbitrary in some cases such as risk reversals where the screen can quote it in terms of ccy1 calls over or ccy 2 calls over; valid values:

					1 = Bought 2 = Sold
555	NoLegs		Y*	Num InGroup	Number of legs
Repeating Group	600	LegSymbol	Y	String	Six-character string that represents the currency pair e.g. "EURUSD"; the first ccy is always the base and the second is always the term
	6714	LegStrategyInd	C	Int	Required if execution is on a specific and non- differential; valid values: 1 = Call 2 = Put 3 = Up 4 = Down
	9125	Cutoff	Y*	Int	Time of expiration of FX option and fixing index for NDF; cutoff value will indicate the expiration or fixing date of trade; see appendix for valid values
	6215	Tenor	C	String	Option maturity; required for FXO if LegMaturityDate (611) is not provided; see appendix for valid values
	611	LegMaturityDate	C	Local MktDate	Date of expiration in YYYYMMDD; required if Tenor (6215) is not provided
	6203	FixingDate	C	Local MktDate	Required for NDFs (YYYYMMDD)
	9121	FarFixingDate	C	Local MktDate	Fixing date for the second leg of FX or NDF
	743	DeliveryDate	C	Local MktDate	Delivery/settlement date (YYYYMMDD)
	8014	FarDeliveryDate	C	Local MktDate	Far delivery/settlement date; required for deliverable forward and NDF spreads
	6035	LegDelta	C	Float	Delta; required if execution is on the standard pods; indicates if a risk reversal or butterfly is quoted as 25 delta or 10 delta; see appendix
	624	LegSide	C	Char	Side of leg; required if execution is on a specific and non differential; indicates aggressor direction; valid values: 1 = Bought 2 = Sold
	5678	Volatility	C	Float	Volatility rate (%); not required if execution is on a differential pods or specific interest
	8588	FwdPrice	C	Price	All-in rate All-in rate for NDFs Points for FX products; required for deliverable forwards and NDFs
	9127	LegDeliveryType	C	Char	Required when Structure (9126)=65 (FX) to indicate deliverable or non-deliverable; valid values: 1 = Cash (indicates non-deliverable) 2 = Physical (indicates deliverable)
	556	LegCurrency	C	Currency	Leg size currency; three-character string; required if execution is on a specific and non differential
	687	LegQty	C	Qty	Size of leg; not required if execution is on a differential pod or specific interest; the entry is million based i.e. an entry of 1 equals 1,000,000
	7940	LegID	C	String	Identifies leg of Quote Request (R) message; required if execution is on a specific and non-differential; the max length is 50 bytes
33	LinesOfText		Y	Num InGroup	Identifies number of lines of text body
	58	Text	Y	String	Free format text string; will read: "Execution Alert"
9102	FXMarketType		C	Int	NDF indicator; valid values: 1 = Onshore 2 = Offshore
5475	PremDeliveryType		C	Char	Premium delivery; assume both if not indicated; valid values: S = Spot F = Forward
5331	IsOwner		C	Boolean	Indicates "Y" if the execution belongs to the receiving client's trading group; this can be a price which is entered in using the FIX API or entered in using the GUI; note Side (54) is always from aggressor's perspective and may not necessarily reflect the owner's side; valid value: Y = Your Trading Groups Price
33	Standard Trailer				

Y=FIX required; Y*=API required; C=Conditional; N=Not required

5.23 Security List Request (x)

API clients submitting orders to CLOB will be expected to request the security list in addition to requesting market data subscription upon connecting to FIX API. The security list provides complete output of all contracts for the parameters specified.

API clients will have the ability to request all securities for which they are permissioned; or to specify currency pair and/or product (FXO or FX). It is recommended that API users subscribe to the same markets in their Security List Request (x) message as in their market data request so to be able to obtain complete picture of the market.

To request for all permissioned securities, client will specify SecurityListRequestType (559) to 4 (All Securities). To specify a subset of the contracts, they may do so by specifying currency pair and/or product. When requesting a subset, the SecurityListRequestType is set to Symbol (0) and the Symbol (55) must be populated along with SecurityType (167) as optional.

Below are responses to Security List Request (x) with the following tags populated:

1. FIX API returns complete security lists to client for which client is permissioned.
 - a. SecurityListRequestType (559) = 4
2. FIX API returns security list of specified currency pair for all assets the client is permissioned.
 - a. SecurityListRequestType (559) = 0
 - b. Symbol (55) = desired ccy pair
3. FIX API returns security list of currency pair and asset specified by client.
 - a. SecurityListRequestType (559) = 0
 - b. Symbol (55) = desired ccy pair
 - c. SecurityType (167) = FXO or FX
4. FIX API returns security list of contracts by asset class
 - a. SecurityListRequestType (559) = 0
 - b. Symbol (55) = 'ALL_ALL'
 - c. SecurityType (167) = FXO or FX

To unsubscribe from previously requested security list message, the client will specify the parameters to which he no longer wishes to subscribe, and set the SubscriptionRequestType (263) = 2 (Unsubscribe the Snapshot + Update Request).

5.23.1 Security List Request (x) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			x (Security List Request)
320	SecurityReqID	Y	String	Unique identifier for security list request
559	SecurityListRequestType	Y	Int	Valid values: 0 = Symbol 4 = All Securities
55	Symbol	C	String	Required when SecurityListReqType (559) = 0
167	SecurityType	N	String	Valid values: - FXO - FX
263	SubscriptionRequestType	Y	Int	Valid values:Valid values 1=Subscribe to Snapshot + Updates 2= Unsubscribe the Snapshot + Update Request
	Standard Trailer			

5.24 Security List (y)

The FIX API server will respond to a valid Security List Request (x) message with full snapshot of its standard contracts via the Security List (y) message. The message provides list of contracts that meet the parameters specified in the request.

When responding to a valid request, FIX API will reply with SecurityRequestResult (560) set to 0 (valid request) and SecurityReqID (320) populated with the client generated request ID. The two fields may be left blank when the security list is sent as result of security update (and not security list request).

The FIX API will reply to an invalid security list request by sending 1 (Invalid Request) in the SecurityRequestResult (560) along with the identifying Security List Request (x) message (320).

The FIX API breaks down its security list messages by currency pair and product. For example, if a security list request for all contracts (SecurityListReqType=4) is received by the FIX API from a client who is entitled to ten currency pairs for product class FXO; the FIX API will respond with ten separate Security List (y) messages.

In addition to the Security List (y) messages sent in response to a request, the FIX API will send security list messages on a nightly basis (between 22:30 to 23:00 (UTC)) for parameters specified in the initial Security List Request (x) message. Any intraday updates to contracts will result in real-time Security List (y) messages to be sent to subscribed API clients.

5.24.1 Full versus Incremental Snapshot

Security list messages sent in response to Security List Request (x) and those sent nightly upon cutting over to new business day will list out all the identifying attributes of a contract (Full Snapshot).

Messages resulting from intraday contract changes are sent as incremental updates, reflecting only the SecurityResponseID (322), NoTenorGroup (8715), FXAPISecurityID (8528), TradeDate (75), TransactTime (60), and the attributes that have been updated.

5.24.2 Security List (y) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			y (Security List)
320	SecurityReqID	C	String	Required when in response to a Security List Request (x)
322	SecurityResponseID	Y	String	Unique identifier for security list message
560	SecurityRequestResult	C	Int	Required when in response to Security Request (x); valid values: 0 = Valid request 1 = Invalid request
167	SecurityType	Y	String	Valid values: FXO = FX Options FX = Spot, Forward and NDF
393	TotNoRelatedSym	N	Int	Provided only in response to Security List Request (x); used to indicate the total number of securities being returned for this request
893	LastFragment	N	Boolean	Provided only in response to Security List Request (x); indicates the last message in the sequence of messages; valid value: Y = Yes
146	NoRelatedSym	Y	NumIn	Number of related symbols (instruments) that follow

				Group	
	55	Symbol	Y	String	Currency pair
	9126	Structure	C	Int	Strategy identifier
	8505	RegulationVenueType	C	Int	Valid values: 1 = SEF 3 = MiFID 7 = OTC 8 = IB
	811	PriceDelta	C	Float	Required for FXO
	9125	Cutoff	C	Int	Time of expiration of FX option; valid values: see appendix; for NDFs the cutoff value will indicate the fixing index
	5159	Clearer	C	String	Specifies the clearer LEI; for bilateral, value = 0
	9127	LegDeliveryType	C	Char	Required for FX; valid values: 1 = Cash 2 = Physical
	9102	FXMarketType	C	Int	Valid values: 1 = Onshore 2 = Offshore 3 = Both (Utilized when passing security attribute that do not differentiate between onshore and offshore markets)
	9072	CallOrPutOver	C	Int	Required for risk reversals; valid values: 1 = CallsOverPuts 2 = PutsOverCalls
	15	Currency	C	Currency	Call/put currency; required for risk reversals
	5830	PremiumCcy	N	Currency	Premium Currency. Applicable for FXO only.
	8715	NoTenorGroup	Y	NumIn Group	Number of tenor groups that follow
	8528	FXAPISecurityID	Y	String	
	6215	Tenor	C	Tenor	Tenor display name; required for standard contracts except in EOD update
	432	ExpiryDate	C	LocalMkt Date	Expiration for FXO; required for FXO
	6203	FixingDate	C	LocalMkt Date	NDF fixing date; required for NDFs
	9121	FarFixingDate	C	LocalMkt Date	Far fixing date for NDF spread; required for NDF spreads
	743	DeliveryDate	C	LocalMkt Date	Settlement/delivery date
	8014	FarDeliveryDate	C	LocalMkt Date	Far delivery date for NDF; required for NDF spreads
	969	MinPriceIncrement	N	Float	Minimum price increase for the Instrument
	454	NoSecurityAltId	N	NumIn Group	Number of security ID groups that follow; required for MiFID II-relevant securities
	455	SecurityAltId	N	String	ISIN of the security; required for MiFID II-relevant securities
	456	SecurityAltIdSource	N	String	Required for MiFID II-relevant securities ; valid value: 4 = ISIN Identifier
	461	CFICode	N	String	Indicates the type of security; required for MiFID II-relevant securities
	24201	IlliquidInstrument	N	Boolean	Designated liquidity type based on components and size; required for MiFID II-relevant securities
	24202	SizeThresholdLIS	N	Qty	LIS size threshold; required for MiFID II-relevant securities when IlliquidInstrument (24201) = N
	24203	SizeThresholdSSTI	N	Qty	SSTI size threshold; required for MiFID II-relevant securities when IlliquidInstrument (24201) = N
75	TradeDate		Y	LocalMkt Date	The trade date for which the expiry/fixing and delivery dates apply
60	TransactTime		Y	UTC Timestamp	Timestamp in microseconds for the message in YYYYMMDD-HH:MM:SS.ssssss format. Timestamps can be in non microsecond format (based on client preference for MiFID tags)
	Standard Trailer				

6 Appendix A: Introduction Broker Venue

The FIX API introduced a venue that offers clients a single liquidity pool for price discovery, with execution venue routing determined at the time of trade arrangement. The brokers managing the IB liquidity pool will introduce a trade arrangement into the appropriate venue for execution.

An arranged trade match will be routed to the appropriate venue for electronic and voice trade execution using the following guidelines:

- If any of the persons/entities involved in the trade match (including brokers managing a customer order) are located in the United States, or qualify as US persons, the arranged trade will be routed to the SEF execution venue.
- In the absence of a US person/entity, if any of the persons/entities involved in the arranged trade match are a European Union person/entity (including brokers managing a customer order on the system), the trade will be routed to the MIFID execution venue.
- In the absence of either a US or EU person/entity being involved in the arranged trade match, the trade will be routed to the OTC execution venue.

Further checks may also determine the outcome of an execution, such as venue membership (if applicable) and credit restrictions (if configured).

6.1.1 Security List (y) Message

The API server will respond to a valid Security List Request (x) message with a full refresh snapshot of its standard contracts via the Security List (y) message. The Security List (y) message provides a list of contracts that meet the parameters specified in the request. The FIX API breaks down its security list messages by currency pair and product. For example, if a security list request for all contracts (SecurityListReqType (559) = 4 (all securities)) is received by the FIX API from a client who is entitled to ten currency pairs for product class FXO, the FIX API will respond with ten separate Security List (y) messages.

The API will send daily Security List (y) messages on a nightly basis (between 21:30 to 23:30 (UTC), depending on daylight savings) for parameters specified in the initial Security List Request (x) message. Additionally, intraday updates to tradable contracts will result in real-time Security List (y) messages sent to subscribed API clients.

When responding to a valid request, the application will reply with SecurityRequestResult (560) = 0 (valid request) and SecurityReqID (320) populated with the client generated request ID. These two fields may be left blank when the Security List (y) message is sent as result of security update in the trading platform (and not as a result of a Security List Request (x) message).

The FIX API will reply to an invalid Security List Request (x) message by sending SecurityRequestResult (560) = 1 (invalid request), along with the request ID in SecurityReqID (320).

The IB venue type is in RegulationVenueType (8505) and will be applied to the construction of the FXAPISecurityID (8528) for determining the liquidity pool on the API messaging. Other venues will be

active and can potentially have liquidity depending on regulatory conditions for the products supported on the FIX API.

6.1.2 Market Data (W) – Snapshot/Full Refresh Message

The Market Data (W) – Snapshot/Full Refresh message is used to electronically stream prices, in terms of volatility for FXO, or spot, forward points or all-in rate for FX products, from the front-end via the FIX API. Published price messages will not require a confirmation of receipt from the client. It will be the responsibility of the heartbeat to ensure a connection is present. Market Data (W) – Snapshot/Full Refresh messages will be published for indirect (anonymous) quotes on specific interests and quotes on standard products in the front-end.

The Market Data (W) – Snapshot/Full Refresh message will reference the venue where the liquidity is active:

- SEF prices are referenced with RegulationVenueType (8505) = 1 (SEF)
- MIFID prices are referenced with RegulationVenueType (8505) = 3 (MiFID)
- OTC prices are referenced with RegulationVenueType (8505) = 7 (OTC)
- IB prices are referenced with RegulationVenueType (8505) = 8 (IB)

The EligibleVenues (6003), within the NoMDEntries (268) repeating group after the IsOwner (5331) field, denotes the eligible venues that the order can trade based on the market maker's (initiator's) entitlements, permissions and location.

The FXAPISecurityID (8528) on the Market Data (W) message will also refer to the IB contracts as detailed in the Security List (y) message.

6.1.3 Quote Request (R) Message

The Quote Request (R) message is used to request prices for specific bespoke contracts. The requestor will send the message to a target audience, and the target audience will respond to the requestor's Quote Request (R) message by sending tradable prices, rejecting the message, or not responding, which will indicate no interest.

The Quote Request (R) message will reference the venue requested:

- SEF requests are referenced with RegulationVenueType (8505) = 1 (SEF)
- MIFID prices are referenced with RegulationVenueType (8505) = 3 (MiFID)
- OTC prices are referenced with RegulationVenueType (8505) = 7 (OTC)
- IB prices are referenced with RegulationVenueType (8505) = 8 (IB)

6.1.4 New Order – Single (D) Message

The FIX API will support the use of the New Order – Single (D) message to submit and aggress on standing orders in the IB venue.

6.1.4.1 Submit IB Order

When submitting unsolicited orders to the IB venue, the New Order – Single (D) message must provide the FXSecurityID (8528) associated for that venue. A list of security IDs will be provided by the FIX API to clients upon receiving their Security List Request (x) message.

6.1.4.2 Aggressing a Standing IB Order

New Order – Single (D) messages are also used to aggress a standing CLOB order. API clients will specify the appropriate FXSecurityID (8528) for the IB venue, along with the other required parameters.

6.1.5 Execution Report (8) Message

The Execution Report (8) message is sent in response to transaction events. Such events are acknowledgements of order submission, in response to an Order Status Request (H) message and arranged trade matching of resting orders and tradable quotes. ExecType (150) will specify the type of Execution Report (8) message the FIX API client is receiving.

6.1.5.1 Order Acknowledgement

When responding to order submission via the New Order – Single (D) message, the FIX API will send an Execution Report (8) message with OrderStatus (39) = 0 (new) or 8 (rejected).

The Execution Report (8) message will reference the order it is responding to via ClOrdID (11). OrderID (37) is generated by the system and will be the identifier used in subsequent Execution Report (8) messages associated to the accepted order.

The order acknowledgement will reference the venue specified on the New Order(D) message:

- SEF orders are referenced with RegulationVenueType (8505) = 1 (SEF) on the Execution Report (8) acknowledgment
- MIFID orders are referenced with RegulationVenueType (8505) = 3 (MiFID) on the Execution Report (8) acknowledgment
- OTC orders are referenced with RegulationVenueType (8505) = 7 (OTC) on the Execution Report (8) acknowledgment
- IB orders are referenced with RegulationVenueType (8505) = 8 (IB) on the Execution Report (8) acknowledgment

6.1.5.2 Arranged Trade Match

When communicating an arranged trade match, the FIX API will send an Execution Report (8) message with OrderStatus (39) = 1 (partially filled) or 2 (filled). This Execution Report (8) message will reference the appropriate RegulationVenueType (8505) when responding to an arranged trade match event:

- SEF matches are referenced with RegulationVenueType (8505) = 1 (SEF) on the Execution Report (8) message
- MIFID matches are referenced with RegulationVenueType (8505) = 3 (MiFID) on the Execution Report (8) message
- OTC matches are referenced with RegulationVenueType (8505) = 7 (OTC) on the Execution Report (8) message

6.1.6 Trade Capture Report (AE) Message

Trade Capture Report (AE) messages are sent electronically by the FIX API to report clients' trade details of the execution. The message identifies trade details by leg of each executed trade, including spot, premium and hedge. In some instances, such as corrections and cancels, multiple Trade Capture Report (AE) messages will be sent in response to a single execution. In addition, Trade Capture Report (AE) messages will contain the necessary details of STP messages.

The Trade Capture Report (AE) messages will reference the appropriate RegulationVenueType (8505) that the trade was booked to:

- SEF executions are referenced with RegulationVenueType (8505) = 1 on the Trade Capture Report (AE) message
- MIFID executions are referenced with RegulationVenueType (8505) = 3 (or 5 for Fenics Direct) on the Trade Capture Report (AE) message
- OTC executions are referenced with RegulationVenueType (8505) = 7 on the Trade Capture Report (AE) message

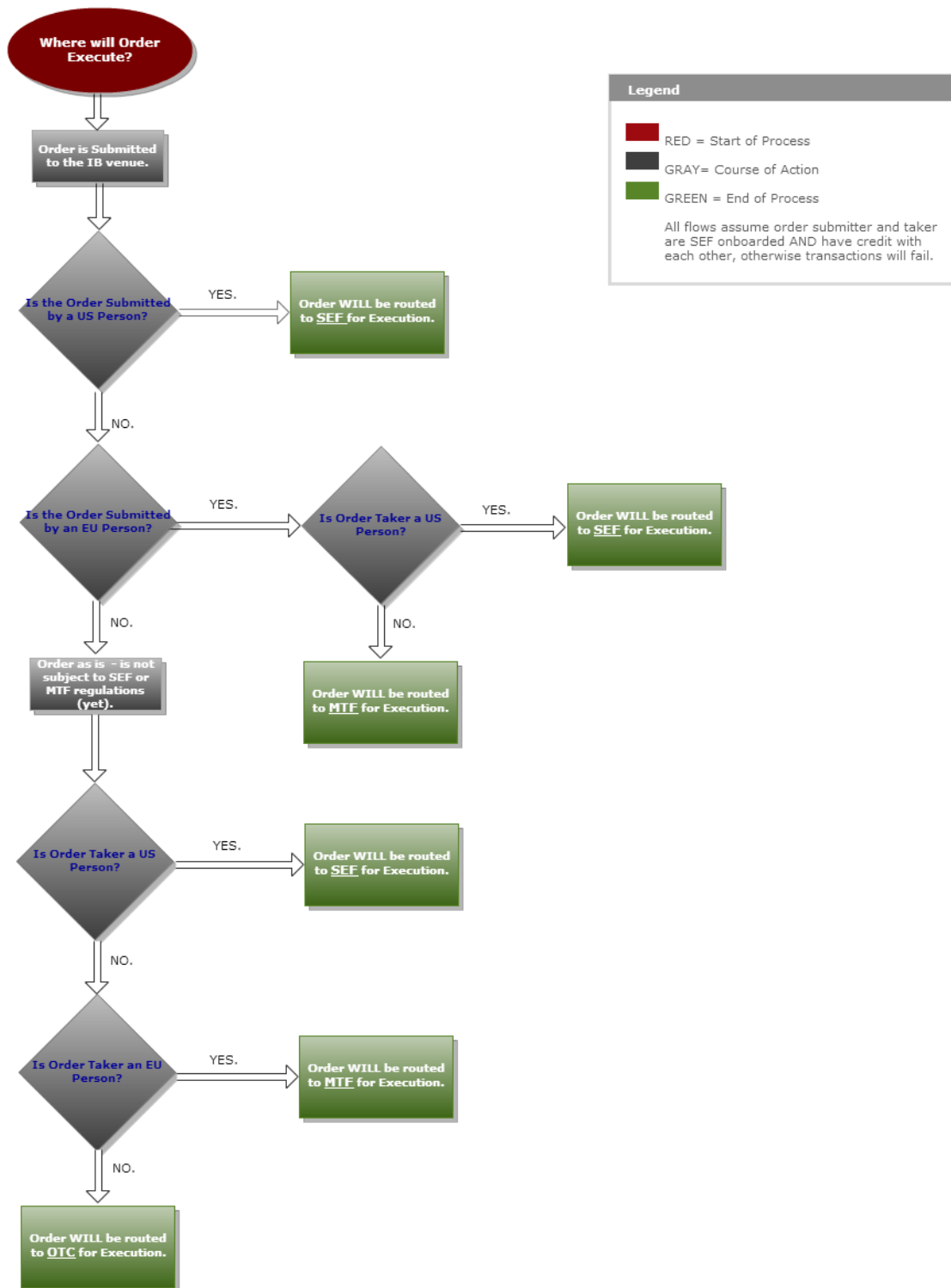
6.1.7 News (B) Message

The News (B) message is a general message which indicates high level arranged trade details in the market. The News (B) message does not provide any client specific information, only the currency pair, structure, cutoff, direction, price and size of the trade match arranged.

The News (B) message will reference the venue of arranged trade match event:

- SEF arranged matches are referenced with RegulationVenueType (8505) = 1 (SEF)
- MIFID arranged matches are referenced with RegulationVenueType (8505) = 3 (MiFID)
- OTC arranged matches are referenced with RegulationVenueType (8505) = 7 (OTC)
- IB arranged matches are referenced with RegulationVenueType (8505) = 8 (IB)

6.1.8 Trade Determination in the IB Venue



7 Appendix B: Fenics Direct

7.1 Mass Quote (i)

The Mass Quote (i) message is used to electronically send multiple unsolicited quotes for a number of standard contracts. In order for a client to RFQ a liquidity bank, the bank must provide an indicative surface to the client using the Mass Quote (i) message. The liquidity Bank must send an indicative surface to the FIX API, which will then forward the surface to on-boarded and configured clients.

To submit indicative prices, liquidity banks must provide for a structure bid and offer volatilities, expiry date, and delta. If QuoteType (537) = 0 (Indicative), then BidSize (134) OfferSize (135) will be ignored. If QuoteType (537) is not provided, then the message will be considered indicative. Subsequent volatility contributions by liquidity banks will overwrite previously ones.

Mass Quote (i) messages will be processed if the messages have the following fields populated:

- MQSymbol (7177)
- CompleteSurface (9854)
- Tenor (6215) or ExpireDate (432)
- BidPx (132)
- OfferPx (133)

If a liquidity bank sends CompleteSurface (9854) = 1, then all volatilities will be cleared for that currency pair, and, if the liquidity bank is feeding depots, then all swap and depo values for that currency pair will also be cleared. Volatilities will be validated and imported into its system based on the following rules:

- All acceptable volatilities must have an ATM straddle
- Risk reversals and butterflies must be in the same message, and must have the same delta values present; the risk reversal bid and offer are used to find the mid rates used as the volatilities
- Acceptable cases:
 1. Straddle
 2. Straddle, 25Δ Risk Reversal, 25Δ Butterfly, 10Δ Risk Reversal, 10Δ Butterfly
 3. Straddle, 25Δ Butterfly, and 25Δ Risk Reversal
 4. Straddle, 32.5Δ Butterfly, and 32.5Δ Risk Reversal
- Unacceptable cases (will be rejected):
 1. Straddle, 25Δ Risk Reversal
 2. Straddle, 25Δ Butterfly, 10Δ Risk Reversal
 3. 25Δ Butterfly, 25Δ Risk Reversal (no ATM Straddle)
 4. Straddle, 25Δ Risk Reversal
 5. Straddle, 25Δ Butterfly
- All messages will have at least one of the following date fields: expiry or maturity
- ForwardPoints (195) and DepoRate (9115) will only be accepted for currency pairs with USD

7.1.1 Market Data Publications Provided by Liquidity Banks

The Mass Quote (i) message may be used by liquidity banks to send market data to Fenics Direct. Fenics Direct will use the market data to price options for authorized clients that are interested in the liquidity bank's prices. To provide market data to Fenics Direct, a liquidity bank will populate:

- ForwardPoints (195)
- DepotRate (9115)
- DepoRateCcy (9073)

Liquidity banks are not required to provide volatility prices for market-data-only publications.

7.1.2 Market Maker Bank API Disconnect

A liquidity bank's Mass Quote (i) messages will be automatically canceled by the FIX API when it disconnects from the FIX API. The FIX API will not recover quotes after they are canceled. Additionally, the FIX API will not resend to reconnected API clients mass quotes sent during, or prior to, the disconnect. As such, liquidity banks are expected to periodically republish Mass Quote (i) messages throughout the trading day. Clients that process mass quotes will expect to receive a complete surface for every active currency pair at intervals no less than once every 15 minutes.

7.1.3 Client Mass Quote Message Identifier

A unique Mass Quote identifier, QuoteID (117), should be generated by the client for any quote submitted to the FIX API. The QuoteID (117) should conform to the following rules:

1. All new messages will have a unique value in this field for that day, regardless of the trading session the report was submitted.

The value will only contain upper-case letters and/or numbers up to 24 characters in length.

7.1.4 Mass Quote (i) Message Configuration

Tag	Field Name	Req	Type	Description
	Standard Header			
35	MsgType			i (Mass Quote)
117	QuoteID	Y	String	Unique identifier for the quote
537	QuoteType	C	Int	Valid value: 0 = Indicative (default)
301	QuoteResponseLevel	N	Int	Level of response requested from receiver of quote messages; valid values: 0 = No Acknowledgement (Default) 1 = Acknowledge only negative or erroneous quotes 2 = Acknowledge each quote message
8505	RegulationVenueType	C	Int	Conditionally required if trading on SEF or MIFID market; indicates the market type of the quote; valid values: 1 = SEF 3 = MiFID 7 = OTC (applicable only when Structure (9126) = 65 and LegDeliveryType (9127) = 2)
8519	OriginatingLEI	C	String	LEI of the beneficial owner of swap; required when quote is for SEF; will be rejected if LEI has not been SEF on-boarded for the respective venue
5159	Clearer	C	String	Specifies the clearer LEI; omission indicates a bilateral market
6661	PBReuterCode	C	String	Assigned Reuter Code used to identify the associated prime broker or FCM; conditionally required if the provider uses a prime broker or FCM
7177	MQSymbol	Y*	String	Six-character string that represents the currency pair e.g. EURUSD; the first ccy is always the base and the second is always the term ccy
9854	CompleteSurface	Y*	Int	Valid value: 1 = Complete Surface
9125	Cutoff	Y*	Int	Time of expiration; for NDF, the value will indicate the fixing index; expected value for NDF is "25"; see appendix for all valid values
296	NoQuoteSets	Y	NumIn	The number of sets of quotes in the message

				Group	
	302	QuoteSetID	Y	String	Sequential number for the quote set; for a given QuoteID (117), it is assumed to start at "1"; must be the first field in the repeating group
	304	TotNoQuoteEntries	Y	Int	Total number of quotes for the quote set across all messages; should be the sum of all NoQuoteEntries (295) in each message that has repeating quotes that are part of the same quote set
	9102	FXMarketType	N	Int	Valid values: 1=Onshore 2=Offshore
	432	ExpireDate	C	LocalMkt Date	Date of option expiration; required for FXO if there is no Tenor (6215)
	6215	Tenor	C	String	Required if there is no ExpireDate (432); see appendix for valid values
	9072	CallOrPutOver	C	Int	Call or put indication; required for risk reversals; valid values: 1 = Calls Over Puts 2 = Puts Over Calls
	5232	CallOrPutOverCcy	C	Currency	Three-character string that represents the call or put currency; required for risk reversals
	9126	Structure	Y*	Int	Primary structure information; see appendix for valid values
	195	LastForwardPoints	C	Price Offset	Swap rate; market data field; assume industry standards for swap points per currency pair e.g. EURUSD Call can be 1.2809
	9115	DepoRate	C	Float	Depo rate (%); market data field e.g. EURUSD Call can be 5.04
	9073	DepoRateCcy	C	Currency	Depo rate currency; market data field; required if DepoRate (9115) is provided
	295	NoQuoteEntries	Y	NumIn Group	The number of quotes for the quote set; not required for market-data-only publications
	299	QuoteEntryID	Y	String	Uniquely identifies the quote as part of the quote set; must be used if NoQuoteEntries (295) is used; this field will be unique across messages for live Mass Quote (i) messages
	811	PriceDelta	Y	Float	Delta; only required when NoQuoteEntries (295) > 0
	132	BidPx	C	Price	Bid price
	133	OfferPx	C	Price	Offer price
	134	BidSize	C	Qty	Bid size
	135	OfferSize	C	Qty	Offer size
	743	DeliveryDate	C	LocalMkt Date	Near term settlement date
	8014	FarDeliveryDate	C	LocalMkt Date	Far term settlement date; required for NDF spreads
	9127	LegDeliveryType	C	Char	Required when Structure (9126)=65 (FX) to indicate deliverable or non-deliverable; valid values: 1 = Cash (indicates non-deliverable) 2 = Physical (indicates deliverable)
	9536	Trader	N	String	Trader who submitted the price
	60	TransactTime	N	UTC Timestamp	Timestamp in microseconds of quote creation by client in YYYYMMDD-HH:MM:SS.ssssss format. Timestamps can be in non microsecond format (based on client preference for MiFID tags)
	453	NoPartyID	N	NumIn Group	Number of party distinctions as required for MiFID II-relevant securities
	448	PartyID	N	String	Value for party role; required for MiFID II-relevant securities
	447	PartyIDSource	N	Char	Required for MiFID II-relevant securities; Valid value: D = Proprietary ID
	452	PartyRole	N	Int	Required for MiFID II-relevant securities; valid values: 11 = Order Origination Trader, execution within firm 16 = Executing System, algorithmic system identifier 122 = Investment Decision Maker
	2593	NoOrderAttributes	N	NumIn Group	Number of regulatory order attributes; required for MiFID II-relevant securities.
	2594	OrderAttributeType	N	Int	Optionally required for MiFID II-relevant securities if NoOrderAttributes (2593) > 0; valid values:

						2 = Liquidity provision activity order 4 = Algorithmic order
		2595	OrderattributeValue	N	String	Attribute value; required for MiFID II-relevant securities securities ; valid values: Y = Yes N = No
		Standard Trailer				

Y = Standard FIX Required; Y* = FIX API Required; C = Conditional; N = Not Required

7.1.5 Canceling Entire Mass Quote Messages

Liquidity banks will be able to cancel their own volatility surfaces. To cancel volatility surfaces, they can send a Quote Cancel (Z) message to the FIX API and will set QuoteCancelType (298) = 3 (Cancel mass quote request). Liquidity banks will be able to cancel entire mass quote surfaces for a specified symbol by setting the Symbol (55) to the symbol to be canceled. To cancel all mass quotes, they can set Symbol (55) = ALL_ALL.

The FIX API will receive the Quote Cancel (Z) message from the client and will send a Quote Status Report (AI) message to the client to indicate the validity of the message. If the Quote Status Report (AI) message returns “canceled by client”, then the FIX API will route the cancel message to the appropriate targets. The appropriate targets will be all of the clients that received the original Mass Quote (i) message sent by the client. The API client will send the Quote Cancel (Z) message to the appropriate targets by setting SenderCompID (49) = broker CompID, TargetCompID (56) = target CompID, and OnBehalfOfCompID (115) = sender CompID. The FIX API will cancel live mass quotes for:

- All API clients before the FIX API weekly recycle
- All API clients at the end of each trading day
- Liquidity banks that logout of the FIX API

The FIX API will cancel all live mass quotes by sending Quote Cancel (Z) messages to all clients that received the original Mass Quote (i) messages. The FIX API will send the cancel messages for all products by sending Symbol (55) = ALL_ALL. Note if at least one of following fields are set in the Quote Cancel (Z) message, then an entire volatility surface will not be canceled: Structure (9126) and/or PriceDelta (811).

7.2 Mass Quote Acknowledgement (b)

The FIX Mass Quote Acknowledgement message is used as an acknowledgement in response to the Mass Quote message. Mass Quote Acknowledgement messages will be sent only in response to a Mass Quote that has the QuoteResponseLevel field (tag 301) set to 1 (acknowledge only negative quotes) or 2 (acknowledge each quote).

7.2.1 Accepting and Rejecting Mass Quotes

The Mass Quote Acknowledgement message will indicate whether the entire Mass Quote message was accepted or rejected and will identify rejected bid and offers. If a Mass Quote message is rejected for a defined QuoteRejectReason (see tag 300), then the QuoteStatus field (tag 297) will equal 5 (reject). If a Mass Quote message is rejected, then none of the individual quotes from the Mass Quote will enter the ForexMatch system.

All rejected bid and offer pairs will appear in the NoQuoteEntries repeating group with a rejection code in the QuoteEntryRejectReason field (tag 368). If the QuoteResponseLevel field is set to 2 (acknowledge all quotes), then all valid quotes will also appear in the NoQuoteEntries repeating group. Valid quotes will not have a QuoteEntryRejectReason. All quotes in the NoQuoteEntries repeating group will be identified by QuoteEntryID.

7.2.2 Mass Quote Acknowledgement (b) Message Configuration

Tag	Field Name		Req	Type	Description
	Standard Header				
35	MsgType				b (Mass Quote Acknowledge)
117	QuoteID		Y	String	Unique identifier for quote
297	QuoteStatus		Y	Int	Status of the Mass Quote Acknowledgement (b) message; valid values: 0 = Accepted 5 = Rejected
300	QuoteRejectReason		N	Int	The rejection code for QuoteStatus (297) = 5 (Rejected); valid values: 1 = Unknown Symbol 6 = Duplicate Quote 7 = Invalid bid/ask spread 8 = Invalid price 9 = Not authorized to quote security See appendix for more error codes
8505	RegulationVenueType		C	Int	Indicates the market type of the quote; valid values: 1 = SEF 3 = MiFID 7 = OTC
5159	Clearer		C	String	Specifies the clearer LEI
6661	PBReuterCode		C	String	Specifies the Reuter Code provided in the Mass Quote (i) message
8519	OriginatingLEI		C	String	LEI of beneficial owner of swap
301	QuoteResponseLevel		N	Int	Level of response requested from receiver of mass quote messages; valid values" 1= Acknowledge only negative or erroneous quotes 2 = Acknowledge each quote message
537	QuoteType		N	Int	Valid value: 0 = Indicative
58	Text		N	String	Text field used to describe the reason an entire Mass Quote (i) message is rejected
296	NoQuoteSets		Y	NumIn Group	The number of sets of quotes in the message
	302	QuoteSetID	Y	String	Sequential number for the quote set; for a given QuoteID (117), assumed to start at "1"; must be the first field in the repeating group
	304	TotNoQuoteEntries	Y	Int	Total number of quotes for the quote set across all messages; should be the sum of all NoQuoteEntries (295) in each message that has repeating quotes that are part of the same quote set
	295	NoQuoteEntries	Y	NumIn Group	The number of quotes for this quote set that follow in this message
	299	QuoteEntryID	Y	String	Uniquely identifies the quote as part of a quote set; must be used if NoQuoteEntries (295) is used
	368	QuoteEntryRejectReason	C	Int	The reason quote entry was rejected; required if a QuoteEntryID (299) is rejected; valid values: 1 = Unknown symbol 6 = Duplicate quote 7 = Invalid bid/ask spread 8 = Invalid price 9 = Not authorized to quote security 99 = Other
	Standard Trailer				

7.2.3 1st Generation Exotics

Following Trigger fields have been added to QuoteRequest, Quote, ExecutionReport and TradeCaptureReport messages

5167	LoTrigger	C	Price	Identifies trigger price in absolute terms for the single or low trigger; required on single or double trigger options
6192	HiTrigger	C	Price	Identifies trigger price in absolute terms for the high trigger; required on double trigger options

The following table gives field usage information about the twelve new Exotic classes:

Exotic Class	LoTrigger (5167)	HiTrigger (6192)	Strike (612)	Leg Strategy (6714)
Knockin	Yes	No	Yes	Call/Put
Knockout	Yes	No	Yes	Call/Put
Reverse Knockin	Yes	No	Yes	Call/Put
Reverse Knockout	Yes	No	Yes	Call/Put
Double Knockin	Yes	Yes	Yes	Call/Put
Double Knockout	Yes	Yes	Yes	Call/Put
Digital	No	No	Yes	Call/Put
Instant One Touch	Yes	No	No	Up/Down
No Touch	Yes	No	No	Up/Down
One Touch	Yes	No	No	Up/Down
Double No Touch	Yes	Yes	No	Up/Down
Double One Touch	Yes	Yes	No	Up/Down

8 Appendix C: Connection Details

The connections details for the production and test (staging) sites are listed below.

For production, the preferred connectivity is Client network or Equinix (through Lucera).



Connection Details

Environment	Production	Beta (Staging)
Internet URL	fx-api.bgctrader.com (Primary) fx-api-dr.bgctrader.com(DR)	fx-api-beta.bgctrader.com
Internet IP	75.98.77.127 (Primary) 148.106.153.204 (DR)	74.217.171.119
Ports	443(SSL only), 20000	443(SSL only), 20000
BGC Client Network URL for Clients	fx-api-secure.bgctrader.com (Primary) fx-api-dr-secure.bgctrader.com(DR)	N/A
Coming Over BGC Client Network IP	148.106.165.127(Primary) 148.106.160.66 (DR)	N/A



Connection Details

Environment	Production	Beta 2 (Staging 2)
Internet URL	api.gfifx.com(Primary) api-dr.gfifx.com(DR)	quotes.stage2.gfifx.com
Internet IP	75.98.77.83(Primary) 148.106.153.187(DR)	74.217.171.24
Ports	443 (SSL Only), 20000	443(SSL only), 20000
Client Network URL for Clients	api-secure.gfifx.com (Primary) api-dr-secure.gfifx.com(DR)	N/A
Coming Over Client Network IP	148.106.165.83(Primary) 148.106.160.96 (DR)	N/A

9

Appendix D: Execution Message Qty Type Scenarios

Scenario	Non-Aggressor (NA) / Aggressor (A) Price / Size	Non-Aggressor	Aggressor	Aggressor
		Execution Report	Execution Report	Execution Cancel
Non-aggressor's quote fully executed and aggressors order fully executed.	(NA: 1/50) / (A: 1/50)	(38) = 50	(38) = 50	N/A
		(151) = 0	(151) = 0	
		(14) = 50	(14) = 50	
		(5359) = 50	(5359) = 50	
Non-aggressor's quote fully executed and aggressors order partially executed	(NA: 1/50) / (A: 1/100)	(38) = 50	(38) = 100	(38) = 100
		(151) = 0	(151) = 50	(151) = 0
		(14) = 50	(14) = 50	(14) = 50
		(5359) = 50	(5359) = 50	(5359) = 50
Non-aggressor quote partially filled and aggressors order fully filled	(NA: 1/100) / (A: 1/50)	(38) = 100	(38) = 50	N/A
		(151) = 50	(151) = 0	
		(14) = 50	(14) = 50	
		(5359) = 50	(5359) = 50	
2 Non-aggressor quotes fully filled and aggressor order fully filled via trade through.	(NA: 1/50;1/50) / (A: 1/100)	<u>Execution Report #1</u>	<u>Execution Report #1</u>	N/A
		(38) = 50	(38) = 100	
		(151) = 0	(151) = 0	
		(14) = 50	(14) = 100	
		(5359) = 50	(5359) = 100	
		<u>Execution Report #2</u>	N/A	N/A
		(38) = 50		
		(151) = 0		
		(14) = 50		
		(5359) = 50		
2 Non-aggressor quotes fully filled and aggressor order partially filled via trade through.	(NA: 1/50;1/50) / (A: 1/150)	<u>Execution Report #1</u>	<u>Execution Report #1</u>	<u>Cancel</u>
		(38) = 50	(38) = 150	(38) = 150
		(151) = 0	(151) = 50	(151) = 0
		(14) = 50	(14) = 100	(14) = 100
		(5359) = 50	(5359) = 100	(5359) = 50
		<u>Execution Report #2</u>	N/A	N/A
		(38) = 50		
		(151) = 0		
		(14) = 50		
		(5359) = 50		
2 Non-aggressor quotes fully filled at different price points and aggressor order fully filled via trade through.	(NA: 1/50;.95/50) / (A: 85/100)	<u>Execution Report #1</u>	<u>Execution Report #1</u>	N/A
		(38) = 50	(38) = 100	
		(151) = 0	(151) = 50	
		(14) = 50	(14) = 50	
		(5359) = 50	(5359) = 50	
		<u>Execution Report #2</u>	<u>Execution Report #2</u>	N/A
		(38) = 50	(38) = 100	
		(151) = 0	(151) = 0	

2 Non-aggressor quotes fully filled at different price points and aggressor order partially filled via trade through.	(NA: 1/50;.95/50) / (A: 85/150)	(14) = 50	(14) = 50	
		(5359) = 50	(5359) = 50	
		<u>Execution Report #1</u>	<u>Execution Report #1</u>	<u>Cancel</u>
		(38) = 50	(38) = 150	(38) = 150
		(151) = 0	(151) = 100	(151) = 0
		(14) = 50	(14) = 50	(14) = 100
		(5359) = 50	(5359) = 50	(5359) = 50
		<u>Execution Report #2</u>	<u>Execution Report #2</u>	N/A
		(38) = 50	(38) = 150	
		(151) = 0	(151) = 50	
		(14) = 50	(14) = 50	
		(5359) = 50	(5359) = 50	

10 **Appendix E: Appendix Removed**

11 Appendix F: Delta Conventions

11.1 Right Handed vs. Left Handed Delta

Delta will always be the left handed delta unless the USD currency is the term currency. If USD is the term currency, Delta will be right hand delta.

11.2 Straddle Delta Configuration

The API will not recognize 50 delta straddles, 50 delta straddles will always be considered 0 (zero) Delta.

11.3 Risk Reversal Delta Configuration

The Risk Reversal strategy will be either 25 or 10 Delta.

11.4 Butterfly Delta Configuration

The Butterfly strategy will be either 25 or 10 Delta.

11.5 Forward Delta Currency Pairs

All currency pairs that have tenor points of 1Y or less will provide the Spot Delta. All currency pairs that have tenor points greater than 1Y will provide the Forward Delta regardless of premium delivery.

12 Appendix G: Structure Entry

FIX Entry (9126 and 6258)	Structure Name	Supported Message Types
1	Call	Market Data Snapshot Full Refresh, Market Data Request, Quote Request, Quote, Quote Status Report, Quote Cancel, New Order Multileg, Execution Report, Trade Capture Report, News
2	Put	Market Data Snapshot Full Refresh, Market Data Request, Quote Request, Quote, Quote Status Report, Quote Cancel, New Order Multileg, Execution Report, Trade Capture Report, News
3	Straddle	Market Data Snapshot Full Refresh, Market Data Request, Quote Request, Quote, Quote Status Report, Quote Cancel, New Order Multileg, Execution Report, Trade Capture Report, News
4	Strangle	Market Data Snapshot Full Refresh, Market Data Request, Quote Request, Quote, Quote Status Report, Quote Cancel, New Order Multileg, Execution Report, Trade Capture Report, News
5	Risk Reversal (Non-differential)	Market Data Snapshot Full Refresh, Market Data Request, Quote Request, Quote, Quote Status Report, Quote Cancel, New Order Multileg, Execution Report, Trade Capture Report, News
	Risk Reversal (Differential)	Market Data Snapshot Full Refresh, Market Data Request, New Order Multileg, Trade Capture Report, News
6	Butterfly	Trade Capture Report, News, Quote Request (for anonymous RFQs), Quote, Market Data Snapshot Full Refresh, Market Data Request, New Order Multileg, Execution Report
7	Generic Spread	Trade Capture Report
8	Call Spread (Non-differential)	Market Data Snapshot Full Refresh, Market Data Request, Quote Request, Quote, Quote Status Report, Quote Cancel, New Order Multileg, Execution Report, Trade Capture Report, News
	Call Spread (Differential)	Trade Capture Report, News, Market Data Snapshot Full Refresh, Market Data Request, New Order Multileg, Execution Report
9	Put Spread (Non-differential)	Market Data Snapshot Full Refresh, Market Data Request, Quote Request, Quote, Quote Status Report, Quote Cancel, New Order Multileg, Execution Report, Trade Capture Report, News
	Put Spread (Differential)	Trade Capture Report, News, Market Data Snapshot Full Refresh, Market Data Request, New Order Multileg, Execution Report
10	Straddle vs. Call	Trade Capture Report, News, Market Data Snapshot Full Refresh, Market Data Request, New Order Multileg, Execution Report
11	Straddle vs. Put	Trade Capture Report, News, Market Data Snapshot Full Refresh, Market Data Request, New Order Multileg, Execution Report
12	Straddle vs. Straddle	Trade Capture Report, News, Market Data Snapshot Full Refresh, Market Data Request, New Order Multileg, Execution Report
13	Strangle vs. Call	Trade Capture Report, News, Market Data Snapshot Full Refresh, Market Data Request, New Order Multileg, Execution Report
14	Strangle vs. Put	Trade Capture Report, News, Market Data Snapshot Full Refresh, Market Data Request, New Order Multileg, Execution Report
15	Strangle vs. Strangle	Trade Capture Report, News, Market Data Snapshot Full Refresh, Market Data Request, New Order Multileg, Execution Report
16	Straddle vs. Strangle	Trade Capture Report, News, Market Data Snapshot Full Refresh, Market Data Request, New Order Multileg, Execution Report
55	Calendar Put Spread (Non-differential)	Market Data Snapshot Full Refresh, Market Data Request, Quote Request, Quote, Quote Status Report, Quote Cancel, New Order Multileg, Execution Report, Trade Capture Report, News
56	Calendar Call Spread (Non-differential)	Market Data Snapshot Full Refresh, Market Data Request, Quote Request, Quote, Quote Status Report, Quote Cancel, New Order Multileg, Execution Report, Trade Capture Report, News
65	FX (forward)	Market Data Snapshot Full Refresh, Market Data Request, Quote Request, Quote, Quote Status Report, Quote Cancel, New Order Multileg, Execution Report, Trade Capture Report, News
66	Structure (Exotics) Fenics Direct only	Quote Request, Quote, Quote Status Report, Quote Cancel, New Order MultiLeg, Execution Report, Trade Capture Report

13 Appendix H: Tenors

The spread tenors are designated for use with FX Spreads.

FIX Entry (6215)	Tenor Name
OD	Odd Date
ON	Overnight (for FXO with expiry at Today + 1 Day)
O/N	Today vs. Today + 1 Day
SP	Spot
SN	Spot vs. Spot + 1 Day
CT	Cash Tomorrow
TN	Tomorrow vs. Tomorrow + 1 Day
TF	1-Month outright as a differential between Today vs. 1-Month (USDCNY, USDIDR, USDINR, USDMYR, USDPHP, USDTWD)
Tod Fix	Today vs. 1-Month
TDPR	Today vs. 1-Month Promedio (USDCOP only)
TDY1M	Today vs. 1-Month
TDY2M	Today vs. 2-Month
Tom Fix	Tomorrow vs. 1-Month (USDCOP, USDCLP, USDARS, USDPEN)
TMF	Tomorrow vs. Tomorrow's 1-Month
TOM1M	Tomorrow vs. 1-Month
MAR	1-Month outright as a differential between Today vs. 1-Month (USDKRW only Market-Agreed Rate)
TMAR	Tomorrow vs. Tomorrow's 1-Month (USDKRW only Market-Agreed Rate)
1W SWAP	Spot vs. 1-Week
2W SWAP	Spot vs. 2-Week
1M SWAP	Spot vs. 1-Month
2M SWAP	Spot vs. 2-Month
3M SWAP	Spot vs. 3-Month
6M SWAP	Spot vs. 6-Month
9M SWAP	Spot vs. 9-Month
1Y SWAP	Spot vs. 1-Year
IMM	First IMM Date
IMM1	First IMM Date
IMM2	Second IMM Date
IMM3	Third IMM Date
IMM4	Fourth IMM Date
IMMROLL	IMM Roll of first IMM date into second IMM date
IMM1x2	First IMM date vs. second IMM date
IMM1x3	First IMM date vs. third IMM date
IMM1x4	First IMM date vs. fourth IMM date
IMM2x3	Second IMM date vs. third IMM date
IMM3x4	Third IMM date vs. fourth IMM date
BMF	First BMF (USDBRL only)
BMF2	Second BMF (USDBRL only)
BMFR	BMF Roll of first BMF into second BMF (USDBRL only)
PBMF	PTAX vs. BMF (USDBRL only)
TOMPBMF	Tomorrow's PTAX vs. BMF (USDBRL only)
BMFX1	BMF vs. 1-Month (USDBRL only)
BMFX2	BMF vs. 2-Month (USDBRL only)
BMFX3	BMF vs. 3-Month (USDBRL only)
BMFX6	BMF vs. 6-Month (USDBRL only)

BMFX9	BMF vs. 9-Month (USDBRL only)
BMFX12	BMF vs. 12-Month (USDBRL only)
BMFX18	BMF vs. 18-Month (USDBRL only)
BMFX24	BMF vs. 24-Month (USDBRL only)
BMFX36	BMF vs. 36-Month (USDBRL only)
ONOFFR	ON/OFF Roll of first onshore/offshore into second onshore/offshore (USDBRL only)
ONOFF1F	First Onshore vs. Offshore (USDBRL only)
ONOFF2F	Second Onshore vs. Offshore (USDBRL only)
CASADO	Casado (USDBRL only)
LOC1X6	Local 1-Month vs. 6-Month (USDCOP only)
LOC1X9	Local 1-Month vs. 9-Month (USDCOP only)
LOC1X12	Local 1-Month vs. 12-Month (USDCOP only)
1W	1-Week
2W	2-Week
3W	3-Week
28D	28-Day
1M	1-Month
2M	2-Month
3M	3-Month
4M	4-Month
5M	5-Month
6M	6-Month
7M	7-Month
8M	8-Month
9M	9-Month
10M	10-Month
11M	11-Month
12M	12-Month
1Y	1-Year
13M	13-Month
18M	18-Month
1.5Y	1.5-Year
21M	21-Month
24M	24-Month
2Y	2-Year
27M	27-Month
30M	30-Month
33M	33-Month
36M	36-Month
3Y	3-Year
39M	39-Month
42M	42-Month
45M	45-Month
48M	48-Month
4Y	4-Year
51M	51-Month
54M	54-Month
57M	57-Month
60M	60-Month
5Y	5-Year
63M	63-Month

66M	66-Month
69M	69-Month
72M	72-Month
6Y	6-Year
7Y	7-Year
8Y	8-Year
9Y	9-Year
10Y	10-Year
11Y	11-Year
12Y	12-Year
13Y	13-Year
14Y	14-Year
15Y	15-Year
16Y	16-Year
17Y	17-Year
18Y	18-Year
19Y	19-Year
20Y	20-Year
21Y	21-Year
22Y	22-Year
23Y	23-Year
24Y	24-Year
25Y	25-Year
26Y	26-Year
27Y	27-Year
28Y	28-Year
29Y	29-Year
30Y	30-Year
35Y	35-Year
40Y	40-Year
50Y	50-Year
1Wx2W	1-Week vs. 2-Week
1Wx3W	1-Week vs. 3-Week
1Wx1M	1-Week vs. 1-Month
1Wx2M	2-Week vs. 2-Month
2Wx3W	2-Week vs. 3-Week
2Wx1M	2-Week vs. 1-Month
3Wx1M	3-Week vs. 1-Month
1Mx2M	1-Month vs. 2-Month
1Mx3M	1-Month vs. 3-Month
1Mx4M	1-Month vs. 4-Month
1Mx5M	1-Month vs. 5-Month
1Mx6M	1-Month vs. 6-Month
1Mx7M	1-Month vs. 7-Month
1Mx8M	1-Month vs. 8-Month
1Mx9M	1-Month vs. 9-Month
1Mx10M	1-Month vs. 10-Month
1Mx11M	1-Month vs. 11-Month
1Mx12M	1-Month vs. 12-Month
1Mx1Y	1-Month vs. 1-Year
1Mx18M	1-Month vs. 18-Month

1Mx24M	1-Month vs. 24-Month
2Mx3M	2-Month vs. 3-Month
2Mx5M	2-Month vs. 5-Month
2Mx6M	2-Month vs. 6-Month
2Mx9M	2-Month vs. 9-Month
2Mx12M	2-Month vs. 12-Month
2Mx1Y	2-Month vs. 1-Year
3Mx4M	3-Month vs. 4-Month
3Mx6M	3-Month vs. 6-Month
3Mx9M	3-Month vs. 9-Month
3Mx12M	3-Month vs. 12-Month
3Mx1Y	3-Month vs. 1-Year
4Mx5M	4-Month vs. 5-Month
4Mx6M	4-Month vs. 6-Month
4Mx9M	4-Month vs. 9-Month
5Mx6M	5-Month vs. 6-Month
6Mx9M	6-Month vs. 9-Month
6Mx12M	6-Month vs. 12-Month
6Mx1Y	6-Month vs. 1-Year
6Mx24M	6-Month vs. 24-Month
9Mx12M	9-Month vs. 12-Month
9Mx1Y	9-Month vs. 12-Month
11Mx12M	11-Month vs. 12-Month
12Mx15M	12-Month vs. 15-Month
12Mx18M	12-Month vs. 18-Month
12Mx24M	12-Month vs. 24-Month
1Yx15M	1-Year vs. 15-Month
1Yx18M	1-Year vs. 18-Month
1Yx2Y	1-Year vs. 2-Year
18Mx24M	18-Month vs. 24-Month
2Yx3Y	2-Year vs. 3-Year
2Yx5Y	2-Year vs. 5-Year
3Yx5Y	3-Year vs. 5-Year

14 Appendix I: Market Cutoff Times & Associated Currencies

FIX Entry (9125)	Associated Currency	Cut Code	Cutoff	Local Time	Product	Platform
*1	Majority	NY	New York	10:00:00	FXO	bgcforex/ForexMatch/Fenics Direct
*2	Majority Asian	TK	Tokyo	15:00:00	FXO	bgcforex/ForexMatch/Fenics Direct
*3	MXN	MX	New York	12:30:00	FXO	bgcforex/ForexMatch/Fenics Direct
*4	PLN	WA	Warsaw	11:00:00	FXO	bgcforex/ForexMatch/ Fenics Direct
*5	HUF	BD	Budapest	12:00:00	FXO	bgcforex/ForexMatch/ Fenics Direct
*6	ILS (Friday Expiry)	TF	Tel Aviv – Fri	12:00:00	FXO	bgcforex/ForexMatch/ Fenics Direct
*8	IDR	JKT	Jakarta	16:15:00	FXO	bgcforex/ForexMatch/ Fenics Direct
*9	THB	BK	Bangkok	13:00:00	FXO	Fenics Direct
*10	KRW	SL	Seoul	15:30:00	FXO	bgcforex/ForexMatch/ Fenics Direct
*11	TWD	TP	Taipei	11:00:00	FXO	bgcforex/ForexMatch/ Fenics Direct
*12	Majority Asian	HK	Hong Kong	14:00:00	FXO	Fenics Direct
*13	INR	MB	Mumbai	13:30:00	FXO	bgcforex/ForexMatch/ Fenics Direct
*14	PHP	MN	Manila	11:30:00	FXO	bgcforex/ForexMatch/ Fenics Direct
*15	MYR	KL	Kuala Lumpur	11:00:00	FXO	bgcforex/ForexMatch/ Fenics Direct
*16	SGD	SG	Singapore	14:00:00	FXO	Fenics Direct
*17	CNY	BJ	Beijing	9:15:00	FXO	bgcforex/ForexMatch/ Fenics Direct
*18	TRY	LD	London	12:00:00	FXO	bgcforex/ForexMatch/ Fenics Direct
19	RUB	EM	EMTA	0:00:00	FXO	bgcforex/ForexMatch
*20	ILS	TL	Tel Aviv	15:00:00	FXO	bgcforex/ForexMatch/ Fenics Direct
*21	Precious Metals NY	NB	New York Bullion	9:30:00	FXO	bgcforex/ForexMatch/ Fenics Direct
22	KZT	AF	AFINEX01	0:00:00	FXO	bgcforex/ForexMatch
*23	ROM	BC	Bucharest	13:00:00	FXO	bgcforex/ForexMatch/ Fenics Direct
24	CLP	CH	Chile	0:00:00	FXO	bgcforex/ForexMatch
25	Emerging Markets NDFs	-	-	-	FXO	bgcforex/ForexMatch
*26	VND	HN	Hanoi	11:00:00	FXO	Fenics Direct
*27	RUB	6CB37	EMTA/ RUB	-	FXO	Fenics Direct
*28	JPY	TKSL	TOKYO	14:00:00	FXO	bgcforex/ForexMatch/ Fenics Direct
*29	RSD	BG	Belgrade	13:00:00	FXO	Fenics Direct
30	RUB	EMTA/ECB37	EMTA/ECB37	0:00:00	FXO	bgcforex/ForexMatch/G Fenics FI Direct
*31	UAH	KievEM	Europe/Kiev	11:00:00	FXO	Fenics Direct
*32	UAH	KievUA	Europe/Kiev	11:00:00	FXO	Fenics Direct
33	ARS	BA	Buenos Aires	13:00:00	FXO	bgcforex/ForexMatch/ Fenics Direct
34	CLP	SA	Santiago	10:30:00	FXO	bgcforex/ForexMatch
*35	MXN	NYEM	New York	12:30:00	FXO	bgcforex/ForexMatch/ Fenics Direct
36	COP	BT	Bogota	10:30:00	FXO	bgcforex/ForexMatch

37	PEN	LM	Lima	0:00:00	FXO	bgcforex/ForexMatch/ Fenics Direct
*38	AUD	SY	Sydney	15:00:00	FXO	Fenics Direct
*39	NZD	WL	Wellington	15:00:00	FXO	Fenics Direct
40	KRW	KFTC30	Seoul NDF Fixing	14:00:00	FXO	bgcforex/ForexMatch
41	AUDKRW	AUSL	Sydney/ Seoul	15:00:00	FXO	bgcforex/ForexMatch Asia
42	AUDTWD	AUTP	Sydney/ Taipei	13:00:00	FXO	bgcforex/ForexMatch Asia
43	AUDINR	AUMB	Sydney/ Mumbai	16:00:00	FXO	bgcforex/ForexMatch Asia
44	AUDMYR	AUKL	Sydney/ Kuala Lumpur	13:00:00	FXO	bgcforex/ForexMatch Asia
45	AUDPHP	AUMN	Sydney/ Manila	13:00:00	FXO	bgcforex/ForexMatch Asia
46	RUB	MS	Moscow	13:30:00	FXO	bgcforex/ForexMatch
*47	EURRUB	CET	Europe/ Moscow	13:30:00	FXO	Fenics Direct
49	BRL	SP	Sao Paulo	13:15:00	FXO	bgcforex/ForexMatch
*50	CNY	BE15	Beijing	15:00:00	FXO	bgcforex/ForexMatch/ Fenics Direct
*51	Majority Asian	HK	Hong Kong	11:15:00	FXO	Fenics Direct
*53	KRW	KRSE	Seoul	15:30:00	FXO	Fenics Direct
54	BRLCNY	SPBJ	Sao Paulo/ Beijing	8:00:00	FXO	bgcforex/ForexMatch Asia
55	INR	TKMB	Tokyo/ Mumbai	15:30:00	FXO	bgcforex/ForexMatch
56	MYR	TKKL	Tokyo/ Kuala Lumpur	12:00:00	FXO	bgcforex/ForexMatch
*57	AUD	HSRA	Sydney	10:00:00	FXO	bgcforex/ForexMatch/ Fenics Direct
103	RUB	MOS	Moscow	10:00	FXO	bgcforex/ForexMatch
999	-	SL14	2pm Seoul	2:00 PM	FXO	Fenics Direct
999	-	TOC	4pm Tokyo	4:00 PM	FXO	Fenics Direct
156	Majority	BFIXFFT1400A	BFIX FFT 1400 auto expiry	14:00:00	FXO	Fenics Direct
157	Majority	BFIXLDN1300A	BFIX London 1300 Auto expiry	13:00:00	FXO	Fenics Direct
158	Majority	BFIXLDN1400A	BFIX London 1400 auto expiry	14:00:00	FXO	Fenics Direct
159	Majority	BFIXLDN1600A	BFIX London 1600 auto expiry	16:00:00	FXO	Fenics Direct
160	Majority	BFIXNY1000A	BFIX New York 1000 auto expiry	10:00:00	FXO	Fenics Direct
161	Majority	BFIXTK1500A	BFIX Tokyo 1500 auto expiry	15:00:00	FXO	Fenics Direct
162	Majority	BFIXTK1700A	BFIX Tokyo 1700 auto expiry	17:00:00	FXO	Fenics Direct
163	Majority	BUNCPB	Budapest New Change PB	12:00:00	FXO	Fenics Direct
164	Majority	BDNC	Budapest New Change	12:00:00	FXO	Fenics Direct
166	Majority	EURKRW	EURKRW BFIX 2PM KFTC30 2pm	14:00:00	FXO	Fenics Direct
167	Majority	JPYKRW	JPYKRW BFIX 2PM KFTC30 2PM	14:00:00	FXO	Fenics Direct
168	Majority	LDBFIX	London BFIX	16:00:00	FXO	Fenics Direct
169	Majority	LDNC1200PB	London New Change 12:00 PB	12:00:00	FXO	Fenics Direct
170	Majority	LDNC1200	London New Change 12:00	12:00:00	FXO	Fenics Direct

171	Majority	LDWMR1	London WMR 1pm	13:00:00	FXO	Fenics Direct
172	Majority	LDWMR4	London WMR 4pm	16:00:00	FXO	Fenics Direct
175	Majority	NYNC1230PB	New York New Change 12:30 PB	12:30:00	FXO	Fenics Direct
173	Majority	NYNC1230	New York New Change 12:30	12:30:00	FXO	Fenics Direct
174	Majority	NYNCPB	New York New Change PB	10:00:00	FXO	Fenics Direct
176	Majority	NYNC	New York New Change	10:00:00	FXO	Fenics Direct
177	Majority	NYWMR	New York WMR	10:00:00	FXO	Fenics Direct
178	Majority	TKBFX	Tokyo BFIX	15:00:00	FXO	Fenics Direct
179	Majority	TKNCPB	Tokyo New Change PB	15:00:00	FXO	Fenics Direct
180	Majority	TKNC	Tokyo New Change	15:00:00	FXO	Fenics Direct
181	Majority	WANCPB	Warsaw New Change PB	11:00:00	FXO	Fenics Direct
182	Majority	WANC	Warsaw New Change	11:00:00	FXO	Fenics Direct

15 Appendix J: Session Level Error Messages

The following error message can be received in the Text field (58) of the Reject message.

This list of error messages is for informational purposes only and may not be complete.

FIX Engine Session Error Messages	
Error Code (373)	Text Description (58)
0	Invalid tag number
1	Required tag missing
2	Tag not defined for this message type
3	Undefined Tag
4	Tag specified without a value
5	Value is incorrect (out of range) for this tag
6	Incorrect data format for value
7	Decryption problem
8	Signature problem
9	CompID problem
10	SendingTime accuracy problem
11	Invalid MsgType
12	XML Validation error
13	Tag appears more than once
14	Tag specified out of required order
15	Repeating group fields out of order
16	Incorrect NumInGroup count for repeating group
17	Non "data" value includes field delimiter (SOH character)
99	Other

16 Appendix K: Application Level Error Messages

16.1 Business Message Reject Messages

The following error messages can be received in the Text field (58) of the Business Message Reject message.

Business Reject Reason (380)	Text Description (58)
0	Other
1	Unknown ID
2	Unknown Security
3	Unsupported Message Type (receive a valid, but unsupported MsgType (35)
4	Application not available
5	Conditionally Required Field Missing
6	Not authorized
7	DeliverTo firm not available at this time

16.2 Quote Status Report & Execution Report Error Messages

The following error messages can be received in the Text field (58) of the Quote Status Report or Execution Report messages.

This list of error messages is for informational purposes only and may not be complete.

Reject / Cancel Codes (9466, 300, 658, or 103)	Text Description (58)
99	Other
1000	Generic reason code for different errors. E.g. "The request was timed out", "Allow inverted Market" etc.
1001	User not found.
1002	The Username and/or Password entered does not match our records.
1003	This is an invalid Structure.
1004	This is an invalid Currency Pair. A slash ("/") is not permitted in the format or the Currency Pair is not supported.
1005	This is an invalid Price for this Structure.
1006	This is an invalid Tenor for this Structure.
1007	This is an invalid Size for this Structure.
1008	This is an invalid Size Currency for this Structure.
1009	The Size Currency must be specified for this Structure.
1010	This Quote entry produces an inverted market. Inverted markets are not allowed for this product.
1011	This user cannot act on behalf of this API client.
1012	This user does not have permission to trade this Structure.
1013	The Identifier provided is not unique or is incorrectly formatted.
1014	This QuoteID does not exist.
1015	This APIQuoteID does not exist.
1016	This CIOrdID does not exist.
1017	This OrderID does not exist.
1018	This order is below the minimum quantity needed for execution.
1019	There are no prices at this level to execute on.
1020	An execution cannot occur on your own price.
1021	This is an invalid Cutoff for this Structure.
1022	This is an invalid Delta for this Structure.
1023	This Structure does not allow negative prices.
1024	This user is not permitted to enter prices.
1025	This Quote cannot be canceled.
1026	This is an invalid Calls Over / Puts Over for this Structure.
1027	This Quote is no longer active and may not be modified.
1028	This is an invalid QuoteID.

1029	This is an invalid ClOrdID.
1030	This Quote was canceled by the client.
1031	This Quote has been canceled because the OCO price protection parameter was triggered.
1032	This Quote has been rejected because one or more components of the message had an incorrect value.
1033	This Quote was Removed from Market because it was executed.
1034	This Quote has been Expired due to the application running its end of day process.
1035	This is an invalid OrigQuoteID
1036	This is an invalid Prime Broker.
1037	Cannot trade due to credit restrictions.
1038	A Quote may only be modified for a size decrease or an OCO indicator change.
1039	This Quote was canceled due to a disconnect between your application and the API.
1040	This Quote has been canceled because the market was referred.
1041	No quotes exist for the specified Quote Cancel parameters.
1042	This QuoteReqID does not exist.
1043	Market doesn't exit.
1044	Target client offline
1045	This particular specific interest already exists in the System
1046	Only acceptable value is 1
1047	Qty submitted is below the minimum acceptable size
1048	Target bank has no interest in responding to this RFQ
1049	Insufficient quantity to execute in the market
1050	SDR/Clearer with value : "value" not found
1051	Quote is canceled because the bid/offer of a two-way quote was executed.
1052	User/LEI is not permitted to transact in this market.
1053	LEI is required

16.3 Market Data Request Reject Error Messages

Text Description (58)	Error Reason
No permission to request ccypair: PNLHUF.	API client requesting market data on a currency pair that is not supported
Structure with value: 100 not found.	API client requesting market data on an incorrect structure value
Subscription can only be done using symbol (field:55) and structure (field:9126). Please unset field: 6215.	Market Data Request may be sent using valid currency pair and/or structure only; making requests utilizing any other tags will result in this error

16.4 CheckSum Calculation

The checksum of a FIX message is calculated by summing every byte of the message up to but not including the checksum field itself. This checksum is then transformed into a modulo 256 number for transmission and comparison.

For transmission, the checksum must be sent as printable characters, so the checksum is transformed into three ASCII digits.

For example, if the checksum has been calculated to be 274 then the modulo 256 value is 22. This value would be transmitted as CheckSum (10) = 022.

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